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VISCIDI et al.(10) **Pub. No.: US 2025/0276056 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **CYTOLYTIC T CELL IMMUNOTHERAPY
FOR HIGHLY PATHOGENIC
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(57)

ABSTRACT

Compositions and methods to induce cytolytic T lymphocytes (CD8+) response, that is, MHC class I restricted T cell responses, to pathogenic and common cold coronaviruses, including a delivery platform for antigens consisting of a polyionic papillomavirus virus-like particle (VLP), with contiguous, negatively charged amino acids flanked by a cysteine residue inserted in the HI loop of the papillomavirus L1 protein. Antigens to be paired with the VLP include fusion peptide/proteins derived from a pathogenic coronavirus, and from the genetically most closely related human coronaviruses that commonly circulate in human populations, with N-terminal or C-terminal amino acids consisting of contiguous, positively charged amino acids preceded and/or followed by a cysteine residue and a C-terminal proteolytic processing sequence (AAYY) to enhance presentation of MHC class I epitopes.

Specification includes a Sequence Listing.

Figure 1: Alignment of bovine papillomavirus (BPV) type 1 L1 protein with the L1 protein of the closely related BPV isolate NY8385: HI loop is aa344-357

	300	310	320	330	340	350
BPV1	SLVSTDNQIFNRPYWLFR	QAQGMNNGIAWNNLL	FLTVDNTRG	TNLTIS	<u>VASDGTPLTEYD</u>	
		:::: :::: :::		:::	: :	::
BPVX	SLVTSSNQLFNKPYWIR	RAQGNNNGILWGNQC	FVTVD	TTRG	TNFDISVPTAGG	--TQYK
	300	310	320	330	340	350
	360	370	380	390	400	410
BPV1	<u>SSKF</u> NVYHRHMEEYKL	AFIL	ELCSVEITAQT	VSHLQGLMPSV	LENWEIGV	QPPTSSILED
			 : : .		.	
BPVX	ATEFNQYKRHVVEEYEL	QFVLQACVVRLT	PETVAHLYQ	MSRILDDWN	LGVVTSQ	NGALAE
	360	370	380	390	400	410

Figure 2: Alignment of M proteins of OC43 and SARS-CoV-2

Waterman-Eggert score: 588; 138.6 bits; $E(1) < 9.6e-38$

40.8% identity (73.7% similar) in 213 aa overlap (9-220:14-226)

	10	20	30	40	50	60	
SARS-2	TVEELKKLLEQWN	LVI	GFLFTWICLLQ	FAYANRNRFLY	IIKLIFLWLLWP	VT	LACFVLA
	...	...	...	...	...	...	..
OC43	TADEAIKFLKEWN	FSLGIILLFIT	IILQFGYTSR	SMFVYVIKMI	ILWLMWPLTI	ILTIFN	
	20	30	40	50	60	70	
	70	80	90	100	110	120	
SARS-2	AVYRINWITGGIA	IAMACLVGLMW	LSYFIASFRLF	ARTRSMWSENP	ETNILLNVPLH	GTI	
	::	::	.		..		::
OC43	CVYALNNVYLGL	SIVFTIVAIIM	WIVYFVNSIR	LFIRTSFWSFN	PETNNLMCID	MDKGMTM	
	80	90	100	110	120	130	
	130	140	150	160	170	180	
SARS-2	LTRPLLESELVIG	AVILRGHLRI	AGHHLGRC-DI	KDLPKEITVAT	SRTL	SSYKLGASQ	RV
		..		:	:	::	..
OC43	YVRPIIEDYHTL	TVTIIRGHLYI	QGIKLTGYSL	ADLPAYMTVA	KVTHLCTYKR	GFLDRI	
	140	150	160	170	180	190	
	190	200	210	220			
SARS-2	AGDSGFAAYSRY	RIGNYKLNTDH	SSSSDNIAL	L			
	.		.		.	...	::
OC43	SDTSGFAVYVK	SKVGNRYRLP	STQKGS	GMDTALL			
	200	210	220				

Figure 3. Alignment of M proteins of OC43 and HKU1

Waterman-Eggert score: 1193; 246.6 bits; $E(1) < 2.9e-70$
80.9% identity (95.8% similar) in 215 aa overlap (9-222:13-227)

	10	20	30	40	50	60	
HKU	FTSDQAVTFLKEWNFSLGVILLFITIILQFGYTSRSMFVYLIKMIILWLMWPLTITLTIF						
							
OC43	WTADEAIKFLKEWNFSLGIILLFITIILQFGYTSRSMFVYVIKMIILWLMWPLTIILTIF						
	20	30	40	50	60	70	
	70	80	90	100	110	120	
HKU	NCFYALNNAFLAFSIVFTIISIVIWILYFVNSIRLFIRTGSWWSFNPETNNLMCIDMKGK						
	::						
OC43	NCVYALNNVYLGLSIVFTIVAIIMWIVYFVNSIRLFIRTGSFWSFNPETNNLMCIDMKGT						
	80	90	100	110	120	130	
	130	140	150	160	170	180	
HKU	MFVRPVIEDYHTLTATVIRGHLYIQGVKLGTGYTLSDLPVYVTVAKVQVLCTYKRAFLDK						
							
OC43	MYVRPIIEDYHTLTVTIIRGHLYIQGIKLGTGYSLADLPAYMTVAKVTHLCTYKRGFLDR						
	140	150	160	170	180	190	
	190	200	210	220			
HKU	<u>LDVNSGFAVFVKSKVGN</u> YRLPSS-KPSGMDTALLR						
	..						
OC43	ISDTSGFAVYVKSKVGNYRLPSTQKSGMDTALLR						
	200	210	220				

Alignment of antigenic regions of N proteins of OC43 and HKU1

Figure 5AWaterman-Eggert score: 165; 54.8 bits; $E(1) < 5.1e-14$ **50.0% identity** (92.5% similar) in 40 aa overlap (1-40:1-40)

	10	20	30	40
OC43	PHAKDQYGT	DIDGVYWVAS	NQADVNTPA	DIVDRDPSSDEA
		...		
HKU	PYANASYGE	SLEGVFWV	ANHQADTST	PSDVSSRDPTTQEA
	10	20	30	40

Figure 5BWaterman-Eggert score: 67; 18.6 bits; $E(1) < 0.0025$ **43.8% identity** (62.5% similar) in 32 aa overlap (1-32:1-32)

	10	20	30
OC43	RSTSRTSS	RASSAGSR	SRANSNGNRTPTSGVTP
	:		
HKU	RPGSRSQ	SRGPNNR	SLSRSNSNFRHSDSIVKP
	10	20	30

Figure 5CWaterman-Eggert score: 32; 15.1 bits; $E(1) < 0.0098$ **33.3% identity** (80.0% similar) in 15 aa overlap (1-15:1-15)

	10
OC43	SRLELAKVQNL
	
HKU	SKLDLVKRDSEADSP
	10

Figure 6A

Waterman-Eggert score: 897; 230.9 bits; $E(1) < 2.1e-65$

52.7% identity (81.2% similar) in 260 aa overlap (1-256:1-260)

	10	20	30	40	50	60
OC43	SKASSRSAIEDLLFDKVKLSDVGFVEAYNNCTGGAEIRDLCVQSYKGIKVLPPLLSENQ					
	:: :::: :					
SARS-2	SKPSKRSFIEDLLFNKVTLADAGFIKQYGDCLGDIARDLICAQKFNGLTVLPPLLTDEM					
	10	20	30	40	50	60
	70	80	90	100	110	
OC43	ISGYTLAATSASLFPPWTAAAG----VPFYLVNQYRINGLGVMTDVLSQLNQLIANAFNN					
	:: :: : ... :: :					
SARS-2	IAQYTSALLAGTITSGWTFGAGAALQIPFAMQMAYRFNGIGVTQNVLLENQKLIANQFNS					
	70	80	90	100	110	120
	120	130	140	150	160	170
OC43	ALYAIQEGFDATNSALVKIQAVVNANAELNLLQQLSNRFGAISASLQEILSRLDALEA					
	:: :					
SARS-2	AIGKIQDSLSSTASALGKLQDVVNQNAQALNTLVKQLSSNFGAISSVLNDILSRDKVEA					
	130	140	150	160	170	180
	180	190	200	210	220	230
OC43	EAQIDRLINGRLTALNAYVSQQLS DSTLVKFSAAQAMEKRVNECVKSQS SRINF CGNGNHI					
	: :					
SARS-2	EVQIDRLITGR LQSLQTYVTQQLIRAAEIRASANLAATKMSECVLGQSKRVDFCGKGYHL					
	190	200	210	220	230	240
	240	250				
OC43	ISLVQNAPYGLYFIHFSYVP					
	:: :					
SARS-2	MSFPQSAPHGVVFLHVTTYVP					
	250	260				

Figure 6B

Waterman-Eggert score: 277; 83.2 bits; $E(1) < 5.1e-22$

51.3% identity (82.9% similar) in 76 aa overlap (1-75:1-76)

```

          10          20          30          40          50
OC43    PNLPDFKEELDQWFKNQTSVAPDLS-LDYINVTFDLQVEMNRLQEAIKVLNQSYINLKD
        :: :::::::::::::: ::. . ::.. .::: :::::::::: : ::: ::..
SARS-2  PELDSFKEELDKYFKNHTSPDVDLGDISGINASVVNIQKEIDRLNEVAKNLNESLIDLQE
          10          20          30          40          50          60
        60          70
OC43    IGTYEYYVKWPWYVWL
        : :: ::::::::::::
SARS-2  LGKYEQYIKWPWYIWL
          70

```

Figure 7. Alignment of S proteins of HKU1 and OC43 within a variable antigenic regionWaterman-Eggert score: 356; 119.0 bits; $E(1) < 1.5e-31$

70.7% identity (88.0% similar) in 75 aa overlap (1-73:1229-1303)

	10	20	30	40	50
OC43	PNLPDFKEELDQWFKNQTSVAPDLSLDY--	INVTFLDLQVEMNRLQEAIKVLNQSYINLK			
	::: ::: ::::::::::::::::::::::	::::::::: ::: ::::: :::::::::::			
HKU	PKLSDFESELSHWFKNQTSIAPNLTLNLHTINATFLDLYYEMNLIQESIKSLNNSYINLK				
	1230	1240	1250	1260	1270
					1280
	60	70			
OC43	DIGTYEYYVKWPWYV				
	::::::::::::::::::::				
HKU	DIGTYEMYVKWPWYV				
	1290	1300			

Alignment of antigenic regions of ORF1ab of OC43 with SARS-CoV-2 nsp3 protein

Figure 8A

Waterman-Eggert score: 425; 118.2 bits; E(1) < 1.9e-30
52.8% identity (84.0% similar) in 106 aa overlap (1-106:2368-2473)

	10	20	30	40	50	60
nsp3	MVRMYIFFASFYVWKS	YVHVVDGCNSSTC	MMCYKRN	RATRVECTTIV	NGVRRSFYVYAN	
			... :			... :
OC43	FMRFYIIIIASFIKLF	SLFRHVAYGCSKSG	CLFCYKRN	RLRVK	STIVGGMIRYYD	VMAN
	2370	2380	2390	2400	2410	2420

	70	80	90	100
nsp3	GGKGFCKLHNWNCV	NCDTFCAGSTFIS	DEVARDSLQFKR	PINPTD
	::			
OC43	GGTGFCSKHQWNC	IDCDSYKPGNTF	ITVEAALDLSKEL	KRPIQPTD
	2430	2440	2450	2460

Figure 8B

Waterman-Eggert score: 167; 60.1 bits; E(1) < 3.6e-13
41.0% identity (80.3% similar) in 61 aa overlap (2542-2602:1-61)

	2550	2560	2570	2580	2590	2600
OC43	NAAVFYAQSLFRPIL	MVDKNLIT	TANTGT	SVTETMFDVY	VDTF	FLSMFDV
						
SARS	SASVYYSQ	LMCQPILL	LDQALVSD	VGDSAEVAV	KMFDA	YVNTFS
	10	20	30	40	50	60

OC A
 :
SARS A

Figure 8C

Waterman-Eggert score: 156; 55.1 bits; E(1) < 1e-11
43.6% identity (74.5% similar) in 55 aa overlap (2651-2705:1-55)

	2660	2670	2680	2690	2700
OC43	ELTDESCNNLVPT	YLKSDNIVAADL	GVLIQNSAKH	VQGNVAKIAG	VSCIWSVDAF
	::				
SARS	EVTGDSCNNYML	TYNKVENMT	PRDLGACID	CSARHINAQ	VAKSHNIALI
	10	20	30	40	50

Alignment of antigenic regions of ORF1ab of OC43 with SARS-CoV-2 nsp4 protein

Figure 9A

Waterman-Eggert score: 174; 60.7 bits; E(1) < 1.7e-13

53.3% identity (75.6% similar) in 45 aa overlap (2875-2919:1-45)

	2880	2890	2900	2910
OC43	SADGVQCYTPHSQISYSNIFYASGCVLSSACTMFTMADGSPQPYCY			
	:: : :::: . : :::: : :::: : :::: : :::: : ::::			
nsp4	SAVGNICYTPSKLIEYTDFAISACVLAAECTIFKDASGKPVPCY			
	10	20	30	40

Figure 9B

Waterman-Eggert score: 446; 134.8 bits; E(1) < 2.5e-35

47.4% identity (76.7% similar) in 133 aa overlap (2932-3064:1-132)

	2940	2950	2960	2970	2980	2990
OC43	SLVPHVRYNLANAKGFIRFPEVLREGLVRIVRTRSMSYCRVGLCEEADGICFNFNGSWV					
	:: : :::: : ... : :::: : :::: : : . :::: : :::: : :::: : ::::					
nsp4	SLRPDTRYVLM DGS-IIQFPNTYLEGSRVVTTFDSEYCRHGTCEASEAGVCVSTSGRWV					
	10	20	30	40	50	

	3000	3010	3020	3030	3040	3050
OC43	LNNDYYRSLPGTFCGRDVF DLIYQLFKGLAQPVDFLALTASSIAGAILAVIVVLVFYYLI					
	:: ::					
nsp4	LNNDYYRSLPGVFCGVDAVNLLTNMFTPLIQPIGALDISASIVAGGIVAIVVTC LAYYEM					
	60	70	80	90	100	110

	3060
OC43	KLKRAFGDYTSVV
	:: ::
nsp4	RFRRAFGEYSHVV
	120 130

Figure 9C

Waterman-Eggert score: 249; 88.5 bits; E(1) < 7.8e-22

72.0% identity (92.0% similar) in 50 aa overlap (3187-3236:1-49)

	3190	3200	3210	3220	3230
OC43	NRYLSLYNKYRYYSKMDTAAYREAACSQLAKAMDTFTNNNGSDVLYQPP				
	:: ::				
nsp4	NRYLALYNKYKYFSGAMDTTSYREAACCHLAKALNDFSNS-GSDVLYQPP				
	10	20	30	40	

Figure 10A

54.8% identity (82.1% similar) in 84 aa overlap (3753-3836:1-84)

	3820	3830
OC43	EALMLNFKLLGIGGVPIIEVSQFQ	
	... ::::::::::: : ::. :	
nsp6	DAFKLNIKLLGVGGKPCIKVATVQ	
	70	80

Waterman-Eggert score: 95; 34.7 bits; $E(1) < 9.7e-06$

	3610	3620	3630
OC43	SLAMLLVKHKHLYLTMYITPVLF	TLLYNNYL	
	:.	...:.	:..
nsp6	AFAMMFVKHKHAF	LCFLPLSLATV	AYFNMV
	10	20	30

Waterman-Eggert score: 235; 82.3 bits; $E(1) < 8.1e-20$

	3840	3850	3860	3870	3880	3890
OC43	SKLTDVKCANVLLNCLQHLHVASNSKLWHYCSTLHNEILATSDLSVAFEKLAQLLIVLF :::..... :... :... : :::::					
nsp7	SKMSDVKCTSVVLLSVLQQLRVESSSKLWAQCVQLHNDILLAKDTTEAFEKMVSLLSVLL 10 20 30 40 50 60					

	3900
OC43	ANPAAVD
	
nsp7	SMQGAVD

Figure 12.

capsid protein L1 [Deltapapillomavirus 4]

NCBI Reference Sequence: NP_056744.1

GenPept Identical Proteins Graphics

>NP_056744.1 capsid protein L1 [Deltapapillomavirus 4]

MALWQQGQKLYLPPTPVSKVLCSETYVQRKSIFYHAETERLLTIGHPYYPVSIGAKTVPKVSANQYRVFK
IQLPDPNQFALPDRTVHNPSKERLVWAVIGVQVSRGQPLGGTVTGHPTFNALLDAENVNRKVTTQTTDDR
KQTGLDAKQQQILLGCTPAEGEYWTTARPCVTDRLENGACPPLELKNKHIEDGDMMEIGFGAANFKEIN
ASKSDLPLDIQNEICLYPDYLKMAEDAAGNSMFFFARKEQVYVRHIWTRGGSEKEAPTTFYLNKNGDA
TLKIPSVHFGSPSGSLVSTDNQIFNRPYWLFRAQGMNNGIAWNNLLFLTVGDNTRGTNLTISVASDGTPL
TEYDSSKFNVYHRHMEEYKLAFILELCSVEITAQTVSHLQGLMPSVLENWEIGVQPPTSSILEDTRYIE
SPATKCASNVIPAKEDPYAGFKFWNIDLKEKLSLDLDQFPLGRRFLAQQGAGCSTVRKRRISQKTSSKPA
KKKKK

Figure 13A: Alignment of OC43 ORF1ab with nsp12 protein of SARS-CoV-2

61.5% identity in 91 residues overlap; Score: 324.0; Gap frequency: 0.0%

```

OC43          4525 YFTKKDWYDFVENPDIINVYKKLGPIFNALVSATEFADKLVEVGLVGLVTLTDNQLNGK
SARS-CoV2      1 YFNKKDWYDFVENPDILRVYANLGERVRQALLKTVQFCDAMRNAGIVGVLTLDNQLNGN
                ** ***** ** ** ** * * * *****

```

```

OC43          4585 WYDFGDYVIAAPGCGVAIADSYYSYIMPLT
SARS-CoV2      61 WYDFGDFIQTPGSGVPVVDSSYSLMPILT
                ***** ** ** ***** ** **

```

Figure 13B: Alignment of OC43 ORF1ab with nsp12 protein of SARS-CoV-2

67.3% identity in 107 residues overlap; Score: 419.0; Gap frequency: 0.0%

```

OC43          4634 DLVQYDFTDYKLELFNKYFKHWSMPYHPNTVDCQDDRCIIHCANFNILFSMVLNPTCFGP
SARS-CoV2      1 DLLKYDFTEERLKLFDRYFKYWDQTYHPCNVNCLDDRCILHCANFNVLESTVFPTSFPG
                ** **** * ** *** * ***** * * * * * * * * * *

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```

OC43          4694 LVRQIFVDGVFPFVVSIGYHYKELGIVMNMVDVTHRYRLSLKDLLLYA
SARS-CoV2      61 LVRKIFVDGVFPFVSTGYHFRELGVVHNQDVNLHSSRLSFKELLVYA
                *** ***** *** * * * * * * * * * *

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Figure 13C Alignment of OC43 ORF1ab with nsp12 protein of SARS-CoV-2

76.8% identity in 341 residues overlap; Score: 1396.0; Gap frequency: 0.0%

```

OC43          4894 AYTKRNVLPRTLQMNLYAISAKNRARTVAGVSILSTMTGRMFHQKCLKLSIAATRGPVW
SARS-CoV2      1 AYTKRNVIPITITQMNLYAISAKNRARTVAGVSICSTMTNRQFHQKLLKSIAATRQATV
                ***** ** ***** ***** * * * * * * * * * *

```

```

OC43          4954 IGTTFYGCWDDMLRRLIKVDNPNVLMGWDYPKCDRAMPNLLRIVSSLVLARKHETCCSQ
SARS-CoV2      61 IGTSKFYGCWHNMLKTVYSDEVNPHLMGWDYPKCDRAMPNMLRIMASLVLARKHTTCCSL
                *** ***** ** ** * * ***** * * * * * * * *

```

```

OC43          5014 SDRFYRLANCAQVLSEIVMCGGCYVVKPGGTSSGDATTAFANSVFNICQAVSANVCALM
SARS-CoV2      121 SHRFYRLANCAQVLSEVMCGGSLYVKPGGTSSGDATTAYANSVFNICQAVTANVNALL
                * ***** * * * * * * * * * * * * * *

```

```

OC43          5074 SCNGNKIEDLSIRALQKRLYSHVYRSDKVDSTFVTEYYEFLNKHFSMMILSDDGVVCYNS
SARS-CoV2      181 STDGINKIADKYVRNLQHRLEYECLYRNRDVTDFVNEFYAYLRKHFSMMILSDDAVVCFNS
                * **** * * ** * * * * * * * * * * * * * *

```

```

OC43          5134 DYASKGYIANISAFQQVLYYQNNVFMSESKCWVEHDINNGPHEFCSQHTMLVKMDGDDVY
SARS-CoV2      241 TYASQGLVASIKNFKSVLYYQNNVFMSEAKCWTETDLTKGPHEFCSQHTMLVKQGGDDVY
                *** * * * * * ***** * * * * * * * * * *

```

```

OC43          5194 LPYPNPSRILGAGCFVDDLKTDVLLIERFVSLAIDAYPL
SARS-CoV2      301 LPYPDPSRILGAGCFVDIVKTDGTLMIERFVSLAIDAYPL
                **** ***** ** * *****

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CYTOLYTIC T CELL IMMUNOTHERAPY FOR HIGHLY PATHOGENIC CORONAVIRUSES

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to compositions and methods for inducing cellular immune responses to coronaviruses.

Description of the Background

[0002] Coronaviruses (CoVs) are enveloped viruses with a single-strand, positive-sense RNA genome approximately 26-32 kilobases in size. SARS-CoV-2, like all coronaviruses, shares common features in the organization and expression of its genome; nonstructural proteins encoded by open reading frame (ORF) 1a/b, are followed by the principal structural proteins spike(S), envelope (E), membrane (M), and nucleocapsid (N) (Chen et al., 2020). The S glycoprotein mediates attachment to the host receptor. The S glycoprotein is cleaved by a host protease into two separate polypeptides designated S1 and S2. S1 makes up the receptor binding domain of the S protein and is the principal target of neutralizing antibodies, while S2 forms the stalk of the spike molecule. The M protein is the most abundant viral protein and participates in the formation of the core viral particle. The N protein is the only nucleocapsid protein and forms the portion of the core particle that interacts with the viral genome. The CoVs are separated into four genera based on phylogeny: α -CoV, β -CoV, γ -CoV and δ -CoV. Within the beta-CoV genus, four lineages (A, B, C, and D) are recognized.

[0003] Coronaviruses cause a large variety of diseases in animals. In humans, CoV infections primarily involve the upper respiratory tract and the gastrointestinal tract, and principally cause mild, self-limiting disease, such as the common cold. Two of these human coronaviruses are α -coronaviruses, HCoV-229E and HCoV-NL63, while the other two are β -coronaviruses, HCoV-OC43 and HCoV-HKU1. HCoV-229E and HCoV-OC43 were isolated nearly 50 years ago, while HCoV-NL63 and HCoV-HKU1 have only recently been identified. These viruses are endemic in the human populations, causing 15-30% of respiratory tract infections each year. Seroprevalence studies suggest that exposure to these viruses is nearly universal in humans (Severance et al., 2008). The existence of additional human coronaviruses is plausible but unknown. The first highly pathogenic human coronavirus was SARS-CoV, a β -coronavirus. It was identified as the causative agent of the Severe Acute Respiratory Syndrome (SARS) outbreak that occurred in 2002-2003 in the Guangdong Province of China. A second novel, highly pathogenic human CoV emerged in the Middle East in 2012. This virus, named Middle East Respiratory Syndrome-CoV (MERS-CoV), is also a β -coronavirus and was found to be the causative agent of highly pathogenic respiratory tract infections in Saudi Arabia and other countries in the Middle East. The third and most recent highly pathogenic human coronavirus, SARS-CoV-2, first appeared in China in late 2019, is also a β -coronavirus, and is currently responsible for an ongoing pandemic. The highly pathogenic coronaviruses are believed to circulate in zoonotic reservoirs, principally in bat species, with occa-

sional spillover into the susceptible human population, possibly via an intermediate host species. The occurrence of 3 cross species transmission events during the past 17 years raises the prospect that similar viruses may emerge in the future. The defining feature of highly pathogenic coronaviruses is their ability to cause serious morbidity and mortality in infected individuals, with crude estimates of 2.3%, 9.5%, and 34% for SARS-CoV-2, SARS-CoV-1 and MERS, respectively (Petrosillo et al., 2020). The commonly circulating human coronaviruses rarely if ever cause serious morbidity or mortality.

[0004] Treatment of pathogenic coronaviruses is primarily supportive. Despite ongoing efforts, there are no highly active anti-viral drugs.

SUMMARY OF THE INVENTION

[0005] The present invention relates to treatment and prevention of disease caused by the highly pathogenic coronaviruses, SARS-CoV-1, MERS and SARS-CoV-2, and other related pathogenic coronaviruses.

[0006] The approach of the present invention to treatment of coronaviruses is an immunotherapy which harnesses the natural immune defenses of the body. For clearance of an existing viral infection the paradigm of immune defense is the MHC class 1-restricted CD8+ cytotoxic T lymphocyte, which has the ability to destroy and clear virally infected cells. However, it is problematic whether induction of T cells directed against an infecting virus can clear that infection before the host is overwhelmed, because a maximal response to immunization typically requires weeks, perhaps months to achieve. Therefore, in the case of highly pathogenic coronaviruses, a strategy is needed to rapidly mobilize an efficacious anti-viral cytolytic T cell response to control or clear acute infection. The proposed approach is to stimulate cross reactive memory T cells. Memory T cells, which are more responsive to stimulation than naive cells, can be clonally expanded very rapidly. Recent studies have identified cross-reactive T cells that recognize SARS-CoV-2 antigens in blood samples obtained prior to the appearance of the virus in 2019 (Grifoni et al., 2020; Weiskopf et al., 2020; Braun et al., 2020; Le et al., 2020). Since the common cold coronaviruses, OC43 and HKU1, share amino acid similarities with SARS-CoV-2, these cross-reactive T cells may be induced by prior infections with common cold human coronaviruses. In fact, a recent study that mapped the epitopes recognized by cross reactive T cells demonstrated that pre-existing memory CD4+ T cells are cross reactive with SARS-CoV-2 and common cold coronaviruses (Mateus et al., 2020). The detailed molecular and structural basis for cross reactive T cells is poorly understood but the current paradigm of immune recognition predicts the requirement for highly conserved amino acid sequences within MHC class I-restricted T cell epitopes (contiguous sequences 8-13 amino acids in length) that are shared by SARS-CoV-2 and the common cold coronaviruses. Alignments of SARS-CoV-2 amino acid sequences with homologous regions of known common human coronaviruses show only limited regions of high amino acid identity in putative T cell epitopes. However, the notion that cross reactivity requires substantial amino acid identity between epitopes from heterologous viruses is based on the clonal selection theory of immune recognition, which postulates individual lymphocytes are specific for a single antigen, one clone-one specificity, and requires close amino acid identity between cross

reactive epitopes. Several scientists have called this theory into question and proposed a theory of T cell immune recognition that postulates T cells recognize multiple specificities, one clone-millions of specificities (Mason, 1998; Wilson et al., 2004; Kersh and Allen, 1996; Sewell, 2012). The theory is grounded in the mathematic consideration that there are only 1012 T cells in humans and $<10^8$ distinct T cell receptors in the human naïve T cell pool, while the theoretical limit of possible peptides of 20 amino acids that can bind to MHC molecules is vast ($>10^{15}$) and likely even greater if peptides that contain post-translational modifications are considered. This theory of T cell cross reactivity leads to a novel approach to both treat and prevent highly pathogenic coronaviruses. Simultaneous induction of T cell responses to a highly pathogenic coronavirus and to one or more genetically related common human coronaviruses to which the majority of individuals have prior exposure and thus have memory T cells will generate two categories of cellular immunity, therapeutic and prophylactic. The strategy will recall cross-reactive partially or fully protective memory T cells and induce naïve T cells specific for the highly pathogenic coronavirus. The dual action of these two classes of T cell response will be therapeutic by rapidly controlling and eradicating the acute highly pathogenic coronavirus infection in the host, preventing severe symptoms and death, and at the same time be prophylactic by providing long term protection against future infection with the highly pathogenic coronavirus.

[0007] The present invention further relates to compositions and methods to induce cytolytic T lymphocytes (CD8+) response, that is, MHC class I restricted T cell responses, to pathogenic and common cold coronaviruses.

[0008] The invention relates to a delivery platform for antigens consisting of a polyionic papillomavirus virus-like particle (VLP), wherein contiguous, negatively charged amino acids flanked by a cysteine residue are inserted in the HI loop of the papillomavirus L1 protein.

[0009] The invention further relates to antigens to be paired with the VLP comprising fusion peptide/proteins with N-terminal or C-terminal amino acids consisting of contiguous, positively charged amino acids preceded and/or followed by a cysteine residue, here after designated the TAG. The invention further relates to the TAG having a C-terminal proteolytic processing sequence (AAYY) to enhance presentation of MHC class I epitopes.

[0010] In particular embodiments, the invention relates to antigens that are derived from a pathogenic coronavirus, and from the genetically most closely related human coronaviruses that commonly circulate in human populations. The invention further relates to the choice of antigens for induction of a cytolytic T cell response based on the abundance of expression of the viral protein in virally infected cells, the density and position of predicted MHC class I-restricted T cell epitopes in the protein, and empirical studies identifying cytolytic T cells directed toward specific viral proteins. The invention further relates to the choice of antigens comprising two separate and distinct classes of targets that are derived from viral structural proteins and viral non-structural proteins, respectively. Since the first proteins expressed in virally infected cells are non-structural viral proteins, targeting these proteins provides for an efficacious cellular immune response at the earliest stage of viral infection. The invention further relates to the choice of antigens of the common nonpathogenic human coronavirus based on the

above criteria and the additional criterion that the antigen encompasses regions of the viral protein that share at least 40% identity to the corresponding viral protein of the pathogenic coronavirus. In particular embodiments of VLP-antigen compositions, the antigens are either short peptides 25-30 amino acids in length, extended peptides approximately 45-55 amino acids in length, short proteins approximately 110-140 amino acids in length, or full-length amino acid sequences of target antigens.

[0011] In a particular embodiment, the pathogenic coronavirus is SARS-CoV-2 and the genetically related human coronaviruses are OC43 and HKU1. In a particular embodiment the antigens of SARS-CoV-2 are derived from the following viral structural proteins, the membrane protein (M), the nucleocapsid protein (N), ORF3a, ORF7a, and the S2 region of the spike(S) envelope protein and the antigens of the non-structural proteins are derived from the nsp6, nsp7 and nsp12. In a particular embodiment, the antigens of the structural proteins of OC43 and HKU1 are derived from the M and N proteins and the S2 region of the S protein, and the antigens of the non-structural proteins are derived from the nsp3, nsp4, nsp6, nsp7 and nsp12 proteins of OC43, and variable regions of HKU1 with respect to OC43 from the same viral proteins.

[0012] The invention also relates to the method of preparing polyionic papillomavirus VLPs paired with coronavirus antigens. The invention further relates to the method of delivery of the VLP-antigen composition by the intravenous, intramuscular or intradermal route. In a particular embodiment, the VLP-antigen composition is delivered intranasally or by inhalation to stimulate lung and nasopharyngeal tissue resident memory T cells and generate cytotoxic T lymphocytes that preferentially traffic to sites into the respiratory tract.

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1 shows alignment of bovine papillomavirus (BPV) type 1 L1 protein with the L1 protein of the closely related BPV isolate NY8385: HI loop is aa344-357.

[0014] FIG. 2 shows alignment of M proteins of OC43 and SARS-CoV-2.

[0015] FIG. 3 shows alignment of M proteins of OC43 and HKU1.

[0016] FIGS. 4A and 4B show alignments of antigenic regions of N proteins of OC43 and SARS-CoV-2.

[0017] FIGS. 5A, 5B and 5C show alignment of antigenic regions of N proteins of OC43 and HKU1.

[0018] FIGS. 6A and 6B show alignments of antigenic regions of S proteins of OC43 and SARS-CoV-2 within the S2 region.

[0019] FIG. 7 shows alignment of S proteins of HKU1 and OC43 within a variable antigenic region.

[0020] FIGS. 8A, 8B and 8C show alignment of antigenic regions of ORF1ab of OC43 with SARS-CoV-2 nsp3 protein.

[0021] FIGS. 9A, 9B and 9C show Alignment of antigenic regions of ORF1ab of OC43 with SARS-CoV-2 nsp4 protein.

[0022] FIGS. 10A and 10B show alignment of antigenic regions of ORF1ab of OC43 with SARS-CoV-2 nsp6 protein.

[0023] FIG. 11 shows alignment of antigenic region of ORF1ab of OC43 with SARS-CoV-2 nsp7 protein.

[0024] FIG. 12 shows the sequence of Bovine Papillomavirus type 1 capsid protein L1.

[0025] FIG. 13A shows alignment of a first set of antigenic regions of ORF1ab of OC43 with SARS-CoV-2 nsp12 protein.

[0026] FIG. 13B shows alignment of a second set of antigenic regions of ORF1ab of OC43 with SARS-CoV-2 nsp12 protein.

[0027] FIG. 13C shows alignment of a third set of antigenic regions of ORF1ab of OC43 with SARS-CoV-2 nsp12 protein.

DETAILED DESCRIPTION

[0028] The present invention provides compositions and methods relating to genetically engineered papillomavirus L1 virus-like particles (VLPs) comprising negatively charged amino acid sequences and a cysteine residue in the HI loop of the L1 protein, hereafter referred to as polyionic VLPs. The invention further relates to an L1 protein from any papillomavirus species, including human, bovine, equine, murine, ovine, porcine, cervine, canine, feline, or leporine. In one embodiment, the L1 papillomavirus protein is from bovine papillomavirus type 1 (BPV1) (FIG. 12).

[0029] In particular embodiments, the HI loop of the genetically engineered BPV1 L1 protein comprises 4 to 10 contiguous, negatively charged amino acids, flanked by a cysteine residue at the N- or C-terminus or both termini. The HI loop of BPV1 L1 is here defined as amino acid positions 344 to 357, with a span of 14 amino acids and the tip defined as the proline (P) residue at position 349 (FIG. 1). The negatively charged amino acids of the VLPs of the present invention can be glutamic acid, aspartic acid, or both. In various embodiments, the amino acid sequence of 5 to 12 amino acids in length (4-10 contiguous, negatively charged amino acids and one or two flanking cysteine residues) may be inserted in the HI loop, and replace none, or one, or more than one of the native amino acids. In specific exemplary embodiments, the amino acid sequence inserted in the HI loop is 9 residues in length, comprising 8 negatively charged amino acids, glutamic acid or aspartic acid or alternating aspartic and glutamic acids, and a C- or N-terminal cysteine residue, and replaces the 9 native amino acids at positions 347-355 of the HI loop of the bovine papillomavirus L1 protein (Table 1, inserts 1-4). In other specific exemplary embodiments, the amino acid sequence inserted in the HI loop is 5, 6, 7, 8, 10, or 11 amino acids in length, comprising 4, 5, 6, 7, 9, or 10 glutamic acids, respectively, and a C-terminal cysteine residue. The respective inserts replace the equivalent number of native amino acids at positions, 347-351, 347-352, 347-353, 347-354, 346-355 or 345-355, respectively, of the HI loop of the bovine papillomavirus L1 protein (Table 1 insert 5-10). In a specific exemplary embodiment, the amino acid sequence inserted in the HI loop is 9 residues in length, comprises 8 negatively charged glutamic acids and a C- and N-terminal cysteine residue, and replaces the 10 native amino acids at positions 346-355 of the HI loop of the bovine papillomavirus L1 protein (Table 1, insert 11). In other various embodiments the negatively charged amino acids and cysteine(s) may be inserted in the HI loop and replace fewer native amino acids than the number comprising the insert. In specific exemplary embodiments a 9-amino acid sequence, comprising 8 glutamic acids and a C-terminal cysteine residue replaces 7, 5 or 3 native amino acids at positions 348-354, or 348-352, or

348-350, respectively, of the HI loop of the bovine papillomavirus L1 protein (Table 1, inserts 12-14). In other various embodiments the negatively charged amino acids and cysteine(s) may be inserted in the HI loop and replace up to 2 more native amino acids than the number comprising the insert. In a specific exemplary embodiment, a 9-amino acid sequence, comprising 8 glutamic acids and a C-terminal cysteine residue, replaces the 11 native amino acids at position 346-356 of the HI loop of the bovine papillomavirus L1 protein (Table 1, insert 15). In various embodiments, the glutamic acid-cysteine amino acid sequence inserted in the HI loop may replace the proline at the putative tip of the loop or may be inserted between the proline residue and the immediate C-terminal leucine residue without removing any native amino acids. In some embodiments, the inserted amino acid replacing proline or located between the proline and leucine residues may be flanked by a glycine-serine-serine-glycine (GSSG) linker amino acid sequence. In specific exemplary embodiments, the amino acid sequence inserted in the HI loop comprises 8 glutamic acids and a C-terminal cysteine residue, with or without flanking GSSG amino acids, and either replaces the proline at position 349 or is located between amino acid positions 349 and 350 amino acid of the HI loop of the bovine papillomavirus L1 protein (Table 1, inserts 16-19).

[0030] The invention further relates to antigens to be linked to the VLP, comprising fusion peptide/proteins with N-terminal or C-terminal amino acids consisting of 4-10 contiguous, positively charged amino acids preceded by and/or followed by a cysteine residue, here after designated the TAG. The TAG allows the antigen to be attached to the polyionic papillomavirus VLP by the combined action of electrostatic interactions and an oxidation reduction reaction between cysteine residues on the VLP and the cysteine residue in the TAG. The invention further relates to the TAG having a C-terminal proteolytic processing sequence (AAYY) to enhance presentation of MHC class I epitopes. Where the TAG is appended to the C-terminus of the antigen, the proteolytic processing sequence (AAYY) is placed at the N-terminus of the peptide/protein antigen.

[0031] In specific exemplary embodiments the TAG comprises the group of positively charged amino acids, arginine (R), lysine (K) or histidine (H), and the sequence is 8 amino acids in length, (Table 2, Tag-1, -2, -3). In other specific exemplary embodiments, the TAG comprises a repeating motif of RKHRKHRK, 8 amino acids in length (Table 2, TAG-4). In other specific exemplary embodiments, the TAG comprises an amino acid sequence of 4, 5, 6, or 7 contiguous arginines followed by a cysteine residue (Table 2, TAG-5, -6, -7, and -8) or an amino acid sequence of 9 or 10 arginines followed by a cysteine residue (Table 2 TAG-9 and -10). In other specific exemplary embodiments, the TAG consists of 8 positively arginines preceded by a cysteine residue or flanked at the N and C terminus by cysteine residues (Table 2, TAG-11 and -12).

[0032] The invention further relates to the composition of polyionic papillomavirus VLPs and target antigens with a TAG sequence for the purpose of inducing cytolytic (MHC class I restricted) T cell responses. In specific embodiments, the invention relates to coronavirus antigens linked to polyionic VLPs for the purpose of inducing cytolytic T cell responses to common cold coronaviruses and pathogenic coronaviruses.

[0033] The entire proteome of a virus can be a target of T cell responses. However, structural proteins are particularly effective targets for antigen-specific T cell responses because they are the most abundantly expressed viral proteins in infected cells, thus allowing for efficient presentation to MHC molecules. Empirical studies of cellular immune responses to coronaviruses also support the importance of structural proteins as principal targets of the cellular immune response (reviewed in (Liu et al., 2017)). Cytolytic T cell responses are directed toward MHC class I restricted T cell epitopes. These epitopes are most commonly 9 amino acids in length and can range from 8-13 amino acids in length. The genetic heterogeneity of MHC class I alleles in humans and other mammals allows for numerous possible epitopes within a viral protein. In order to encompass the universe of possible epitopes, an antigen needs to be the full-length amino acid sequence of a target protein or sets of shorter fragments of the antigen that together include the entire amino acid sequence. The preferred length of antigenic fragments linked to polyionic VLPS is an empirical function of efficiency of epitope presentation for induction of a T cell response via the alternative antigen presentation pathway. In addition, manufacturing considerations may influence the preferred choice of length of an antigen.

[0034] The preferred embodiment for a polyionic papillomavirus VLP composition of matter for the induction of cytotoxic T cell response to SARS-CoV-2 comprises antigens from the membrane (M), nucleocapsid (N), S2 region of the spike(S), ORF3a, and ORF7a structural proteins and the nsp6, nsp7 and nsp12 nonstructural proteins.

[0035] The M protein is 222 amino acids in length and the antigenic region spans aa6-221. The N protein is 419 amino acids in length. Based on empirical studies and predictive algorithms for MHC class I-restricted T cell epitope the antigenic region is aa51-369. The Spike(S) protein is 1270 amino acids in length and comprises two regions, S1 (aa1-661) containing the receptor binding domain (aa330-583) and S2 (aa662-1270). T cell epitopes are widely distributed across the S protein. The S1 region contains the majority of virion surface exposed amino acids and is the principal target of humoral immune responses, including neutralizing antibodies. The region is preferably excluded from a vaccine intended to induce cellular immunity to avoid unintended induction of antibody responses with possible deleterious effects, such as induction of antibodies mediating antibody dependent enhancement (ADE). ORF3a is 275 amino acids in length. Empirical studies support the choice of ORF3a as a target antigen for T cell response. Prediction algorithms (MHC-NP: prediction of peptides naturally processed by MHC, developed by Sébastien Giguère, Alexandre Drouin, Alexandre Lacoste, Mario Marchand, Jacques Corbeil and François Laviolette; <http://tools.iedb.org/mhcnp/>) show that the C terminus, as compared to the N terminus, has a greater density of putative MHC class I restricted epitopes for 5 representative alleles (67 spanning the C-terminal 171 amino acids versus 20 spanning the 104 N-terminal amino acids). Empirical studies support choice of ORF7a, a protein of 121 amino acids, as a target of cytolytic T cell responses. The ORF1ab of SARS-CoV-2 is a polyprotein of 7096 amino acids. The polyprotein is proteolytically processed within virally infected cells to yield multiple non-structural proteins that serve diverse functions in the viral life cycle. The proteins are present in low abundance in virally infected cells and for this reason are not major targets of cellular

immune responses, but recent studies have shown that cytolytic T cell responses to several nsp proteins can be detected in the blood of SARS-CoV-2 infected patients and also in blood obtained from some healthy blood donors in the pre-COVID19 era (Grifoni et al., 2020; Le et al., 2020). Because these proteins are the first to be expressed in infected cells they are attractive targets for cytolytic T cells. Rapid killing of these virally infected cells can prevent the cell from producing infectious virus. In terms of frequency of antigen specific T cells and percentage of responding subjects, the principal targets of CD8+ T cell responses are the nsp6, nsp7 and nsp12 proteins.

[0036] The amino acid length of antigens linked to polyionic VLPs can range from a defined epitope of 8-14 amino acids or longer amino acid sequences up to the full-length amino acid sequence of the antigenic region of a viral protein or fusions of several viral proteins. In exemplary embodiments, the antigens are short peptides approximately 25-30 amino acids in length, extended peptides approximately 45-55 amino acids in length, short proteins approximately 110-140 amino acids in length, or the full-length amino acid sequence of the antigenic region of a targeted viral protein. For antigenic fragments shorter than the full-length amino acid sequence of the antigenic region, the specific embodiment comprises a set of antigens that includes all the amino acids of the antigenic region of the target viral protein. Antigens for the M protein of SARS-CoV-2, embodied as short peptides, extended peptides, short proteins, or the entire amino acid sequence of the antigenic region, are shown in Table 3. Comparable antigens for the N protein of SARS-CoV-2 are shown in Table 4. Comparably designed antigens for the S2, ORF3a, and ORF7a structural proteins of SARS-CoV-2 are shown in Tables 5-7. Antigens for the nonstructural proteins nsp6 and nsp7 are shown in Table 8 and antigens for the nsp12 protein are shown in Table 9 and 10. Short and extended peptides are overlapping by 10 amino acids and short protein antigens are overlapping by 11 amino acids. In the final formulation, peptide, protein, or full-length antigens have a TAG, as described above.

[0037] The preferred embodiment for a polyionic papillomavirus VLP composition of matter to stimulate memory T cells induced by prior exposure to common cold coronaviruses comprises antigens derived from both structural and nonstructural proteins. The multiple specificities theory of T cell recognition does not define at the molecular level the structural basis for T cell cross reactivity. We define herein viral proteins or sub regions of viral proteins containing cross-reactive T cell epitopes as those with an average amino acid identity of >40% across homologous amino acid sequences of a common cold coronavirus and a specific pathogenic coronavirus.

[0038] In particular embodiments, the common coronaviruses are OC43 and HKU1, and the antigens for induction of cross-reactive T cell responses are from the M, N and S2 structural proteins and the nsp3, nsp4, nsp6, nsp7 and nsp12 nonstructural proteins. Where the amino acids sequences of OC43 and HKU1 share on average >40-80% identity across homologous amino acids of the target antigen, only the amino acid sequences of OC43 are used as the target antigens. In other particular embodiments, the amino acid sequences of HKU1 are the antigens.

[0039] The M protein of OC43 is 230 aa in length and shares overall identity of 40.8% with the M protein of SARS-CoV-2 within the region of aa14-226 (FIG. 2). Align-

ment of OC43 M protein with that of HKU1 shows no significant areas of variability (amino acid identity <80%) (FIG. 3). The N protein of OC43 is 448 aa in length and on average shares 36% amino acid identity with the N protein of SARS-CoV-2. The antigenic region of OC43 from aa99-400, excluding non-aligned regions aa266-269, aa341-349, and aa382-387, shares 43.5% identity with the homologous region of the N protein of SARS-CoV-2, after excluding aa221-224 that do not align with the amino acid sequence of the OC43 N protein (FIG. 4A). The N-terminus amino acids of the OC43 N protein at positions aa64-88 share 52% identity with the corresponding region of N of SARS-CoV-2 (FIG. 4B). Alignment of OC43 N protein with that of HKU1 shows 3 antigenic regions with low levels of amino acid identity between the two viral amino acid sequences, ranging from 33%-50%. (FIGS. 5A, 5B, and 5C). The S protein of OC43 is 1353 amino acids and composed of 2 sub-regions, S1 and S2. The S1 extends from aa 1-789 and shows low amino acid identity (24%) with the S1 region of SARS-CoV-2 in the aligned region between aa70-552. In contrast, the S2 region of the viruses contains two antigenic regions, aa898-1153 and aa1228-1302, that share an average of 52.72% and 51.3% amino acid identity, respectively (FIG. 6). Alignment of the homologous region of HKU1 with aa898-1153 of OC43 showed overall identity of 83.5% without regions of high variability, while the antigenic region of OC43 between aa1228-1303 shares 70.7% identity with the homologous S2 region of HKU1 (FIG. 7).

[0040] In exemplary embodiments, the antigens are short peptides approximately 25-30 amino acids in length, extended peptides approximately 45-55 amino acids in length, short proteins approximately 110-140 amino acids in length, or the full-length amino acid sequence of the antigenic region of a targeted viral protein. For antigenic fragments shorter than the full-length amino acid sequence of the antigenic region, the specific embodiment comprises a set of antigens that includes all the amino acids of the antigenic region of the target viral protein. Antigens for the M protein of OC43 and variable regions of HKU1, embodied as short peptides, extended peptides, short proteins, or the entire amino acid sequence of the antigenic region, are shown in Table 11. Comparable antigens for the N protein of OC43 and variable regions of HKU1 are shown in Table 12, and antigens for the S2 protein of OC43 and variable regions of HKU1 in Table 13. Short and extended peptides are overlapping by 10 amino acids and short protein antigens are overlapping by 11 amino acids. In the final formulation peptide, protein or full-length amino acid sequences include a TAG, as described above.

[0041] The ORF1ab of OC43 coronaviruses encodes a polyprotein of 7095 amino acids. The polyprotein is proteolytically processed within virally infected cells to yield multiple non-structural proteins that serve diverse functions in the viral life cycle. In terms of frequency of antigen specific T cells and percentage of responding subjects, the principal targets of CD8+ T cell responses are the nsp3, nsp4, nsp6, nsp7 and nsp12 proteins. The choice of antigens from these nonstructural proteins is also based on the additional consideration of an average of >40% identity between amino acid sequences of common cold viruses and SARS-CoV-2.

[0042] The nsp3 protein is 1,945 amino acids in length. The overall identity between SARS-CoV-2 and OC43 nsp3 amino acids sequences is 26%. However, the C-terminal

region of 384 amino acids shares 39% identity and contains within it 3 regions of continuous amino acids with 52.8%, 41%, and 43.6% identity between the amino acid sequence of OC43 and the homologous amino acid sequence of SARS-CoV-2 (FIG. 8A-C). The nsp4 protein is 500 amino acids in length; the nsp4 proteins of SARS-CoV-2 and OC43 share 42% amino acid identity. Three sub-regions in the mid portion or C terminus of the nsp4 protein, which are 45, 133 and 50 amino acids in length, share 53.3%, 47.4% and 72% amino acid identity, respectively, between homologous amino acid sequences of SARS-CoV-2 and OC43 (FIG. 9A-C). The nsp6 protein is 290 amino acids in length and the homologous amino acid sequences of SARS-CoV-2 and OC43 share 30.5% identity. The C-terminal 84 amino acids share 54.8% identity and a 31 amino acid region at the N-terminus shares 41.9% identity (FIG. 10A-B). The nsp7 protein is 83 amino acids in length and the amino acid sequences of the SARS-CoV-2 and OC43 nsp7 share 46% identity. Excluding the C-terminal 13 amino acids of SARS-CoV-2, the amino acid identity of SARS-CoV-2 and OC43 nsp7 proteins is 55.2% (FIG. 11). The nsp12 protein is 932 amino acids in length. An empirical study of antigen specific cellular immune responses in SARS-CoV-2 infected patients located the majority of T cell epitopes in the regions encompassed by aa 125-375 and aa 520-920 of the protein (Grifoni et al, 2021) Two amino acid sequences within OC43 ORF1ab share 61.5% and 67% identity with corresponding amino acids within the aa125-275 fragment of the SARS-CoV-2 nsp12 protein (FIGS. 13A-B) and 341 aa of OC43 ORF1ab shares 76.8% identity with corresponding amino acids within the aa520-920 fragment of the SARS-CoV-2 nsp12 protein (FIG. 13C). The nonstructural proteins of OC43 and HKU1 share ~90% identity and have no extended regions of amino acid variability (identity <80%).

[0043] In exemplary embodiments, the antigens are short peptides approximately 25-30 amino acids in length, extended peptides approximately 45-55 amino acids in length, short proteins approximately 110-140 amino acids in length, or the full-length amino acid sequence of the antigenic region of a targeted viral protein. For antigenic fragments shorter than the full-length amino acid sequence of the antigenic region, the specific embodiment comprises a set of antigens that includes all the amino acids of the antigenic region of the target viral protein. Antigens for the nsp3 and nsp4 proteins, embodied as short peptides, extended peptides, short proteins, or the entire amino acid sequence of the antigenic region are shown in Table 14. Comparably designed antigens for the nsp6 and nsp7 proteins of OC43 are shown in Table 15 and for nsp12 in Tables 16 and 17. Short and extended peptides are overlapping by 10 amino acids and short protein antigens are overlapping by 11 amino acids. In the final formulation peptide, protein or full-length amino acid sequences include a TAG, as described above.

EXAMPLES

Example 1: Generation of Recombinant Baculoviruses with Genetically Engineered BPV L1 Genes

[0044] The entire open reading frame (ORF) of BPV L1 with a Kozak consensus and unique restrictions sites at each end (EcoRI/NotI) is artificially engineered by PCR-based gene synthesis and cloned in a pUC18 vector. The entire

ORF is codon-modified using *Drosophila melanogaster* preferred codons, for efficient expression in insect cells. The ORF contains insertion of peptides with aspartic or glutamic acid residues and a cysteine residue and inserted into the HI loop as described in Table 1, and designated insert-1 to insert-19.

[0045] The modified BPV L1 genes are subcloned between the EcoRI/NotI sites of the pORB baculovirus transfer vector. The transfer vectors are co-transfected with a linear baculovirus DNA in *Spodoptera frugiperda* sf9 cells using a preferred commercially available transfection reagent, as suggested by the manufacturer. Five days post-transfection, the recovered recombinant baculoviruses are further amplified by large scale infections of sf9 cells. Small scale infections to confirm expression of the modified L1 proteins are conducted with 2×10^6 *Trichoplusia ni* (High Five) cells, growing in 6-well plates and infected with serial dilutions of Baculovirus stocks. 72 hrs post-infection, the cells are lysed in 500 μ l of RIPA buffer and the clarified lysates are subjected to SDS-PAGE analysis to detect over-expression of a protein of the expected molecular weight of 55 kDa.

Example 2: Production of Polyionic VLPs from Recombinant Baculoviruses

[0046] For production of VLPs, approximately 2×10^9 *Trichoplusia ni* (High Five) cells growing in spinner flasks are infected with a pre-determined amount of a high-titer recombinant baculovirus stock in 500 ml of TNM-FH/10% FBS. After 96 h of incubation at 27° C., the cells are harvested, and collected by centrifugation at 2,000 rpm (Sorvall FH18/250 rotor) for 5 min. Cell pellets are resuspended in extraction buffer (20 mM phosphate buffer, pH 6.5, 1 M NaCl, 0.1 mM CaCl_2 , 50 μ M FeCl_2) containing protease inhibitors (Roche Complete ULTRA, 1 tablet per 10 ml) and subjected to 5 cycles of thawing at 37° C. and freezing in a -80° C. ethanol bath. The lysate is spun 1 h at 8000 rpm to remove baculovirus particles. The clarified lysate is extracted for 10 min with an equal volume of Vertrel DF (Fisher Scientific). The aqueous layer is loaded onto a 40% sucrose cushion and centrifuged in a SW32Ti rotor at 32,000 rpm for 1.5 h. The sucrose pellet is resuspended in 20 mM phosphate pH 8.0, 0.5 M NaCl, 5 mM MgCl_2 , and incubated 30 min at 37° C. with 250 U/ml of Salt Active Nuclease (Arcticzymes). After dialysis in 20 mM phosphate pH 6.5, 0.5 M NaCl, the VLP solution is adjusted to 0.01% Tween 80, 0.05% carboxymethyl cellulose, 50 μ M FeCl_2 and stored at 4° C. Purity is assessed by SDS-PAGE gel analysis and protein concentration is measured by Bradford dye method and uv spectroscopy. To facilitate direct visualization of VLPs, an aliquot of diluted particles is placed on 300-mesh formvar/carbon-coated copper grids, negatively stained with 2% phosphotungstic acid (pH=7.0) and examined by transmission electron microscopy (TEM). Chimeric VLPs with inserts from Table 1 are shown to yield capsid-like structures ~50 nm in diameter with a variegated appearance consistent with formation from smaller capsomere-like structures and resembling native HPV type 1 VLPs.

Example 3: Conjugation of Antigens to Polyionic VLPs

[0047] For conjugation to the VLP, peptides are solubilized in distilled water at 5 mg/ml (or dissolved at 5 mg/ml

in DMSO, if they are not soluble in water). Peptides at a preferred concentration of 2.5 mg/ml, but lower if less soluble in the buffer, are reduced with 10 mM Bond-breaker TCEP solution (Thermo Fisher Scientific) for 20 min at 50° C. After dialysis in 20 mM phosphate buffer, pH 6.5, 0.15 M NaCl, VLP protein (1 mg/ml), and peptide at peptide: L1 protein molar ratios between 4:1 to 16:1, based on solubility characteristics, are mixed in the presence of 4 mM glutathione disulfide (GSSG) and 0.8 mM reduced glutathione (GSH), and incubated overnight at 37° C. To remove unreacted peptide, reactants are dialyzed against 20 mM phosphate pH 6.5, 0.5 M NaCl, using dialysis tubing with a cut-off of 1 million kDa. The VLP-peptide solutions are adjusted to 0.01% Tween 80, 0.05% carboxymethyl cellulose, 0.5 mM GSSG and 0.05 mM GSH, aliquoted, and stored at -20° C. The amount of peptide bound to the VLP is determined by SDS-PAGE analysis and interpolation of sample-peptide band density from a standard curve of known amounts of peptide. Gels are scanned in a BioRad ChemiDocXR imager, and images are analyzed with NIH ImageJ software.

Example 4: Ability of Exemplary Embodiments of VLPs with Various Inserts in the HI Loop to Induce a Cytolytic CD8+ T Cell Response

[0048] VLPs generated with various exemplary inserts in the L1 protein (Table 1) are linked via a TAG to a representative peptide encoding an MHC class I restricted epitope recognized in the genetic background of C57BL/6 mice. Mice are immunized by the intradermal route, as described below. The immune response is measured as described below. VLPs formulated with different inserts linked to a representative antigen are shown to induce comparable frequencies of antigen specific, interferon- γ secreting CD8+ T cells (no significant differences in mean responses by t-test comparison) cytolytic T cell responses.

Example 5: Ability of VLPs Linked to an Antigen by Exemplary Embodiments of TAGs to Induce a Cytolytic CD8+ T Cell Response

[0049] A representative peptide encoding an MHC class I-restricted epitope recognized in the genetic background of C57BL/6 mice is chemically synthesized to >90% purity with the TAGs listed in Table 2. Peptides with TAGs -1 to -4, and -11 and -12 are linked to a polyionic VLP with Insert-1. Peptides with TAGs 5-10 are linked to VLPs with inserts 5-10, respectively. Mice are immunized by the intradermal route, as described below. The immune response is measured as described below. VLPs linked to a representative antigen formulated with different TAGs are shown to induce comparable frequencies of antigen specific, interferon- γ secreting CD8+ T cells (no significant differences in mean responses by t-test comparison).

Example 6: Immunization with Polyionic VLPs and Detection of Antigen Specific CD8+ T Cell Responses

[0050] C57BL/6J mice 6-8 weeks of age are immunized three times, 1 week apart with between 5-50 μ g each of VLP-peptide, administered by intradermal injection. As a control, C57BL/6J mice are immunized with unlabeled (no peptide) VLP protein. VLP-peptide immunogens are formulated with the set of extended peptides for the difference

SARS-CoV-2 and OC43 antigens described in Table 3-12. For intradermal immunization, the VLP/antigen vaccine is injected in a single, or split doses, into the skin of the back of shaved mice.

[0051] To provide samples for immunological assays, mice are sacrificed 10-14 days after the last dose of vaccine and spleens are dissected. Splenocytes (or suspensions of CD3+ cells from lung tissue), are stimulated with 1 µg/ml of a pool of overlapping peptides 11 amino acids in length (overlapping by 8 amino acids) spanning the full-length amino acid sequence of the target antigen, in the presence of brefeldin A (10 µg/ml) overnight at 37° C. in 5% CO₂. Cells are stained with Zombie Green™ fixable viability dye, treated with Fixation buffer and stored in Cytolast. Cells are permeabilized with Permeabilization buffer and stained with Brilliant Violet BV™ 510-conjugated anti-mouse CD3, clone 17A2, PerCP-Cy5.5-conjugated anti-mouse CD8α, clone 53-6.7, and PE-conjugated anti-mouse IFNγ, clone XMG1.2. Reagents are purchased from a commercial source. Flow cytometry is performed on an LSR-II or comparable flow cytometer and data are analyzed using FACS Diva software or FlowJo software. Gating is done on forward and side scatter parameters to select for lymphocytes and singlets. After exclusion of dead cells, CD8+ T lymphocytes are identified on a CD3/CD8 dot plot of gated lymphocytes, and interferon-γ (IFNγ) secreting cells are identified on a CD8/IFNγ dot plot of gated CD8+ T cells. A minimum of 30,000 CD8+ T cells are analyzed. VLP-peptide immunogens of the M, N, S2, ORF3a, ORF7a, nsp6, nsp7, and nsp12 antigens of SARS-CoV-2, and immunogens of the M, N, S2, nsp6, nsp7, and nsp12 antigens of OC43 are shown to induce detectable antigen specific, interferon-γ secreting CD8+ T cells

Example 7: Ability of Polyionic VLP SARS-CoV-2 Vaccines to Protect Mice Against SARS-CoV-2 Disease in a Challenge Model

[0052] Mice strain: Stable humanized angiotensin converting enzyme-II (ACE2) mice generated using CRISPR/Cas9 knock-in technology to replace the endogenous mouse ACE2 (mACE2) with the human ACE2 in the C57BL/6 strain of mouse will be used for SARS-CoV-2 challenge (Sun et al., 2020).

[0053] Mouse immunization: VLPs with an exemplary insert in the HI loop are formulated with the set of exemplary extended peptides with an exemplary and appropriate TAG for the particular insert. A polyionic VLP vaccine is formulated with M, N, S2, ORF3a, ORF7a, nsp6, nsp7 and/or nsp12 SARS-CoV-2 antigens of SARS-CoV-2 antigens. Peptides are drawn from the list in Tables 3-10 with SEQ ID ending in extension, -E1, -E2, -E3, etc. The precise number of peptides will depend on the antigen as described in the table. The preferred insert is the E8C amino acid sequence at amino acid positions 347 to 355 in the HI loop of the L1 protein of bovine papillomavirus type 1, with replacement of native amino acids (insert-1, Table 1), and the preferred TAG is CRRRRRRRCAAYY (TAG-1, in Table 2). Additional insert and TAG combinations with sets of peptides for one or more SARS-CoV-2 antigens are also tested, as informed by other enabling experiments. VLP/antigen constructs are generated as described above. Mice are immunized by the intradermal route, intranasal/lung route, or both routes simultaneously. For intradermal immunization, the VLP/antigen vaccine is injected in a single, or

split doses, into the skin of the back of shaved mice. For nasal/pulmonary immunization, the VLP/antigen construct is administered dropwise (10 µl) into the nose of a lightly anesthetized mouse. In anesthetized mice intra-nasally administered vaccine is also inhaled by the mouse and thus is delivered to both the lung and nasopharyngeal tissues.

[0054] Mouse challenge: For intranasal infection, aged (30 weeks old) hACE2 mice are anesthetized with Isoflurane delivered with a precision vaporizer, and then intranasally infected with 4×10⁵ pfu of SARSCoV-2. Mice are then weighed and monitored daily and sacrificed on day 6 post infection for serum collection and tissue processing. Spleens and lung tissue are collected for analysis of viral RNA load, histopathology, and measurement of cellular immune response.

[0055] Measurement of viral RNA load: Viral RNA in lung tissue is extracted with a RNeasy Mini kit (QIAGEN) according to the protocols. The viral RNA quantification is performed by RT-qPCR targeting the S gene of SARS-CoV-2. RT-qPCR is performed using One Step PrimeScript RT-PCR Kit (Takara) with the following primers and probes: CoV-F3, CoV-R3 and CoV-P3 (Sun et al., 2020).

[0056] Antigen specific CD8+ T cell responses: CD8+ T cell response of splenocytes and CD3+ T cells recovered from lung homogenates are measured as described above.

[0057] SARS-CoV-2 polyionic VLP vaccines are shown to induce antigen specific CD8+ T cells responses detectable in splenocytes and in CD3+ T cells from lung tissue, to each of the target antigen (M, N S2, ORF3a and ORF7a). Vaccinated mice are further shown to have a significantly reduced viral RNA load and less lung pathology than mock vaccinated mice. Vaccines are also shown to decrease expression of viral RNA in the lung. Of note, the vaccines are not expected to provide sterilizing immunity.

Example 8: Ability of Polyionic VLP SARS-CoV-2 Vaccines to Protect Syrian Hamsters Against SARS-CoV-2 Infection in a Challenge Model

[0058] Animal species: The Syrian hamster is highly susceptible to SARS-CoV-2 infection, making it the most suitable small animal model to evaluate the protective efficacy of vaccines (Rosenke et al., 2020; Chan et al. 2020). Syrian hamsters (*Mesocricetus auratus*), 6-8 weeks of age are purchased from Jackson Laboratories.

[0059] Hamster immunization: VLPs with an exemplary insert in the HI loop are formulated with the set of exemplary extended peptides with an exemplary and appropriate TAG for the particular insert. A polyionic VLP vaccine is formulated with M, N, S2, ORF3a, ORF7a, nsp6, nsp7 and/or nsp12 SARS-CoV-2 antigens. Peptides are drawn from the list in Tables 3-10 with SEQ ID ending in extension, -E1, -E2, -E3, etc. The precise number of peptides will depend on the antigen as described in the table and may include all or fewer antigens with a-E extension designation. The preferred insert is the E8C amino acid sequence at amino acid positions 347 to 355 in the HI loop of the L1 protein of bovine papillomavirus type 1, with replacement of native amino acids (insert-1, Table 1), and the preferred TAG is CRRRRRRRCAAYY (TAG-1, in Table 2). Additional insert and TAG combinations with sets of peptides for one or more SARS-CoV-2 antigens are also tested, as informed by other enabling experiments. VLP/antigen constructs are generated as described above. Hamsters are immunized by the intradermal route, intranasal route, or both routes indi-

vidually or in combination. Intradermal immunization is performed as described above. For nasal immunization, the VLP/antigen construct is administered dropwise (10 µl) into the nose of a lightly anesthetized mouse.

[0060] Hamster challenge: To mimic the natural route of infection, vaccinated animals and controls (contacts) will be exposed to previously infected animals (index) by co-housing in the same cage. For intranasal infection of the index animal, hamsters are anesthetized with Isoflurane delivered with a precision vaporizer, and then intranasally infected with 100 tissue culture dose 50 (TCID₅₀) of SARSCoV-2 virus. SARS-CoV-2 isolate nCoV-WA1-2020 (MN985325.1) from the CDC, or a suitable alternative isolate, is obtained and propagated in Vero E6 cells. The TCID₅₀ dose is determined by titration of the viral stock in VeroE6 cells. The infected hamster and co-housed vaccinated and control naïve hamsters are weighed and monitored daily for clinical signs of disease. To monitor infection by RT-qPCR, nasal washes are collected from lightly anesthetized naïve contact (vaccinated and control) and index (previously infected) animals daily for 10 days by instillation and then collection of 150 µl of PBS/0.3% BSA in both nostrils.

[0061] Measurement of viral RNA load: Viral RNA in lung tissue is extracted with a RNeasy Mini kit (QIAGEN) according to the protocols. The viral RNA quantification is performed by RT-qPCR targeting the S gene of SARS-CoV-2. RT-qPCR is performed using One Step PrimeScript RT-PCR Kit (Takara) with the following primers and probes: CoV-F3, CoV-R3 and CoV-P3 (Sun et al., 2020).

[0062] SARS-CoV-2 polyionic VLP vaccinated hamsters are shown to have a significantly reduced viral RNA load in nasal washes than mock vaccinated mice after exposure to an infected index hamster.

Tables

TABLE 1			
Representative negatively charged amino acid-cysteine sequences in HI loop			
Seq ID	Inserted aa sequence	Replaced aa native aa sequence	Position(s) of replacement in BPV1 L1 ORF
Insert-1	EEEEEEEC	GTPLTEYDS	Aa347-355
Insert-2	CEEEEEEE	GTPLTEYDS	Aa347-355
Insert-3	DDDDDDDC	GTPLTEYDS	Aa347-355
Insert-4	EDEDEDDC	GTPLTEYDS	Aa347-355
Insert-5	EEEEEC	GTPLT	Aa347-351
Insert-6	EEEEEC	GTPLTE	Aa347-352
Insert-7	EEEEEC	GTPLTEY	aa347-353
Insert-8	EEEEEEEC	GTPLTEYD	Aa347-355
Insert-9	EEEEEEEEEC	DGTPLTEYDS	Aa346-355
Insert-10	EEEEEEEEEEEC	DGTPLTEYDSS	Aa346-356
Insert-11	CEEEEEEEEC	DGTPLTEYDS	aa346-355
Insert-12	EEEEEEEC	TPLTEYD	aa348-354
Insert-13	EEEEEEEC	TPLTE	aa348-352

TABLE 1-continued			
Representative negatively charged amino acid-cysteine sequences in HI loop			
Seq ID	Inserted aa sequence	Replaced aa native aa sequence	Position(s) of replacement in BPV1 L1 ORF
Insert-14	EEEEEEEC	TPL	aa347-350
Insert-15	EEEEEEEC	DGTPLTEYDS	aa346-3556
Insert-16	EEEEEEEC	P	aa349
Insert-17	GSSGEEEEEEEC	ECGSSG	aa349
Insert-18	EEEEEEEC	None	Between aa349-350
Insert-19	GSSGEEEEEEEC	ECGSSG	Between aa349-350

TABLE 2		
Representative TAG sequences		
Seq ID	TAG designation	Amino acid sequence ¹
TAG-1	polyR8	RRRRRRRRC
TAG-2	PolyK8	KKKKKKKKC
TAG-3	PolyH8	HHHHHHHHC
TAG-4	MixedRKH8	RKHKRHKRC
TAG-5	PolyR4	RRRRC
TAG-6	PolyR5	RRRRC
TAG-7	PolyR6	RRRRRRC
TAG-8	PolyR7	RRRRRRRC
TAG-9	PolyR9	RRRRRRRRRC
TAG-10	PolyR10	RRRRRRRRRRC
TAG-11	NH-terminal C	CRRRRRRRR
TAG-12	Dual C	CRRRRRRRRRC

¹All TAG amino acid sequences include a C-terminal AAYY proteolytic processing sequence.

TABLE 3

SARS-CoV-2 antigens of the M structural protein.			
Immunogen	Seq ID	aa-position ¹	Amino acid sequence ²
Short peptide	SARS-CoV-2-M-S1	aa6-36	GTITVEELKKLLEQWNLVIGFLPLTWIC LLQ
	SARS-CoV-2-M-S2	aa27-56	LFLTWICLLQFAYANRNRFLYIIKLIFLY WL
	SARS-CoV-2-M-S3	aa48-77	IIKLIFLWLLWPVTLACFVLAAYRIN WIT
	SARS-CoV-2-M-S4	aa68-97	AAVYRINWITGGIAIAMACLVGLMWL SYFI
	SARS-CoV-2-M-S5	aa88-117	VGLMWLSYFIASFRLFARTSMWSFNP ETN
	SARS-CoV-2-M-S6	aa108-138	SMWSFNPETNILLNVPLHGTILTRPLLE SEL
	SARS-CoV-2-M-S7	aa129-159	LTRPLLESELVIGAVILRGHLRIAGHHL GRC
	SARS-CoV-2-M-S8	aa150-179	RIAGHHLGRCDIKDLPKEITVATSRTLS YY
	SARS-CoV-2-M-S9	aa170-200	VATSRTLSYYKLGASQRVAGDSGFAA YSRYR
	SARS-CoV-2-M-S10	aa191-221	SGFAAYSRYRIGNYKLNTDHSSSDNI ALLV
Extended peptide	SARS-CoV2-M-E1	aa6-50	GTITVEELKKLLEQWNLVIGFLPLTWIC LLQFAYANRNRFLYIIK
	SARS-CoV2-M-E2	aa43-85	NRNRFLYIIKLIFLWLLWPVTLACFVL AAVYRINWITGGIAIAMA
	SARS-CoV2-M-E3	aa76-120	ITGGIAIAMACLVGLMWLSYFIASFRLF ARTSMWSFNPETNILL
	SARS-CoV2-M-E4	aa111-155	SFNPETNILLNVPLHGTILTRPLLESELV IGAVILRGHLRIAGHH
	SARS-CoV2-M-E5	aa146-190	RGHLRIAGHHLGRCDIKDLPKEITVAT SRTLSYYKLGASQRVAGD
	SARS-CoV2-M-E6	aa181-221	LGASQRVAGDSGFAAYSRYRIGNYKL NTDHSSSDNIALLV
Short protein	SARS-CoV2-M-P1	aa6-116	GTITVEELKKLLEQWNLVIGFLPLTWIC LLQFAYANRNRFLYIIKLIFLWLLWPV TLACFVLAAYRINWITGGIAIAMACL VGLMWLSYFIASFRLFARTSMWSFNP ET
	SARS-CoV2-M-P2	aa106-221	TRSMWSFNPETNILLNVPLHGTILTRP LLESELVIGAVIL RGHLRIAGHHLGRCDIKDLPKEITVAT SRTLSYYKLGASQRVAGDSGFAAYSRY RIGNYKLNTDHSSSDNIALLV
Full length target	SARS-CoV-2-FL	aa6-221	GTITVEELKKLLEQWNLVIGFLPLTWIC LLQFAYANRNRFLYIIKLIFLWLLWPV TLACFVLAAYRINWITGGIAIAMACL VGLMWLSYFIASFRLFARTSMWSFNP ETNILLNVPLHGTILTRPLLESELVIGA VILRGHLRIAGHHLGRCDIKDLPKEIT VATSRTLSYYKLGASQRVAGDSGFAA YSRYRIGNYKLNTDHSSSDNIALLV

¹aa positions based on SARS-CoV-2 reference genome (NC-06577.2 (Wuhan strain) and reference M protein sequences, YP_009724393.1.

²For conjugation to polyionic VLPs, the peptide/ protein antigens have a N-terminal TAG and AAYV proteolytic processing sequence.

TABLE 4

SARS-CoV-2 antigens of the N structural protein			
Immunogen	Seq ID	aa-position ¹	Amino acid sequence ²
Short peptide	SARS-CoV-2-N-S1	aa51-84	SWFTALTQHGKEDLKFPFGQGVPIINTNSSP DDQI
	SARS-CoV-2-N-S2	aa75-108	NTNSSPDDQIGYYRRATRRIRGGDGKMKD LSPRW
	SARS-CoV-2-N-S3	aa99-132	GKMKDLSRWYFYLLGTGPEAGLPYGAN KDGIW
	SARS-CoV-2-N-S4	aa123-156	YGANKDGIWVATEGALNTPKDHIGTRNP ANNAA
	SARS-CoV-2-N-S5	aa147-180	GTRNPANNAIIVLQLPQGTTLPKGFYAEG SRGGS
	SARS-CoV-2-N-S6	aa171-204	FYAEGSRGGSQASSRSSRSRNSRNSTPGSS RG
	SARS-CoV-2-N-S7	aa195-227	RNSTPGSSRGTSPPARMAGNGGDAALALL LDRLN
	SARS-CoV-2-N-S8	aa219-252	LALLLLDRLNQLESKMSGKGQQGQQT TKKSA
	SARS-CoV-2-N-S9	aa243-276	GQTVTKKSAEASKKPRQKRTATKAYNVT QAFGR
	SARS-CoV-2-N-S10	aa267-300	AYNVTQAFGRRGPEQTQGNFGDQELIRQG TDYKH
	SARS-CoV-2-N-S11	aa291-324	LIRQGTDYKHWPQIAQFAPSASAFFGMSRI GMEV
	SARS-CoV-2-N-S12	aa315-347	FGMSRIGMEVTPSGTWLTYTGAIKLDDKD PNFK
	SARS-CoV-2-N-S13	aa338-369	KLDDKDPNFKDQVILLNKHIDAYKTFPPTPE PK
Extended peptide	SARS-CoV2-N-E1	aa51-104	SWFTALTQHGKEDLKFPFGQGVPIINTNSSP DDQIGYYRRATRRIRGGDGKMKDL
	SARS-CoV2-N-E2	aa95-148	RGGDGKMKDLSRWYFYLLGTGPEAGLPY GANKDGIWVATEGALNTPKDHIGT
	SARS-CoV2-N-E3	aa139-192	LNTPKDHIGTRNPANNAIIVLQLPQGTTL PKGFYAEGSRGGSQASSRSSRSRN
	SARS-CoV2-N-E4	aa183-235	SSRSSRSRNSRNSTPGSSRGTSPPARMAGN GGDAALALLLLDRLNQLESKMS
	SARS-CoV2-N-E5	aa227-280	LNQLESKMSGKGQQGQQTQVTKKSAEA SKKPRQKRTATKAYNVTQAFGRRGPE
	SARS-CoV2-N-E6	aa271-325	TQAFGRRGPEQTQGNFGDQELIRQGTQDYK HWPQIAQFAPSASAFFGMSRIGMEVT
	SARS-CoV2-N-E7	aa316-369	GMSRIGMEVTPSGTWLTYTGAIKLDDKDP NFKDQVILLNKHIDAYKTFPPTPEPK
Short protein	SARS-CoV2-N-P1	aa51-162	SWFTALTQHGKEDLKFPFGQGVPIINTNSSP DDQIGYYRRATRRIRGGDGKMKDLSRWY FYLLGTGPEAGLPYGANKDGIWVATEGA LNTPKDHIGTRNPANNAIIVLQLPG
	SARS-CoV2-N-P2	aa153-265	NNAAIIVLQLPQGTTLPKGFYAEGSRGGSQ ASSRSSRSRNSRNSTPGSSRGTSPPARMAG NGGDAALALLLLDRLNQLESKMSGKGQ QQGQTVTKKSAEASKKPRQKRTAT
	SARS-CoV2-N-P3	aa255-369	SKKPRQKRTATKAYNVTQAFGRRGPEQTQ GNFGDQELIRQGTQDYKHWPQIAQFAPSAS AFFGMSRIGMEVTPSGTWLTYTGAIKLDDK DPNFKDQVILLNKHIDAYKTFPPTPEPK
Full length	SARS-CoV-2-N-FL (319 aa)	aa51-369	SWFTALTQHGKEDLKFPFGQGVPIINTNSSP DDQIGYYRRATRRIRGGDGKMKDLSRWY FYLLGTGPEAGLPYGANKDGIWVATEGA LNTPKDHIGTRNPANNAIIVLQLPQGTTL PKGFYAEGSRGGSQASSRSSRSRNSRNST PGSSRGTSPPARMAGNGGDAALALLLLDRL NQLESKMSGKGQQGQQTQVTKKSAEAS KKPRQKRTATKAYNVTQAFGRRGPEQTQ GNFGDQELIRQGTQDYKHWPQIAQFAPSAS AFFGMSRIGMEVTPSGTWLTYTGAIKLDDK DPNFKDQVILLNKHIDAYKTFPPTPEPK

¹aa positions based on SARS-CoV-2 reference genome (NC-06577.2 (Wuhan strain) and reference N protein sequences, YP_009724397.2.

²For conjugation to polyionic VLPs, the peptide/protein antigens have a N-terminal TAG and AAYY proteolytic processing sequence.

TABLE 5

SARS-CoV-2 antigens of the S2 structural protein.			
Immunogen	Seq ID	aa-position ¹	Amino acid sequence ²
Extended peptide	SARS-CoV2-S2-E1	aa671-725	CASYQTQTNsprrarsvasqsIIAYTMSLGA ENSVAY SNNsIAIPTNFTISVTTE
	SARS-CoV2-S2-E2	aa716-769	TNFTISVTTEILPVSMTKTSVDCTMYICGDST ECSNLL LQYGSFCTQLNRALTG
	SARS-CoV2-S2-E3	aa760-811	CTQLNRALTGIAVEQDKNTQEVFAQVKQIY KTPPIKDF GGFNFSQILPDPSKPSK
	SARS-CoV2-S2-E4	aa805-860	ILPDPSKPSKRSFIEDLLFNKVTLADAGFIKQ YGDCLGDIA ARDLICAQKFNGLTV
	SARS-CoV2-S2-E5	aa851-905	CAQKFNGLTVLPPLLTDEMIAYTSALLAG TITSGWTF GAGAALQIPFAMQMAYR
	SARS-CoV2-S2-E6	aa897-951	PFAMQMAYRFNGIGVTQNVLYENQKLIAN QFNsAIG KIQDSLSTASALGKLQDV
	SARS-CoV2-S2-E7	aa942-996	ASALGKLQDVVNQNAQALNTLVKQLSSNF GAISSVLN DILSRLDKVEAEVQIDRL
	SARS-CoV2-S2-E8	aa987-1041	VEAEVQIDRLITGRQLSLQTYVTQQLIRAAEI RASANLA ATKMSECVLGQSKRVD
	SARS-CoV2-S2-E9	aa1032-1087	CVLGQSKRVDFCGKGHYLMSFPQSAPHGV VFLHVTYV PAQEKNTTAPAI CHDGKA
	SARS-CoV2-S2-E10	aa1078-1132	APAICHDGKAHFPRG VFVSN GTHWFTVQ RNFYEP QIITDNTFVSGNCDVVIGI
	SARS-CoV2-S2-E11	aa1123-1177	SGNCDVVIGIVNNTVYDPLQPELDSFKEELD KYFKNHT SPDVLGDISGINASVV
	SARS-CoV2-S2-E12	aa1168-1222	DISGINASVVNIQKEIDRLNEVAKNLNESLI DLQELGKYE QYIKWPWYIWLGFIA
	SARS-CoV2-S2-E13	aa1213-1268	PWYIWLGFIAGLIAIVMTIMLCMTSCCSC LKGCCSCGS CCKFDEDDSEPVLKGV
Short protein	SARS-CoV2-S2-P1	aa671-798	CASYQTQTNsprrarsvasqsIIAYTMSLGA ENSVAYSNN SIAIPTNFTISVTTEILPVSMTKTSVDCTMYIC GDSTECNLL LQYGSFCTQLNRALTGIAVEQDKNTQEVFA QVKQIYKTPPIKDFG
	SARS-CoV2-S2-P2	aa788-916	IYKTPPIKDFGGFNFSQILPDPSKPSKRSFIED LLFNKVTLADA GFIKQYGDCLGDIAARDLICAQKFNGLTVL PPLLTDEMIAYTSALLAGTITSGWTFGAG AALQIPFAMQMAYRFNGIGVTQNVL
	SARS-CoV2-S2-P3	aa906-1033	FNGIGVTQNVLYENQKLIANQFNsAIGKIQ DSLSTASALGK LQDVVNQNAQALNTLVKQLSSNFGAISSVL NDILSRLDKVE AEVQIDRLITGRQLSLQTYVTQQLIRAAEIR ASANLAATKMSECV
	SARS-CoV2-S2-P4	aa1023-1152	NLAATKMSECVLGQSKRVDFCGKGHYLMS FPQAPHGVV FLHVTYVPAQEKNTTAPAI CHDGKAHFPR EGVFVSN GTHWFTVQ RNFYEPQIITDNTFVSGNCDVVI GIVNNTVYD PLQPELIDSFKEEL

TABLE 5-continued

SARS-CoV-2 antigens of the S2 structural protein.			
Immunogen	Seq ID	aa-position ¹	Amino acid sequence ²
	SARS-CoV2-S2-P5	aa1142-1268	QPELIDSPKEELDKYFKNHTSPDVLGDISGI
			NASVVNIQKE
			DRLNEVAKNLNESLIDLQELGKYEQYIKWP
			WYIWLGFIAI
			LIAIVMVTIMLCCMTSCCSCCLKGCCCGSCC
			KFDEDDSEPVLKGV

¹aa positions based on SARS-CoV-2 reference genome (NC-06577.2 (Wuhan strain) and reference spike protein sequences, YP_009724390.1.

²For conjugation to polyionic VLPs, the peptide/protein antigens have a N-terminal TAG and AAYY proteolytic processing sequence.

TABLE 6

SARS-CoV-2 antigens of the ORF3a structural protein			
Immunogen	Seq ID	aa-position ¹	Amino acid sequence ²
Short peptide	SARS-CoV2-ORF3a-S1	aa105-132	FLYLYALVYFLQSFNVRRII MRLWLCWK
	SARS-CoV2-ORF3a-S2	aa123-147	IIMRLWLCWKCRSKNPLLYD ANYFL
	SARS-CoV2-ORF3a-S3	aa138-165	PLLYDANYFLCWHTNCYDYC IPYNSVTS
	SARS-CoV2-ORF3a-S4	aa156-183	YCIPYNSVTSSIVITSGDGT TSPISEHD
	SARS-CoV2-ORF3a-S5	aa174-201	GTTSPISEHDYQIGGYTEKW ESGVKDCV
	SARS-CoV2-ORF3a-S6	aa192-219	KWESGVKDCVVLHSHYFTSDY YQLYSTQL
	SARS-CoV2-ORF3a-S7	aa210-238	DYYQLYSTQLSTDTGVEHVT FFIYNKIVD
	SARS-CoV2-ORF3a-S8	aa229-257	TFFIYNKIVDEPEEHVQIHT IDGSSGVVN
	SARS-CoV2-ORF3a-S9	aa248-275	TIDGSSGVVNPVMEPIYDEP TTTTSVPL
	SARS-CoV2-ORF3a-E1	aa105-154	FLYLYALVYFLQSFNVRRII MRLWLCWKCRSKNPLLYDAN YFLCWHTNCY

TABLE 6-continued

SARS-CoV-2 antigens of the ORF3a structural protein			
Immunogen	Seq ID	aa-position ¹	Amino acid sequence ²
	SARS-CoV2-ORF3a-E2	aa145-194	YFLCWHTNCYDYCIPYNSVT SSIVITSGDGTTSPISEHDY QIGGYTEKWE
	SARS-CoV2-ORF3a-E3	aa185-234	QIGGYTEKWESGVKDCVVLH SYFTSDYYQLYSTQLSTDTG VEHVTFFIYN
	SARS-CoV2-ORF3a-E4	aa225-275	VEHVTFFIYNKIVDEPEEHV QIHTIDGSSGVVNPVMEPIY DEPTTTTSVPL
	SARS-CoV2-ORF3a-P1	aa105-195	FLYLYALVYFLQSFNVRRII MRLWLCWKCRSKNPLLYDAN YFLCWHTNCYDYCIPYNSVT SSIVITSGDGTTSPISEHDY QIGGYTEKWES
	SARS-CoV2-ORF3a-P2	aa185-275	QIGGYTEKWESGVKDCVVLH SYFTSDYYQLYSTQLSTDTG VEHVTFFIYNKIVDEPEEHV QIHTIDGSSGVVNPVMEPIY DEPTTTTSVPL
	Full-length	SARS-CoV2-ORF3a-FL (171 aa)	FLYLYALVYFLQSFNVRRII MRLWLCWKCRSKNPLLYDAN YFLCWHTNCYDYCIPYNSVT SSIVITSGDGTTSPISEHDY QIGGYTEKWESGVKDCVVLH SYFTSDYYQLYSTQLSTDTG VEHVTFFIYNKIVDEPEEHV QIHTIDGSSGVVNPVMEPIY DEPTTTTSVPL

¹aa positions based on SARS-CoV-2 reference genome (NC-06577.2 (Wuhan strain) and reference ORF3a protein sequences, YP_009724391.1.

²For conjugation to polyionic VLPs, the peptide/protein antigens have a N-terminal TAG and AAYY proteolytic processing sequence.

TABLE 7

SARS-CoV-2 antigens of the ORF7a structural protein			
Immunogen	Seq ID	aa- position ¹	Amino acid sequence ²
Short peptide	SARS-CoV2-ORF7a-S1	aa1-28	MKIILFLALITLATCELYHY QECVRGTT
	SARS-CoV2-ORF7a-S2	aa20-47	YQECVRGTTVLLKEPCSSGT YEGNSPFH
	SARS-CoV2-ORF7a-S3	aa38-65	GTYEGNSPFHPLADNKFALT CFSTQFAF
	SARS-CoV2-ORF7a-S4	aa56-84	LTCFSTQFAFACPDGVKHVY QLRARSVSP
	SARS-CoV2-ORF7a-S5	aa75-103	YQLRARSVSPKLFIRQEEVQ ELYSPIFLI
	SARS-CoV2-ORF7a-S6	aa94-121	QELYSPIFLIVAAIVFITLC FTLKRKTE
Extended peptide	SARS-CoV-2-ORF7a-E1	aa1-46	MKIILFLALITLeATCELYH YQECVRGTTVLLKEPCSSGT YEGNSPFH
	SARS-CoV-2-ORF7a-E2	aa38-84	GTYEGNSPFHPLADNKFALT CFSTQFAFACPDGVKHVYQL RARSVSP
	SARS-CoV-2-ORF7a-E3	aa75-121	YQLRARSVSPKLFIRQEEVQ ELYSPIFLIVAAIVFITLCF TLKRKTE
Full length	SARS-CoV2-ORF7a-FL	aa1-121	MKIILFLALITLATCELYHY QECVRGTTVLLKEPCSSGT EGNSPFHPLADNKFALT TQFAFACPDGVKHVYQLRARS VSPKLFIRQEEVQELYSPI FLIVAAIVFITLCFTLKRKT E

¹aa positions based on SARS-CoV-2 reference genome (NC-06577.2 (Wuhan strain) and reference ORF7a protein sequences, [yp_009724395.1](#)

²For conjugation to polyionic VLPS, the peptide/protein antigens have a N-terminal TAG and AAYY proteolytic processing sequence.

TABLE 8

SARS-CoV-2 antigens of nsp6 and nsp7 nonstructural proteins			
Immunogen	Seq ID	aa- position ¹	Amino acid sequence ²
Short peptide	SARS-CoV-2 nsp6-S1	1-30	SAVKRTIKGTHHLLLTILT SLLVLVQSTQ

TABLE 8-continued

SARS-CoV-2 antigens of nsp6 and nsp7 nonstructural proteins			
Immunogen	Seq ID	aa- position ¹	Amino acid sequence ²
SARS-CoV-2 nsp6-S2	SARS-CoV-2 nsp6-S2	21-50	SLLVLVQSTQWSLFFFLYEN AFLPFAMGII
		41-70	AFLPFAMGIIAMSAFAMMFV KHKHAFCLCLF
		61-90	KHKHAFCLCLFLPSLATVAY FNMVMPASW
		81-110	FNMVMPASWVMRIMTWLDM VDTSLSGFKL
		101-130	VDTSLSGFKLKDCVMYASAV VLLILMTART
		121-150	VLLILMTARTVYDDGARRVW TLMNVLTLYV
		141-170	TLMNVLTLYVKVYGNALDQ AISMWALIIS
		161-190	AISMWALIISVTSNYSGVVT TVMFLARGIV
		181-210	TVMFLARGIVFMCVEYCPFI FITGNTLQCI
		201-230	FITGNTLQCIMLVYCFGLGYF CTCYFGLFCL
		221-250	CTCYFGLFCLLNRYFRLTLG VYDYLVSSTQE
		241-270	VYDYLVSSTQEFYRMSQGLL PPKNSIDAFK
		261-290	PPKNSIDAFKLNKLLGVGG KPCIKVATVQ
		1-28	SKMSDVKCTSVVLLSVLQQL RVESSSKL
SARS-CoV-2 nsp7-S1	SARS-CoV-2 nsp7-S1	19-46	QLRVESSSKLWAQCVQLHND ILLAKDTT
		37-64	NDILLAKDTTEAFEKMSVLL SVLLSMQG
		55-83	LLSVLLSMQGAVDINKLCEE MLDNRTLQ
SARS-CoV-2 nsp7-S4	SARS-CoV-2 nsp7-S4		

TABLE 8-continued

SARS-CoV-2 antigens of nsp6 and nsp7 nonstructural proteins			
Immunogen	Seq ID	aa- position ¹	Amino acid sequence ²
Extended peptide	SARS- CoV-2 nsp6-E1	1-50	SAVKRTIKGTHHLLLLTILT SLLVLVQSTQWSLFFFLYEN AFLPFAMGII
	SARS- CoV-2 nsp6-E2	41-90	AFLPFAMGIIAMSAFAMMFV KHKHAFCLCLFLLPSLATVAY FNMVYMPASW
	SARS- CoV-2 nsp6-E3	81-130	FNMVYMPASWVRIMTWLDM VDTSLSGFKLKDCVMYASAV VLLILMTART
	SARS- CoV-2 nsp6-E4	121-170	VLLILMTARTVYDDGARRVW TLMNVLTLYVKVYGNALDQ AISMWALIIS
	SARS- CoV-2 nsp6-E5	161-210	AISMWALIISVTSNYSGVVT TVMFLARGIVFMCVEYCP FITGNTLQCI
	SARS- CoV-2 nsp6-E6	201-250	FITGNTLQCIMLVYCFGLGYF CTCYFGLFCLLNRYFRLTLG VYDYLVTQ
	SARS- CoV-2 nsp6-E7	241-290	VYDYLVTQEFYRMYNSQGLL PPKNSIDAFKLNKLLGVGG KPCIKVATVQ
Short protein	SARS- CoV-2 nsp7-E1	1-46	SKMSDVKCTSVVLLSVLQQ RVESSSKLWAQCVQLHNDIL LAKDTT
	SARS- CoV-2 nsp7-E2	37-83	NDILLAKDTTEAFKMSVLL SVLLSMQGAVDINKLCEEML DNRATLQ
	SARS- CoV-2 nsp6-SP1	1-104	SAVKRTIKGTHHLLLLTILT SLLVLVQSTQWSLFFFLYEN AFLPFAMGIIAMSAFAMMFV KHKHAFCLCLFLLPSLATVAY FNMVYMPASWVRIMTWLDM VDTSLSGFKLKDCVMYASAV VLLILMTARTVYDDGARRVW TLMNVLTLYVKVYGNALDQ AISMWALIISVTSNYSGVVT TVMFLARGIVFMCVEYCP
	SARS- CoV-2 nsp6-SP2	94-197	IMTWLDMVDTSLSGFKLKDC VMYASAVVLLILMTARTVYD DGARRVWVTLMNVTLYVKVY YGNALDQAISMWALIISVTS NYSGVVTVMFLARGIVFMC VEYC
Full length	SARS- CoV-2 nsp6-FL	1-290	SAVKRTIKGTHHLLLLTILT SLLVLVQSTQWSLFFFLYEN AFLPFAMGIIAMSAFAMMFV KHKHAFCLCLFLLPSLATVAY FNMVYMPASWVRIMTWLDM VDTSLSGFKLKDCVMYASAV VLLILMTARTVYDDGARRVW TLMNVLTLYVKVYGNALDQ AISMWALIISVTSNYSGVVT TVMFLARGIVFMCVEYCP

TABLE 8-continued

SARS-CoV-2 antigens of nsp6 and nsp7 nonstructural proteins			
Immunogen	Seq ID	aa- position ¹	Amino acid sequence ²
			FITGNTLQCIMLVYCFGLGYF CTCYFGLFCLLNRYFRLTLG VYDYLVTQEFYRMYNSQGLL PPKNSIDAFKLNKLLGVGG KPCIKVATVQ
	SARS- CoV-2 nsp7-FL	1-83	SKMSDVKCTSVVLLSVLQQ RVESSSKLWAQCVQLHNDIL LAKDTTEAFKMSVLLSVL SMQGAVDINKLCEEMLNRA TLQ

¹aa positions based on SARS-CoV-2 reference genome (NC-06577.2; Wuhan strain) and reference nsp6 protein (YP_009725302.1) and nsp7 protein (YP_009725303.1).

²For conjugation to polyionic VLPs, the peptide/protein antigens have a N-terminal TAG and AAYY proteolytic processing sequence.

TABLE 9

SARS-CoV-2 antigens of nsp12-1 nonstructural protein fragment			
Immunogen	Seq ID	aa- position ¹	Amino acid sequence ²
Short peptide	SARS- CoV-2 nsp12- 1-S1	126-153	DLVYALRHFDEGNCDTLKEI LVTYNCCD
	SARS- CoV-2 nsp12- 1-S2	144-171	EILVTYNCCDDYFNKKDWY DFVENPDI
	SARS- CoV-2 nsp12- 1-S3	162-189	WYDFVENPDILRVYANLGER VRQALLKT
	SARS- CoV-2 nsp12- 1-S4	180-207	ERVQRALLKTVQFCDAMRNA GIVGVLT
	SARS- CoV-2 nsp12- 1-S5	198-225	NAGIVGVLTLDNQDLNGN WYDFGDFIQ
	SARS- CoV-2 nsp12- 1-S6	216-243	WYDFGDFIQTPGSGVPVVD SYYSLLMP
	SARS- CoV-2 nsp12- 1-S7	234-261	VDSYYSLLMPILTLTRALTA ESHVDTDL
	SARS- CoV-2 nsp12- 1-S8	252-280	TAESHVDTDLTKPYIKWDL LYDFTEERL

TABLE 9-continued

SARS-CoV-2 antigens of nsp12-1 nonstructural protein fragment			
Immunogen	Seq ID	aa-position ¹	Amino acid sequence ²
	SARS-CoV-2 nsp12-1-S9	271-299	LKYDFTEERLKLFDYFKYW DQTYHPNCV
	SARS-CoV-2 nsp12-1-S10	290-317	WDQTYHPNCVNCDDRILH CANFNVLF
	SARS-CoV-2 nsp12-1-S11	308-336	LHCANFNVLFSTVFPPTS PLVRKIFVD
	SARS-CoV-2 nsp12-1-S12	327-355	GPLVRKIFVDGVPFVSTGY HFRELGVVH
	SARS-CoV-2 nsp12-1-S13	346-375	YHFRELGVVHNQDVNLHSSR LSFKELLVYA
Extended peptide	SARS-CoV-2 nsp12-1-E1	126-175	DLVYALRHFDEGNCDTLKEI LVTYNCCDDYFNKKDWYDF VENPDILRVY
	SARS-CoV-2 nsp12-1-E2	166-215	VENPDILRVYANLGERVRQA LLKTQVQCDAMRNAGIVGVL TLDNQDLNGN
	SARS-CoV-2 nsp12-1-E3	206-255	TLDNQDLNGNWDYFGDFIQT TPGSGVPVVDYYSLLMPIL TLTRALTAES
	SARS-CoV-2 nsp12-1-E4	246-295	TLTRALTAESHVDTDLTKPY IKWDLKDYDFTEERLKLFD YFKYWDQTYH
	SARS-CoV-2 nsp12-1-E5	286-335	YFKYWDQTYHPNCVNCDDR CILHCANFNVLFSTVFPPTS FGPLVRKIFV
	SARS-CoV-2 nsp12-1-E6	326-375	FGPLVRKIFVDGVPFVSTG YHFRELGVVHNQDVNLHSSR LSFKELLVYA
	SARS-CoV-2 nsp12-1-SP1	126-255	DLVYALRHFDEGNCDTLKEI LVTYNCCDDYFNKKDWYDF VENPDILRVYANLGERVRQA LLKTQVQCDAMRNAGIVGVL TLDNQDLNGNWDYFGDFIQT TPGSGVPVVDYYSLLMPIL TLTRALTAES
Short protein	SARS-CoV-2 nsp12-1-SP2	246-375	TLTRALTAESHVDTDLTKPY IKWDLKDYDFTEERLKLFD YFKYWDQTYHPNCVNCDDR CILHCANFNVLFSTVFPPTS FGPLVRKIFVDGVPFVSTG YHFRELGVVHNQDVNLHSSR LSFKELLVYA

TABLE 9-continued

SARS-CoV-2 antigens of nsp12-1 nonstructural protein fragment			
Immunogen	Seq ID	aa-position ¹	Amino acid sequence ²
Full length antigen	SARS-CoV-2 nsp12-1-FL	126-375	DLVYALRHFDEGNCDTLKEI LVTYNCCDDYFNKKDWYDF VENPDILRVYANLGERVRQA LLKTQVQCDAMRNAGIVGVL TLDNQDLNGNWDYFGDFIQT TPGSGVPVVDYYSLLMPIL TLTRALTAESHVDTDLTKPY IKWDLKDYDFTEERLKLFD YFKYWDQTYHPNCVNCDDR CILHCANFNVLFSTVFPPTS FGPLVRKIFVDGVPFVSTG YHFRELGVVHNQDVNLHSSR LSFKELLVYA

¹aa positions based on SARS-CoV-2 reference genome (NC-06577.2; Wuhan strain) and reference nsp12 protein (YP_009725307.1).
²For conjugation to polyionic VLPs, the peptide/protein antigens have a N-terminal TAG and AAYY proteolytic processing sequence.

TABLE 10

SARS-CoV-2 antigens of nsp12-2 nonstructural protein fragment			
Immunogen	Seq ID	aa-position	Amino acid sequence
Short peptide	SARS-CoV-2 nsp12-2-S1	520-548	SYEQDALFAYTKRNVIP ITQMNLYAI
	SARS-CoV-2 nsp12-2-S2	538-566	TITQMNLYAISAKNRART VAGVSICSTM
	SARS-CoV-2 nsp12-2-S3	557-586	VAGVSICSTMNRQFHQKLL KSIAATRGAT
	SARS-CoV-2 nsp12-2-S4	576-604	LKSIAATRGATVVIGTSKF YGGWHNMLKT
	SARS-CoV-2 nsp12-2-S5	595-624	YGGWHNMLKTVYSDVENPH LMGWDYPKCDR
	SARS-CoV-2 nsp12-2-S6	615-643	MGWDYPKCDRAMPNMLRIM ASLVLARKHT
	SARS-CoV-2 nsp12-2-S7	634-662	ASLVLARKHTCCSLSHRF YRLANCAQV
	SARS-CoV-2 nsp12-2-S8	653-683	YRLANCAQVLSMVMCGG SLYVKPGGTSSG

TABLE 10-continued

SARS-CoV-2 antigens of nsp12-2 nonstructural protein fragment			
Immunogen	Seq ID	aa-position	Amino acid sequence
Extended peptide	SARS-CoV-2 nsp12-2-S9	674-704	YVKPGGTSSGDATTAYANS VFNICQAVTANV
	SARS-CoV-2 nsp12-2-S10	695-724	NICQAVTANVNALLSTDGN KIADKYVRNLQ
	SARS-CoV-2 nsp12-2-S11	715-744	IADKYVRNLQHRLYECLYR NRDVTDFVNE
	SARS-CoV-2 nsp12-2-S12	735-764	RDVDTFVNEFYAYLRKHF SMMILSDDAVV
	SARS-CoV-2 nsp12-2-S13	755-784	MMILSDDAVVCFNSTYASQ GLVASIKNFKS
	SARS-CoV-2 nsp12-2-S14	775-804	LVASIKNFKSVLYYQNNVF MSEAKCWTETD
	SARS-CoV-2 nsp12-2-S15	795-824	SEAKCWTETDLTKGPHEFC SQHTMLVKQGD
	SARS-CoV-2 nsp12-2-S16	815-844	QHTMLVKQGDDYVYLPYPD PSRILGAGCFV
	SARS-CoV-2 nsp12-2-S17	835-863	SRILGAGCFVDDIVKTDGT LMIERFVSLA
	SARS-CoV-2 nsp12-2-S18	854-882	LMIERFVSLAIDAYPLTKH PNQEYADV FH
	SARS-CoV-2 nsp12-2-S19	873-901	PNQEYADV FHLYLQYIRKL HDELTGHMLD
	SARS-CoV-2 nsp12-2-S20	892-920	HDELTGHMLDMYSVMLTND NTSRYWEPEF
	SARS-CoV-2 nsp12-2-E1	520-498	SYEDQDALFAYTKRNV IPT ITQMNLKYAISAKNRARTV AGVSICSTMTN
	SARS-CoV-2 nsp12-2-E2	559-607	GVSICSTMTNRQFHQKLLK SIAATRGATVVI GTSKFG GWHNMLKTVYS

TABLE 10-continued

SARS-CoV-2 antigens of nsp12-2 nonstructural protein fragment			
Immunogen	Seq ID	aa-position	Amino acid sequence
Short protein	SARS-CoV-2 nsp12-2-E3	598-646	WHNMLKTVYSDVENPHLMG WDYPKCDRAMPNMLRIMAS LVLARKHTTCC
	SARS-CoV-2 nsp12-2-E4	637-685	VLARKHTTCCSLSHRFYRL ANCAQVLSMVMCGGSLY VKPGGTSSGDA
	SARS-CoV-2 nsp12-2-E5	676-724	KPGGTSSGDATTAYANSVF NICQAVTANVNALLSTDGN KIADKYVRNLQ
	SARS-CoV-2 nsp12-2-E6	715-763	IADKYVRNLQHRLYECLYR NRDVTDFVNEFYAYLRKH FSMMILSDDAV
	SARS-CoV-2 nsp12-2-E7	754-802	SMMILSDDAVVCFNSTYAS QGLVASIKNFKSVLYYQNN VFMSEAKCWTE
	SARS-CoV-2 nsp12-2-E8	793-841	FMSEAKCWTETDLTKGPHE FCSQHTMLVKQGDDYVYLP YPDP SRILGAG
	SARS-CoV-2 nsp12-2-E9	832-880	PDP SRILGAGCFVDDIVKT DGTLMIERFVSLAIDAYPL TKHPNQEYADV
	SARS-CoV-2 nsp12-2-E10	871-920	KHPNQEYADV FHLYLQYIR KLHDEL TGHMLDMYSVMLT NDNTSRYWEPEF
	SARS-CoV-2 nsp12-2-SP1	520-629	SYEDQDALFAYTKRNV IPT ITQMNLKYAISAKNRARTV AGVSICSTMTNRQFHQKLL KSIAATRGATVVI GTSKFG GGWHNMLKTVYSDVENPHL MGWDYPKCDRAMPNM
	SARS-CoV-2 nsp12-2-SP2	619-727	YPKCDRAMPNMLRIMASLV LARKHTTCCSLSHRFYRLA NECAQVLSMVMCGGSLYV KPGGTSSGDATTAYANSVF NICQAVTANVNALLSTDGN KIADKYVRNLQHRL
	SARS-CoV-2 nsp12-2-SP3	717-826	DKYVRNLQHRLYECLYRNR DVTDFVNEFYAYLRKHFS MMILSDDAVVCFNSTYASQ GLVASIKNFKSVLYYQNNV FMSEAKCWTETDLTKGPHE FCSQHTMLVKQGDDY
	SARS-CoV-2 nsp12-2-SP4	816-920	HTMLVKQGDDYVYLPYPDP SRILGAGCFVDDIVKTDGT LMIERFVSLAIDAYPLTKH PNQEYADV FHLYLQYIRKL HDELTGHMLDMYSVMLTND NTSRYWEPEF

TABLE 10-continued

SARS-CoV-2 antigens of nsp12-2 nonstructural protein fragment			
Immunogen	Seq ID	aa-position	Amino acid sequence
Full length antigen	SARS-CoV-2 nsp12-2-FL	520-920	SYEDQDALFAYTKRNV IPTQMNLYKAI SAKNRARTVAGV SICSTMTNRQPHQKLLK SIAATRGATV VIGTSK FYGGWHNMLKTVYSDVENPHLMGWDY PKCDRAMPNMLRIMASLVLARKHTTCCSLSHRFYRLANEC AQV LSEMVMCGGSLYVKPGGTSSGDATAYANSVFNICQAVTANVNALLS TDGNKIADKYVRNLQHRLYECLYRNRD VDTDFVNEFYA YLRKHFSMMILSDDAVVCFNSTYASQGLVASIKNFKSVLYYQNNVFMSEAKCWTETD LTKGPHEFCSQHTMLVKQGGDDYVYLPYPDP SRILGAGCFVDDIVKTDGTLMIERFVSLAIDAYPLTKHPNQEYADV FHLYLQYIRKLHDEL TGHLDMYSVMLTNDNTSR YWEP EF

¹aa positions based on SARS-CoV-2 reference genome (NC-06577.2; Wuhan strain) and reference nsp12 protein (YP_009725307.1).

²For conjugation to polyionic VLPs, the peptide/protein antigens have a N-terminal TAG and AAYY proteolytic processing sequence.

TABLE 11

OC43 and HKU1 antigens of the M structural protein			
Immunogen	Seq ID	aa-position ¹	Amino acid sequence ²
Short peptides	OC43-M-S1	aa14-42	TADEAIKFLKEWNFSLGIIL LFITIIILQF
	OC43-M-S2	aa33-60	LLFITIIILQFGYTSRSMFV YVIKMIILW
	OC43-M-S3	aa51-79	VYVIKMIILWLMWPLTIILT IFNCVYALN
	OC43-M-S4	aa70-98	TIFNCVYALNNVYLGLSIVF TIVAIIMWI
	OC43-M-S5	aa89-117	FTIVAIIMWIVYFVNSIRLF IRTGSFWSF
	OC43-M-S6	aa108-136	FIRTGSFWSFNPETNNLMCI DMKGTMYVR
	OC43-M-S7	aa127-154	IDMKGTMYVRPIIEDYHTLT VTIIIRGHL
	OC43-M-S8	aa145-172	LTVTIIIRGHLYIQGIKLTG YSLADLPA
	OC43-M-S9	aa163-190	TGYSLADLPAYMTVAKVTHL CTYKRGFL
	OC43-M-S10	aa181-208	HLCTYKRGFLDRISDTSGFA VYVKS KVG
	OC43-M-S11	aa199-226	FAVYVKS KVGNYRLPSTQKG SGMDTALL

TABLE 11-continued

OC43 and HKU1 antigens of the M structural protein			
Immunogen	Seq ID	aa-position ¹	Amino acid sequence ²
Extended peptides	HKU1-M-S1	aa75-97	NNAFLAFSIVFTIISIVIWI LYF
	HKU1-M-S2	aa174-200	KVQVLCTYKRAFLDKLDVNS GFAV FVK
	OC43-M-E1	aa14-57	TADEAIKFLKEWNFSLGIIL LFITIIILQFGYTSRSMFVYV IKMI
	OC43-M-E2	aa48-91	SMFVYVIKMIILWLMWPLTI ILTIFNCVYALNNVYLGLSI VFTI
	OC43-M-E3	aa82-125	YLGLSIVFTIVAIIMWIVYF VNSIRLFIRTGSFWSFNPET NNLM
	OC43-M-E4	aa116-159	SFNPETNNLMCIDMKGTMFV RPIIEDYHTLTVTIIIRGHLY IQGI
	OC43-M-E5	aa150-194	IRGHLYIQGIKLTGYSLAD LPAYMTVAKVYLTCTYKRGF LDKIS
	OC43-M-E6	aa185-226	YKRGFLDKISDTSGFAVYVK SKVGNRYRLPSTQKSGMDTA LL
	HKU1-M-E1	aa75-97/174-200	NNAFLAFSIVFTIISIVIWI LYF/KVQVLCTYKRAFLDKL DVNSGFAV FVK
	OC43-M-P1	aa14-151	TADEAIKFLKEWNFSLGIIL LFITIIILQFGYTSRSMFVYV IKMIILWLMWPLTIILTIFN C VYALNNVYLGLSIVFTIVA IIMWIVYFVNSIRLFIRTGS FWSFNPETNNLMCIDMKGTM YVRPIIEDYHTLTVTIIIR
	OC43// HKU1-M-P2	aa141-226// aa75-97/ aa174-200	DYHTLTVTIIIRGHLYIQGIK LGTGYSLADLPAYMTVAKVT HLCTYKRGFLDRISDTSGFA VYVKS KVGNYRLPSTQKSG MDTALL/ NNAFLAFSIVFTI ISIVIWI LYF/KVQVLCTYK RAFLDKLDVNSGFAV FVK

TABLE 11-continued

OC43 and HKU1 antigens of the M structural protein			
Immunogen	Seq ID	aa- position ¹	Amino acid sequence ²
Full length	OC43-M//	aa14-226//	TADEAIKFLKEWNFSLGIIL
	HKU1-M-	aa75-97//	LFITIIILQPGYTSRSMFVV
variant-		aa174-200	IKMIILWLMWPLTIILTIFN
FL			CVYALNNVYLGLSIVFTIVA
(263 aa)			IIMWIVYFVNSIRLFIPTGS
			FWSFNPETNNLMCIDMKGTM
			YVRPIIEDYHTLTVTIIRGH
			LYIQGIKLGTYSLADLPAY
			MTVAKVTHLCTYKRGFLDRI
			SDTSGFAVYVKSQVGNRLP
			STQKGSGMDTALL//NNAFL

TABLE 11-continued

OC43 and HKU1 antigens of the M structural protein			
Immunogen	Seq ID	aa- position ¹	Amino acid sequence ²
			AFSIVFTIISIVIWIYF/K VQVLCTYKRAFLDKLDVNSG FAVFK

¹aa positions based on OC43 reference M protein sequences, YP_009555244.1, and HKU1 reference M protein sequence, YP_173241.1.

²For conjugation to polyionic VLPs, the peptide/protein antigens have a N-terminal TAG and AAY proteolytic processing sequence.

/ indicates break in native amino acid sequence

// indicates break between OC43 amino acid sequences and HKU1 amino acid sequences.

TABLE 12

OC43 and HKU1 antigens of N structural protein			
Immunogen	Seq ID	aa positions ¹	Amino acid sequence ²
Short peptides	OC43-N-S1	aa64-88	SWFSGITQFQKGKEFEVGEQGVPI
	OC43-N-S2	aa99-131	GYWYRHNRRSFKTADGNQRQLLPRWYF YYLGTG
	OC43-N-S3	aa122-155	RWYFYLLGTGPHAKDQYGDIDGVYWA SNQAVD
	OC43-N-S4	aa146-177	YWVASNQADVNTPADIVDRDPSSDEAIP TRFP
	OC43-N-S5	aa168-200	SDEAIPTRFPPGTLPQGGYIEGSGRSAPN SRS
	OC43-N-S6	aa191-223	SGRSAPNSRSTSRSSRASSAGSRSRANS NRT
	OC43-N-S7	aa214-245	RSRANSNRTPTSGVTPDMADQIASLVL AKLG
	OC43-N-S8	aa236-265	IASLVLAKLGKDATKPPQVTKHTAKEVR QK
	OC43-N-S9	aa270-302	PRQKRSPNKQCTVQCCFGKRGPNQNF GGEMLK
	OC43-N-S10	aa293-325	QNFGGGEMLKLGTSDDPQPILAEALPTA GAFFF
	OC43-N-S11	aa316-340/ aa350-358	LAPTAGAFFFGSRLELAKVQNLSGN/ELR YNGAIR
	OC43-N-S12	aa350-381	ELRYNGAIRFDSTLSGFETIMKVLNENLN AYQ
	HKU1-N-S1	aa131-156	PYANASYGESLEGVFWVANHQAADTST
	HKU1-N-S2	aa147-170	VANHQAADTSTPSDVSSRDPTTQEA
	HKU1-N-S3	aa258-289	RPGRSRSRGPNNRSLRSNSNFRHSDSIV KP
	HKU1-N-S4	aa325-341	SKLDLVKRDSEADSPVK

TABLE 12-continued

OC43 and HKU1 antigens of N structural protein			
Immunogen	Seq ID	aa positions ¹	Amino acid sequence ²
Extended peptides	OC43-N-1	aa64-88/ aa99-119	SWFSGITQFQKGKEFEFVEGQGVPI/ GYWYRHNRRSFKTADGNQRQL
	OC43-N-2	aa110-155	KTADGNQRQLLPWYFYLLGTGPHAKD QYGTIDIGVYVWASNQADV
	OC43-N-3	aa146-191	YWVASNQADVNTPADIVDRDPSSDEAIP TRFPPGTVLPQGYIEGS
	OC43-N-4	aa182-224	LPQGYIEGSGRSAPNSRSTSRSTSSRAS SAGSRSRANSNGNRP
	OC43-N-5	aa215-260	SRANSNGNRTPTSGVTPDMADQIASLVLA KLGKDATKPQQVTKHTAK
	OC43-N-6	aa251-265/ aa270-300	PQQVTKHTAKEVRQK/ PRQKRSPNKQCTVQQCFGKRGPNQNF GGEM
	OC43-N-7	aa291-336	PNQNFGGGEMKLKGTSDPQFPILAEALP TAGAFFFGSRLELAKVQN
	OC43-N-8	aa327-340/ aa350-381	SRLELAKVQNLSGN/ELRYNGAIRFDSTL SGFETIMKVLNENLNAYQ
Short protein	KHU1-N-1	aa131-170	PYANASYGESLEGVFWVANHQADTSTPS DVSSRDPTTQEA
	KHU1-N-2	aa198-229/ aa325-341	RPGSRSQSRGPNRSLRSNSNFRHSDSIV KP/SKLDLVKRDSEADSPVK
	OC43-N-P1	aa64-88/ aa99-184	SWFSGITQFQKGKEFEFVEGQGVPI/ GYWYRHNRRSFKTADGNQRQLLPWYF YYLGTGPHAKDQYGT DIDGVYVWASNQADVNTPADIVDRDPS SDEAIPTRFPPGTVLPQ
	OC43-N-P2	aa174-265/ 270-283	TRFPPGTVLPQGYIEGSGRSAPNSRSTSR TSSRASSAGSRSRANSNGNRTPTSGVTPDM ADQIASLVLAKLKDATKPQQVTKHTA KEVRQK/PRQKRSPNKQCTVQ
	OC43-N-P3	aa273-381	KRSPNKQCTVQQCFGKRGPNQNFGGGE MLKGTSDPQFPILAEALPTAGAFFFGSRL ELAKVQNLSGNELRYNGAIRFDSTLSGFE TIMKVLNENLNAYQ
	HKU-N-P1	aa131-170/ 198-229/ 325-341	PYANASYGESLEGVFWVANHQADTSTPS DVSSRDPTTQEA/ RPGSRSQSRGPNRSLRSNSNFRHSDSIV KP/SKLDLVKRDSEADSPVK

TABLE 12-continued

OC43 and HKU1 antigens of N structural protein			
Immunogen	Seq ID	aa positions ¹	Amino acid sequence ²
Full length	OC43-	aa64-88/	SWFSGITQFQKGKEFEFVEGQGVPI/
	N/HKU1-	aa99-265/	GYWYRHNNRRSFKTADGNQRQLLPWYF
N-variant-FL (383 aa)	N-variant-FL (383 aa)	aa270-381//	YYLGTGPHAKDQYGDIDGVYVWASNQA
		aa131-170/	DVNTPADIVDRDPSSDEAIPTRFPPPGTVL
		198-229/	PQGYIEGSGRSAPNSRSTSRSSSRASSAG
		325-341	SRSRANSNGNRTPTSGVTPDMADQIASLV
			LAKLGKDATKPKQVTKHTAKEVRQK/
			PRQKRSPNKQCTVQQCFGKRGNQNF
			GGEMCLKGTSDFPFILAEALPTAGAFF
			GSRLLEAKVQNLGSGNELRYNGAIRFDSTL
			SGFETIMKVLNENLNAYQ//
			PYANASYGESLEGVFWVANHQADTSTPS
			DVSSRDPTTQEA/
			RPGSRSQSRGPNRSLRSNSNFRHSDSIV
			KP/SKLDLVKRDSEADSPVK

¹aa positions based on OC43 reference N protein sequences, YP_009555245.1, and HKU1 reference N protein sequence, YP_173242.1.

²For conjugation to polyionic VLPs, the peptide/protein antigens have a N-terminal TAG and AAYY proteolytic processing sequence.

/ indicates break in native amino acid sequence

// indicates break between OC43 amino acid sequences and HKU1 amino acid sequences.

TABLE 13

OC43 and HKU1 S2 antigens of the S2 structural protein			
Immunogen	Seq ID	aa positions ¹	Amino acid sequence ²
Short peptide	OC43-S2-S1	aa898-929	SKASSRSAIEDLLFDKVKLSDVGFVEAY NNCT
	OC43-S2-S2	aa920-951	GFVEAYNNCTGGAEIRDLICVQSYKGI KVLPP
	OC43-S2-S3	aa943-973	SYKGIKVLPPLLSSENQISGYTLAATSASL FPP
	OC43-S2-S4	aa964-995	AATSASLFPPWTAAGVPFFYLVNVQYRI NGLGV
	OC43-S2-S5	aa986-1017	VQYRINGLGVTMDVLSQNQKLIANAF NNALYA
	OC43-S2-S6	aa1008-1039	ANAFNNALYAIQEGFDATNSALVKIQ AVVNAN
	OC43-S2-S7	aa1030-1061	VKIQAVVNANAEALNNLLQQLSNRFG AISASL
	OC43-S2-S8	aa1052-1084	NRFGAISASLQEILSRDLAEAEAQIDRL INGR
	OC43-S2-S9	aa1075-1107	AQIDRLINGRLTALNAYVSQQLSDSTLV KFSAA
	OC43-S2-S10	aa1098-1130	DSTLVKFSAAQAMEKVNECVKSQSSRI NFCGNG
	OC43-S2-S11	aa1120-1153	QSSRINFCGNGNHI ISLVQNAPYGLYFIHFSYVP/
	OC43-S2-S12	aa1228-1258	PNLPDFKEELDQWFKNQTSVAPDLSLD YINVT
	OC43-S2-S13	aa1250-1281	DLSLDYINVTFDLQVEMNRLQEAIKV LNQSY
	OC43-S2-S14	aa1272-1302	EAIKVLNQSYINLKDIGTYEYVVKWPW YVWL

TABLE 13-continued

OC43 and HKU1 S2 antigens of the S2 structural protein			
Immunogen	Seq ID	aa positions ¹	Amino acid sequence ²
Extended peptide	HKU1-S2-S1	aa1229-1258	PKLSDFESELSHWFKNQTSIAPNLTNLHT
	HKU1-S2-S2	aa1249-1280	APNLTNLHTINATFLDLYYEMNLIQESIKSL
	OC43-S2-E1	aa898-942	SKASSRSAIEDLLFDKVKLSDVGFVEAYNNCTGGAEIRDLCIVQS
	OC43-S2-E2	aa933-978	EIRDLCIVQSYKGIVLPPLLENQISGYTLAATSASLPPWTAAA
	OC43-S2-E3	aa969-1013	SLFPPWTAAAGVPFYLVNQYRINGLGV TMDVLSQNQKLIANAFNN
	OC43-S2-E4	aa1004-1048	QKLIANAFNNALYAIQEGFDATNSALV KIQAVVNANAEALNNLLQ
	OC43-S2-E5	aa1039-1083	NAEALNNLLQQLSNRFGAISASLQEILSRLDALEAEAQIDRLING
	OC43-S2-E6	aa1074-1119	EAQIDRLINGRLTALNAYVSQQLSDSL VKFSAAQAMEKVNECKS
	OC43-S2-E7	aa1109-1153	AMEKVNECVKSQSSRINFCGNGNHIISLVQNAPYGLYFIHFSYVP/
	OC43-S2-E8	aa1228-1269	PNLPDFKEELDQWFKNQTSVAPDLSLDYINVTFLDLQVEMNR
Short protein	OC43-S2-E9	aa1260-1302	FLDLQVEMNRLQEAIKVLNQSYINLKD IGTYEYYVKWPWYVWL
	HKU1-S2-E1	aa1229-1280	PKLSDFESELSHWFKNQTSIAPNLTNLHTINATFL MNLIQESIKSL
	OC43-S2-P1	aa898-1033	SKASSRSAIEDLLFDKVKLSDVGFVEAYNNCTGGAEIRDLCIVQSYKGIVLPPLLENQISGYTLAATSASLPPWTAAAGVPFYLVNQYRINGLGV TMDVLSQNQKLIANAFNNALYAIQEGFDATNSALVKIQ
	OC43-S2-P2	aa1022-1153	FDATNSALVKIQAVVNANAEALNNLLQQLSNRFGAISASLQEILSRLDALEAEAQIDRLINGRLTALNAYVSQQLSDSL VKFSAAQAMEKVNECVKSQSSRINFCGNGNHIISLVQNAPYGLYFIHFSYVP/
	OC43/HKU1-variant-S2-P3	aa1228-1302//1229-1280	PNLPDFKEELDQWFKNQTSVAPDLSLDYINVFLDLQVEMNRLQEAIKVLNQSYINLKD IGTYEYYVKWPWYVWL//PKLSDFESELSHWFKNQTSIAPNLTNLHTINATFLMNLIQESIKSL

¹aa positions based on OC43 reference Spike protein sequences, YP_009555241.1, and HKU1 reference spike protein sequence, YP_173238.1.

²For conjugation to polyionic VLPs, the peptide/protein antigens have a N-terminal TAG and AAYY proteolytic processing sequence.

/ indicates break in native amino acid sequence

// indicates break between OC43 amino acid sequences and HKU1 amino acid sequences

TABLE 14

OC43 antigens of nonstructural proteins nsp3 and nsp4			
Immunogen	Seq ID	aa positions ¹	Amino acid sequence ²
Short peptide	OC43-nsp3-S1	aa2368-2399	FMRFYIIIASFIKFLFRHVAYGCSKSGCLF

TABLE 14-continued

OC43 antigens of nonstructural proteins nsp3 and nsp4			
Immunogen	Seq ID	aa positions ¹	Amino acid sequence ²
	OC43-nsp3-S2	aa2390-2422	YGCSKSGCLFYKRNRLRVKCSITVGGMIRYY
	OC43-nsp3-S3	aa2413-2442	TIVGGMIRYYDVMANGGTGFCSKHQWNCID
	OC43-nsp3-S4	aa2433-2463	CSKHQWNCIDCDSYKPGNTFITVEAALDLSK
	OC43-nsp3-S5	aa2454-2473/ aa2542-2553	TVEAALDSLKELKRPIQPTD/NAAVFYAQLFR
	OC43-nsp3-S6	aa2544-2574	AVFYAQLFRPILMVDKNLITTANTGTSVTE
	OC43-nsp3-S7	aa2565-2595	TANTGTSVTETMFDVYVDTFLSMFDVDDKSL
	OC43-nsp3-S8	aa2586-2602/ aa2651-2665	SMFDVDDKSLNALIATA/ELTDESCNNLVPTYL
	OC43-nsp3-S9	aa2656-2686	SCNNLVPTYLKSDNIVAADLGVLIQNSAKHV
	OC43-nsp3-S10	aa2677-2705	VLIQNSAKHVQGNVAKIAGVSCIWSVDAF
Extended peptide	OC43-nsp3-E1	aa2368-2409	FMRFYIIIASFIKLFSLFRHVAYGCSKSGCLFCYKRN RSLRV
	OC43-nsp3-E2	aa2400-2442	CYRNRLRVKCSITVGGMIRYYDVMANGGTGFCSK HQWNCID
	OC43-nsp3-E3	aa2433-2473	CSKHQWNCIDCDSYKPGNTFITVEAALDSLKELKR PIQPTD/
	OC43-nsp3-E4	aa2542-2585	NAAVFYAQLFRPILMVDKNLITTANTGTSVTETM FDVYVDTFL
	OC43-nsp3-E5	aa2576-2602/ aa2651-2673	MFDVYVDTFLSMFDVDDKSLNALIATA/ ELTDESCNNLVPTYLKSDNIVAA
	OC43-nsp3-E6	aa2664-2705	YLKSDNIVAADLGVLIQNSAKHVQGNVAKIAGVS CIWSVDAF
Short protein	OC43-nsp3-P1	aa2368-2473	FMRFYIIIASFIKLFSLFRHVAYGCSKSGCLFCYKRN RSLRVKCSITVGGMIRYYDVMANGGTGFCSKHQWNCIDCDSYK PGNTFITVEAALDSLKELKRPIQPTD/
	OC43-nsp3-P2	aa2542-2602/ aa2651-2705	NAAVFYAQLFRPILMVDKNLITTANTGTSVTETM FDVYVDTFLS MFDVDDKSLNALIATA / ELTDESCNNLVPTYLKSDNIVAADLGVLIQNSAKH VQGNVAKIAG VSCIWSVDAF
Full length	OC43-nsp3-FL	aa2368-2473/ aa2542-2602/ aa2651-2705	FMRFYIIIASFIKLFSLFRHVAYGCSKSGCLFCYKRN RSLRVKCSITVGGMIRYYDVMANGGTGFCSKHQWNCIDCDSYK GNTFITVEAA LDLSKELKRPIQPTD/NAAVFYAQLFRPILMV DKNLITTANTGTSVTETMFDVYVDTFLSMFDVDDK SLNALIATA/ ELTDESCNNLVPTYLKSDNIVAADLGVLIQNSAKH VQGNVAKIAG VSCIWSVDAF
Short peptide	OC43-nsp4-S1	aa2875-2902	SADGVQCYTPHSQISYSNFIYASGCVLSS
	OC43-nsp4-S2	aa2893-2919	FYASGCVLSSACTMFMADGSPQPYCY/

TABLE 14-continued

OC43 antigens of nonstructural proteins nsp3 and nsp4			
Immunogen	Seq ID	aa positions ¹	Amino acid sequence ²
	OC43-nsp4-S3	aa2932-2957	SLVPHVRYNLANAKGFIRPPEVLREG
	OC43-nsp4-S4	aa2948-2973	IRFPEVLREGLVRIVRTRSMSYCRVG
	OC43-nsp4-S5	aa2964-2988	TRSMSYCRVGLCEEDEGICFNFNG
	OC43-nsp4-S6	aa2979-3003	DEGICFNFNNGSWVLNNDYYRSLPGT
	OC43-nsp4-S7	aa2994-3018	NDYYRSLPGTFCGRDVFDFLIYQLFK
	OC43-nsp4-S8	aa3009-3033	VDFLIYQLFKGLAQPVDFLALTASS
	OC43-nsp4-S9	aa3024-3048	VDFLALTASSIAGAILAVIVVLVFI
	OC43-nsp4-S10	aa3039-3064	LAVIVVLVFYYLIKLKRAFGDYTSVV/
	OC43-nsp4-S11	aa3187-3216	NRYSLSYNKYRYYSKGMDTAAYREAACSQL
	OC43-nsp4-S12	aa3207-3236	AYREAACSQLAKAMDTFTNNGSDVLYQPP
Extended peptide	OC43-nsp4-E1	aa2875-2919	SADGVQCYTPHSQISYSNFIYASGCVLSSACTMFTMADGSPQPYCY
	OC43-nsp4-E2	aa2932-2977	SLVPHVRYNLANAKGFIRPPEVLREGLVRIVRTRSMSYCRVGLCEE
	OC43-nsp4-E3	aa2968-3011	SYCRVGLCEEDEGICFNFNNGSWVLNNDYYRSLPGTFCGRDVFDF
	OC43-nsp4-E4	aa3002-3046	GTFCGRDVFDFLIYQLFKGLAQPVDFLALTASSIAGAILAVIVVLV
	OC43-nsp4-E5	aa3038-3064/ aa3187-3203	ILAVIVVLVFYYLIKLKRAFGDYTSVV/ NRYSLSYNKYRYYSKGMDTAAYREAACSQLAKAMDTFTNNN
	OC43-nsp4-E6	aa3195-3236	KYRYYSKGMDTAAYREAACSQLAKAMDTFTNNNGSDVLYQPP
Short protein	OC43-nsp4-P1	aa2875-2919/ aa2932-3006	SADGVQCYTPHSQISYSNFIYASGCVLSSACTMFTMADGSPQPYCY/ SLVPHVRYNLANAKGFIRPPEVLREGLVRIVRTRSMSYCRVGLCEE DEGICFNFNNGSWVLNNDYYRSLPGTFCG
	OC43-nsp4-P2	aa2996-3064/ aa3187-3236	YYRSLPGTFCGRDVFDFLIYQLFKGLAQPVDFLALTASSIAGAILAVIVVLVFYYLIKLKRAFGDYTSVV/ NRYSLSYNKYRYYSKGMDTAAYREAACSQLAKAMDTFTNNNGSDVLYQPP

TABLE 14-continued

OC43 antigens of nonstructural proteins nsp3 and nsp4			
Immunogen	Seq ID	aa positions ¹	Amino acid sequence ²
Full length	OC43-nsp4-FL	aa2875-2919/ aa2932-3064/ aa3187-3236	SADGVQCYTPHSQISYSNFYASGCVLSSACTMFTM ADGSPQPYCY/ SLVPHVRYNLANAKGFIRPPEVLREGLVRIVRTRSM SYCRVGLCEEA DEGIC FNFNGSWVLNNDYYRSLPGTFCGRDVFDLIYQLFK GLAQPVDLAL TASSIA GAILAVIVVLVFYYLIKLRAPGDYTSVV/ NRYLSLYNKYRYSGKMDTAAYREAACSQLAKA MDTFTNNN GSDVLYQPP

¹aa positions based on OC43 reference ORF1ab protein sequence, YP_009555238.1

²For conjugation to polyionic VLPs, the peptide/protein antigens have a N-terminal TAG and AAYY proteolytic processing sequence.

/ indicates break in native amino acid sequence

TABLE 15

OC43 antigens of nonstructural proteins nsp6 and nsp7			
Immunogen	Seq ID	aa positions ¹	Amino acid sequence ²
Short peptide	OC43-nsp6-S1	aa3602-3622	SLAMLLVKHKHLYLTMYITPV
	OC43-nsp6-S2	aa3613-3633	LYLTMYITPVLTLLYNNYLV
	OC43-nsp6-S3	aa3753-3780	IKIVLLCYLFIGYIISCYWGLFSLMNSL
	OC43-nsp6-S4	aa3770-3798	WGLFSLMNSLFRMPLGVYNYKISVQELRY
	OC43-nsp6-S5	aa3789-3816	YKISVQELRYMNANGLRPPKNSFEALML
	OC43-nsp6-S6	aa3807-3836	PKNSFEALMLNPKLLGIGGVPIIEVSQFQ
Extended peptide	OC43-nsp6-E1	aa3602-3633	SLAMLLVKHKHLYLTMYITPVLTLLYNNYLV
	OC43-nsp6-E2	aa3753-3799	IKIVLLCYLFIGYIISCYWGLFSLMNSLFRMPLGVYNYKISVQELRY
	OC43-nsp6-E3	aa3790-3836	YKISVQELRYMNANGLRPPKNSFEALMLNFKLLGIGGVPIIEVSQFQ
Short protein	OC43-nsp6-P1	aa3602-3633/ aa3753-3836	SLAMLLVKHKHLYLTMYITPVLTLLYNNYLV/ IKIVLLCYLFIGYIISCYWGLFSLMNSLFRMPLGVYNYKISVQELRYMNANGLRPPKNSFEALMLNPKLLGIGGVPIIEVSQFQ
Short peptide	OC43-nsp7-S1	aa3837-3862	SKLTDVKCANVVLNCLQHLHVASNS
	OC43-nsp7-S2	aa3853-3878	LQHLHVASNSKLWHYCSTLHNEILAT
	OC43-nsp7-S3	aa3869-3894	STLHNEILATSDLSVAFEKLAQLLIV
	OC43-nsp7-S4	aa3885-3910	FEKLAQLLIVLFANPAVDSKCLTSI
	OC43-nsp7-S5	aa3901-3925	AVDSKCLTSIEEVCCDDYAKDNTVLQ

TABLE 15-continued

OC43 antigens of nonstructural proteins nsp6 and nsp7			
Immunogen	Seq ID	aa positions ¹	Amino acid sequence ²
Extended peptide	OC43-nsp-7-E1	aa3837-3875	SKLTDVKCANVLLNCLQHLHVASNSKLW HYCSTLHNEI
	OC43-nsp-7-E2	aa3866-3903	HYCSTLHNEILATSDLSVAFEKLAQLLIVLF ANPAAVD
Short protein	OC43-nsp7-P1	aa3837-3925	SKLTDVKCANVLLNCLQHLHVASNSKLW HYCSTLHNEILATSDLSVAFEKLAQLLIVLF ANPAAVDSKCLTSIEEVCDDYAKDNTVLQ

¹aa positions based on OC43 reference ORF1ab protein sequence, YP_009555238.1²For conjugation to polyionic VLPS, the peptide/protein antigens have a N-terminal TAG and AAYY proteolytic processing sequence.

/ indicates break in native amino acid sequence

TABLE 16

OC43 antigens homologous to the nsp12-1 nonstructural protein fragment of SARS-CoV-2			
Immunogen	Seq ID	aa-position ¹	Amino acid sequence ²
Short peptide	OC43 nsp12-1-S1	aa4525-4555	YFTKKDWYDFVENPDI INVYKKGPIFNRL
	OC43 nsp12-1-S2	aa4546-4575	KLGPINFNRLVSATEFADKLVEVGLVGVLT
	OC43 nsp12-1-S3	aa4566-4594	VEVGLVGVLTLDNQDLNGKWYDFGDYVIA
	OC43 nsp12-1-S4	aa4585-4615	WYDFGDYVIAAPGCGVAIADSYYSYIMPMLT
	SARS-CoV-2 nsp12-1-S5	aa4634-4663	DLVQYDFTDYKLELFNKYFKHWSMPYHPNT
	OC43 nsp12-1-S6	aa4654-4683	HWSMPYHPNTVDCQDDRCIIHCANFNILFS
	OC43 nsp12-1-S7	aa4674-4703	HCANFNILFSMVLNPNTCFGPLVRQIFVDG
	OC43 nsp12-1-S8	aa4693-4721	PLVRQIFVDGVFPFVVSIGYHYKELGIVMN
	OC43 nsp12-1-S9	aa4712-4740	HYKELGIVMNMDVDTHRYRLSLKDLLLYA
Extended peptide	OC43 nsp12-1-E1	aa4525-4574	YFTKKDWYDFVENPDI INVYKKGPIFNRLVSATEFAD KLVEVGLVGVLT
	OC43 nsp12-1-E2	aa4565-4615	LVEVGLVGVLTLDNQDLNGKWYDFGDYVIAAPGCGVAIA DSYYSYIMPMLT
	OC43 nsp12-1-E3	aa4634-4675	DLVQYDFTDYKLELFNKYFKHWSMPYHPNTVDCQDDRCI IHC
	OC43 nsp12-1-E4	aa4666-4707	CQDDRCIIHCANFNILFSMVLNPNTCFGPLVRQIFVDGVP FVV
	OC43 nsp12-1-E5	aa4698-4740	IFVDGVFPFVVSIGYHYKELGIVMNMDVDTHRYRLSLKDL LLYA
Short protein	OC43 nsp12-1-SP1	aa4525-4615	YFTKKDWYDFVENPDI INVYKKGPIFNRLVSATEFAD KLVEVGLVGVLTLDNQDLNGKWYDFGDYVIAAPGCGVAI ADSYYSYIMPMLT
	OC43 nsp12-1-SP2	aa4634-4740	DLVQYDFTDYKLELFNKYFKHWSMPYHPNTVDCQDDRCI IHCANFNILFSMVLNPNTCFGPLVRQIFVDGVFPFVVSIGY HYKELGIVMNMDVDTHRYRLSLKDLLLYA

TABLE 16-continued

OC43 antigens homologous to the nsp12-1 nonstructural protein fragment of SARS-CoV-2			
Immunogen	Seq ID	aa-position ¹	Amino acid sequence ²
Full length	OC43 nsp12-1-FL	aa4525-4615/ aa4634-4740	YFTKKDWYDFVENPDIINVYKKLGPIFNRLVSATEFAD KLVEVGLVGLTLDNQDLNGKWYDFGDYVIAAPGCGVAI ADSYYSYIMPMLT/DLVQYDFTDYKLELFNKYFKHWSMP YHPNTVDCQDDRCIIHCANFNILFSMVLNPTCFGPLVRQ IFVDGVPFVVSIGYHYKELGIVMMDVDTHRYRLSLKDL LLYA

¹aa positions based on OC43 reference ORF1ab protein (YP_009555238.1).

²For conjugation to polyionic VLPs, the peptide/protein antigens have a N-terminal TAG and AAYY proteolytic processing sequence.

/ indicates break in native amino acid sequence

TABLE 17

OC43 antigens homologous to of the nsp12-2 nonstructural protein fragment of SARS-CoV-2			
Immunogen	Seq ID	aa-position	Amino acid sequence
Short peptide	OC43 nsp12-2-S1	aa4894-4921	AYTKRNVLP TLTQMNLKYAISAKNRART
	OC43 nsp12-2-S2	aa4912-4939	AISAKNRARTVAGVSILSTMTGRMFHQK
	OC43 nsp12-2-S3	aa4930-4957	TMTGRMFHQKCLKSIAATRGPVVGITTT
	OC43 nsp12-2-S4	aa4948-4975	RGVPVVGITTKFYGGWDDMLRRLIKDVD
	OC43 nsp12-2-S5	aa4966-4993	MLRRLIKVDNPNVLMGWDPKCDRAMPN
	OC43 nsp12-2-S6	aa4984-5012	YPKCDRAMPNLLRIVSSSLVLARKHETCCS
	OC43 nsp12-2-S7	aa5003-5030	LARKHETCCSQSDRFYRLANECQVLSE
	OC43 nsp12-2-S8	aa5021-5048	ANECQVLSEIVMCGGCYYVKPGGTSSG
	OC43 nsp12-2-S9	aa5039-5067	YVKPGGTSSGDATTAFANSVFNICQAVSA
	OC43 nsp12-2-S10	aa5058-5085	VFNICQAVSANVCALMSCNGNKIEDLSI
	OC43 nsp12-2-S11	aa5076-5103	NGNKIEDLSIRALQKRLYSHVYRSDKVD
	OC43 nsp12-2-S12	aa5094-5121	SHVYRSDKVDSTFVTEYYEFLNKHFSMM
	OC43 nsp12-2-S13	aa5112-5140	EFLNKHFSMMILSDDGVVCYNVDYASKGY

TABLE 17-continued

OC43 antigens homologous to of the nsp12-2 nonstructural protein fragment of SARS-CoV-2			
Immunogen	Seq ID	aa-position	Amino acid sequence
Extended peptide	OC43 nsp12-2-S14	aa5131-5159	YNSDYASKGYIANISAFQQVLYYQNNVF M
	OC43 nsp12-2-S15	aa5150-5177	VLYYQNNVFMSESKCWVEHDINNGPHEF
	OC43 nsp12-2-S16	aa5168-5196	HDINNGPHEFCSQHTMLVKMDGDDVYLP Y
	OC43 nsp12-2-S17	aa5187-5215	MDGDDVYLPYPNPSRILGAGCFVDDLK T
	OC43 nsp12-2-S18	aa5206-5234	GCFVDDLKTDVLLIERFVSLAIDAYP L
	OC43 nsp12-2-E1	aa4894-4940	AYTKRNVLP TLTQMNLKYAISAKNRART VAGVSILSTMTGRMFHQKC
	OC43 nsp12-2-E2	aa4931-4977	MTGRMFHQKCLKSIAATRGPVVGITTK FYGGWDDMLRRLIKDVDNP
	OC43 nsp12-2-E3	aa4968-5013	RRLIKDVDNPVLMGWDYPKCDRAMPN LLRIVSSLVLARKHETCCSQ
	OC43 nsp12-2-E4	aa5004-5049	ARKHETCCSQSDRFYRLANECAQVLSEI VMCGGCYYVKPGGTSSGD
	OC43 nsp12-2-E5	aa5040-5085	VKPGGTSSGDATTAFANSVFNICQAVSA NVCALMSCNGNKIEDLSI
Short protein	OC43 nsp12-2-E6	aa5076-5121	NGNKIEDLSIRALQKRLYSHVYRSDKVDS TFVTEYYEFLNKHFSMM
	OC43 nsp12-2-E7	aa5112-5158	EFLNKHFSMMILSDDGVVCYNSDYASKG YIANISAFQQVLYYQNNVF
	OC43 nsp12-2-E8	aa5149-5195	QVLYYQNNVFMSESKCWVEHDINNGP HEFCSQHTMLVKMDGDDVYLP
	OC43 nsp12-2-E9	aa5188-5234	KMDGDDVYLPYPNPSRILGAGCFVDDLK KTDSVLLIERFVSLAIDAYPL
	OC43 nsp12-2-SP1	aa4894-5018	AYTKRNVLP TLTQMNLKYAISAKNRART VAGVSILSTMTGRMFHQKCLKSIAATRG VPVVGITTKFYGGWDDMLRRLIKDVDN PVLWGWDYPKCDRAMPNLLRIVSSLVL ARKHETCCSQSDRFY
	OC43 nsp12-2-SP2	aa5008-5132	ETCCSQSDRFYRLANECAQVLSEIVMCG GCYYVKPGGTSSGDATTAFANSVFNICQ AVSANVCALMSCNGNKIEDLSIRALQKR LYSHVYRSDKVDSTFVTEYYEFLNKHFS MMILSDDGVVCYN

TABLE 17-continued

OC43 antigens homologous to of the nsp12-2 nonstructural protein fragment of SARS-CoV-2			
Immunogen	Seq ID	aa-position	Amino acid sequence
	OC43 nsp12-2- SP3	aa5122-5234	ILSDDGVVCYNVDYASKGYIANISAFQQV L YYQNNVFMSESKCWVEHDINNGPHEF CSQHTMLVKMDGDDVLPYPNPSRILG AGCFVDDLLKTDVLLIERFVSLAIDAYP L
Full length antigen	OC43 nsp12-2- FL	aa4894-5234	AYTKRNVLP TLTQMNLYAISAKNRART VAGVSLSTMTGRMFHQKCLKSIAATRG VPVVGITTKFYGGWDDMLRLIKDVND PVLMGWDYPKCDRAMPNLLRIVSSLV ARKHETCCSQSDRFYRLANCAQVLSEI VMCGGCYYVKPGGTSSGDATAFANSV FNICQAVSANVCALMSCNGKIEDLSIR ALQKRLYSHVYRSDKVDSTFVTEYYEFL NKHFSMMILSDDGVVCYNVDYASKGYIA NISAFQQVLYYYQNNVFMSESKCWVEHDI NNGPHEFCSQHTMLVKMDGDDVLPY PNPSRILGAGCFVDDLLKTDVLLIERFVS LAIDAYPL

¹aa positions based on OC43 reference ORF1ab protein (YP_009555236.1).

²For conjugation to polyionic VLPS, the peptide/protein antigens have a N-terminal TAG and AAYF proteolytic processing sequence.

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<400> SEQUENCE: 17

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Gly Thr Pro Leu Thr
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<400> SEQUENCE: 19

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1 5

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<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

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<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

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<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

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1 5

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<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

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<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 27

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<210> SEQ ID NO 28

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<212> TYPE: PRT

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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

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1 5

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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
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<400> SEQUENCE: 31

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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
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Arg Arg Arg Arg Arg Cys
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<210> SEQ ID NO 34

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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

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Cys Arg Arg Arg Arg Arg Arg Arg Arg Cys
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<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 38

Gly Thr Ile Thr Val Glu Glu Leu Lys Lys Leu Leu Glu Gln Trp Asn
1 5 10 15

Leu Val Ile Gly Phe Leu Phe Leu Thr Trp Ile Cys Leu Leu Gln
20 25 30

<210> SEQ ID NO 39
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<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 39

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1 5 10 15

Asn Arg Phe Leu Tyr Ile Ile Lys Leu Ile Phe Leu Tyr Trp Leu
20 25 30

<210> SEQ ID NO 40
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<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

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Ile Ile Lys Leu Ile Phe Leu Trp Leu Leu Trp Pro Val Thr Leu Ala
1 5 10 15

Cys Phe Val Leu Ala Ala Val Tyr Arg Ile Asn Trp Ile Thr
20 25 30

<210> SEQ ID NO 41
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1 5 10 15

Met Ala Cys Leu Val Gly Leu Met Trp Leu Ser Tyr Phe Ile
20 25 30

<210> SEQ ID NO 42
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1 5 10 15

Ala Arg Thr Arg Ser Met Trp Ser Phe Asn Pro Glu Thr Asn
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Ser Met Trp Ser Phe Asn Pro Glu Thr Asn Ile Leu Leu Asn Val Pro
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Leu His Gly Thr Ile Leu Thr Arg Pro Leu Leu Glu Ser Glu Leu
20 25 30

<210> SEQ ID NO 44
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<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 44

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<210> SEQ ID NO 45
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<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

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20 25 30

<210> SEQ ID NO 46
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<400> SEQUENCE: 46

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Arg Val Ala Gly Asp Ser Gly Phe Ala Ala Tyr Ser Arg Tyr Arg
20 25 30

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1 5 10 15

Asn Thr Asp His Ser Ser Ser Ser Asp Asn Ile Ala Leu Leu Val
20 25 30

<210> SEQ ID NO 48
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<400> SEQUENCE: 48

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1 5 10 15

Leu Val Ile Gly Phe Leu Phe Leu Thr Trp Ile Cys Leu Leu Gln Phe
20 25 30

Ala Tyr Ala Asn Arg Asn Arg Phe Leu Tyr Ile Ile Lys
35 40 45

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<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 49

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1 5 10 15

Leu Trp Pro Val Thr Leu Ala Cys Phe Val Leu Ala Ala Val Tyr Arg
20 25 30

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35 40 45

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<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

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1 5 10 15
Trp Leu Ser Tyr Phe Ile Ala Ser Phe Arg Leu Phe Ala Arg Thr Arg
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Ser Met Trp Ser Phe Asn Pro Glu Thr Asn Ile Leu Leu
35 40 45

<210> SEQ ID NO 51
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<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

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Thr Ile Leu Thr Arg Pro Leu Leu Glu Ser Glu Leu Val Ile Gly Ala
20 25 30
Val Ile Leu Arg Gly His Leu Arg Ile Ala Gly His His
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<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 52

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Lys Asp Leu Pro Lys Glu Ile Thr Val Ala Thr Ser Arg Thr Leu Ser
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Tyr Tyr Lys Leu Gly Ala Ser Gln Arg Val Ala Gly Asp
35 40 45

<210> SEQ ID NO 53
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<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

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Leu	Val	Ile	Gly	Phe	Leu	Phe	Leu	Thr	Trp	Ile	Cys	Leu	Leu	Gln	Phe
			20					25					30		
Ala	Tyr	Ala	Asn	Arg	Asn	Arg	Phe	Leu	Tyr	Ile	Ile	Lys	Leu	Ile	Phe
		35					40					45			
Leu	Trp	Leu	Leu	Trp	Pro	Val	Thr	Leu	Ala	Cys	Phe	Val	Leu	Ala	Ala
	50					55					60				
Val	Tyr	Arg	Ile	Asn	Trp	Ile	Thr	Gly	Gly	Ile	Ala	Ile	Ala	Met	Ala
65				70					75						80
Cys	Leu	Val	Gly	Leu	Met	Trp	Leu	Ser	Tyr	Phe	Ile	Ala	Ser	Phe	Arg
			85					90						95	
Leu	Phe	Ala	Arg	Thr	Arg	Ser	Met	Trp	Ser	Phe	Asn	Pro	Glu	Thr	
			100					105					110		

<210> SEQ ID NO 55
 <211> LENGTH: 116
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 55

Thr	Arg	Ser	Met	Trp	Ser	Phe	Asn	Pro	Glu	Thr	Asn	Ile	Leu	Leu	Asn
1				5					10					15	
Val	Pro	Leu	His	Gly	Thr	Ile	Leu	Thr	Arg	Pro	Leu	Leu	Glu	Ser	Glu
			20					25					30		
Leu	Val	Ile	Gly	Ala	Val	Ile	Leu	Arg	Gly	His	Leu	Arg	Ile	Ala	Gly
		35				40					45				
His	His	Leu	Gly	Arg	Cys	Asp	Ile	Lys	Asp	Leu	Pro	Lys	Glu	Ile	Thr
	50				55					60					
Val	Ala	Thr	Ser	Arg	Thr	Leu	Ser	Tyr	Tyr	Lys	Leu	Gly	Ala	Ser	Gln
65				70					75						80
Arg	Val	Ala	Gly	Asp	Ser	Gly	Phe	Ala	Ala	Tyr	Ser	Arg	Tyr	Arg	Ile
			85					90					95		
Gly	Asn	Tyr	Lys	Leu	Asn	Thr	Asp	His	Ser	Ser	Ser	Ser	Asp	Asn	Ile
			100					105					110		
Ala	Leu	Leu	Val												
			115												

<210> SEQ ID NO 56
 <211> LENGTH: 216
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 56

Gly	Thr	Ile	Thr	Val	Glu	Glu	Leu	Lys	Lys	Leu	Leu	Glu	Gln	Trp	Asn
1				5					10					15	
Leu	Val	Ile	Gly	Phe	Leu	Phe	Leu	Thr	Trp	Ile	Cys	Leu	Leu	Gln	Phe
			20					25					30		
Ala	Tyr	Ala	Asn	Arg	Asn	Arg	Phe	Leu	Tyr	Ile	Ile	Lys	Leu	Ile	Phe
		35					40					45			
Leu	Trp	Leu	Leu	Trp	Pro	Val	Thr	Leu	Ala	Cys	Phe	Val	Leu	Ala	Ala
	50					55					60				
Val	Tyr	Arg	Ile	Asn	Trp	Ile	Thr	Gly	Gly	Ile	Ala	Ile	Ala	Met	Ala
65				70					75						80

-continued

Cys Leu Val Gly Leu Met Trp Leu Ser Tyr Phe Ile Ala Ser Phe Arg
 85 90 95
 Leu Phe Ala Arg Thr Arg Ser Met Trp Ser Phe Asn Pro Glu Thr Asn
 100 105 110
 Ile Leu Leu Asn Val Pro Leu His Gly Thr Ile Leu Thr Arg Pro Leu
 115 120 125
 Leu Glu Ser Glu Leu Val Ile Gly Ala Val Ile Leu Arg Gly His Leu
 130 135 140
 Arg Ile Ala Gly His His Leu Gly Arg Cys Asp Ile Lys Asp Leu Pro
 145 150 155 160
 Lys Glu Ile Thr Val Ala Thr Ser Arg Thr Leu Ser Tyr Tyr Lys Leu
 165 170 175
 Gly Ala Ser Gln Arg Val Ala Gly Asp Ser Gly Phe Ala Ala Tyr Ser
 180 185 190
 Arg Tyr Arg Ile Gly Asn Tyr Lys Leu Asn Thr Asp His Ser Ser Ser
 195 200 205
 Ser Asp Asn Ile Ala Leu Leu Val
 210 215

<210> SEQ ID NO 57
 <211> LENGTH: 34
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 57

Ser Trp Phe Thr Ala Leu Thr Gln His Gly Lys Glu Asp Leu Lys Phe
 1 5 10 15
 Pro Arg Gly Gln Gly Val Pro Ile Asn Thr Asn Ser Ser Pro Asp Asp
 20 25 30

Gln Ile

<210> SEQ ID NO 58
 <211> LENGTH: 34
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 58

Asn Thr Asn Ser Ser Pro Asp Asp Gln Ile Gly Tyr Tyr Arg Arg Ala
 1 5 10 15
 Thr Arg Arg Ile Arg Gly Gly Asp Gly Lys Met Lys Asp Leu Ser Pro
 20 25 30

Arg Trp

<210> SEQ ID NO 59
 <211> LENGTH: 34
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 59

Gly Lys Met Lys Asp Leu Ser Pro Arg Trp Tyr Phe Tyr Tyr Leu Gly
 1 5 10 15
 Thr Gly Pro Glu Ala Gly Leu Pro Tyr Gly Ala Asn Lys Asp Gly Ile
 20 25 30

Ile Trp

-continued

<210> SEQ ID NO 60
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 60

Tyr Gly Ala Asn Lys Asp Gly Ile Ile Trp Val Ala Thr Glu Gly Ala
1 5 10 15

Leu Asn Thr Pro Lys Asp His Ile Gly Thr Arg Asn Pro Ala Asn Asn
20 25 30

Ala Ala

<210> SEQ ID NO 61
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 61

Gly Thr Arg Asn Pro Ala Asn Asn Ala Ala Ile Val Leu Gln Leu Pro
1 5 10 15

Gln Gly Thr Thr Leu Pro Lys Gly Phe Tyr Ala Glu Gly Ser Arg Gly
20 25 30

Gly Ser

<210> SEQ ID NO 62
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 62

Phe Tyr Ala Glu Gly Ser Arg Gly Gly Ser Gln Ala Ser Ser Arg Ser
1 5 10 15

Ser Ser Arg Ser Arg Asn Ser Ser Arg Asn Ser Thr Pro Gly Ser Ser
20 25 30

Arg Gly

<210> SEQ ID NO 63
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 63

Arg Asn Ser Thr Pro Gly Ser Ser Arg Gly Thr Ser Pro Ala Arg Met
1 5 10 15

Ala Gly Asn Gly Gly Asp Ala Ala Leu Ala Leu Leu Leu Asp Arg
20 25 30

Leu Asn

<210> SEQ ID NO 64
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 64

Leu Ala Leu Leu Leu Leu Asp Arg Leu Asn Gln Leu Glu Ser Lys Met
1 5 10 15

Ser Gly Lys Gly Gln Gln Gln Gln Gly Gln Thr Val Thr Lys Lys Ser
20 25 30

-continued

Ala Ala

<210> SEQ ID NO 65
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 65

Gly Gln Thr Val Thr Lys Lys Ser Ala Ala Glu Ala Ser Lys Lys Pro
1 5 10 15
Arg Gln Lys Arg Thr Ala Thr Lys Ala Tyr Asn Val Thr Gln Ala Phe
20 25 30

Gly Arg

<210> SEQ ID NO 66
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 66

Ala Tyr Asn Val Thr Gln Ala Phe Gly Arg Arg Gly Pro Glu Gln Thr
1 5 10 15
Gln Gly Asn Phe Gly Asp Gln Glu Leu Ile Arg Gln Gly Thr Asp Tyr
20 25 30

Lys His

<210> SEQ ID NO 67
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 67

Leu Ile Arg Gln Gly Thr Asp Tyr Lys His Trp Pro Gln Ile Ala Gln
1 5 10 15
Phe Ala Pro Ser Ala Ser Ala Phe Phe Gly Met Ser Arg Ile Gly Met
20 25 30

Glu Val

<210> SEQ ID NO 68
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 68

Phe Gly Met Ser Arg Ile Gly Met Glu Val Thr Pro Ser Gly Thr Trp
1 5 10 15
Leu Thr Tyr Thr Gly Ala Ile Lys Leu Asp Asp Lys Asp Pro Asn Phe
20 25 30

Lys

<210> SEQ ID NO 69
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 69

Lys Leu Asp Asp Lys Asp Pro Asn Phe Lys Asp Gln Val Ile Leu Leu

-continued

1	5	10	15
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Asn Lys His Ile Asp Ala Tyr Lys Thr Phe Pro Pro Thr Glu Pro Lys
 20 25 30

<210> SEQ ID NO 70
 <211> LENGTH: 54
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 70

Ser Trp Phe Thr Ala Leu Thr Gln His Gly Lys Glu Asp Leu Lys Phe
 1 5 10 15

Pro Arg Gly Gln Gly Val Pro Ile Asn Thr Asn Ser Ser Pro Asp Asp
 20 25 30

Gln Ile Gly Tyr Tyr Arg Arg Ala Thr Arg Arg Ile Arg Gly Gly Asp
 35 40 45

Gly Lys Met Lys Asp Leu
 50

<210> SEQ ID NO 71
 <211> LENGTH: 54
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 71

Arg Gly Gly Asp Gly Lys Met Lys Asp Leu Ser Pro Arg Trp Tyr Phe
 1 5 10 15

Tyr Tyr Leu Gly Thr Gly Pro Glu Ala Gly Leu Pro Tyr Gly Ala Asn
 20 25 30

Lys Asp Gly Ile Ile Trp Val Ala Thr Glu Gly Ala Leu Asn Thr Pro
 35 40 45

Lys Asp His Ile Gly Thr
 50

<210> SEQ ID NO 72
 <211> LENGTH: 54
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 72

Leu Asn Thr Pro Lys Asp His Ile Gly Thr Arg Asn Pro Ala Asn Asn
 1 5 10 15

Ala Ala Ile Val Leu Gln Leu Pro Gln Gly Thr Thr Leu Pro Lys Gly
 20 25 30

Phe Tyr Ala Glu Gly Ser Arg Gly Gly Ser Gln Ala Ser Ser Arg Ser
 35 40 45

Ser Ser Arg Ser Arg Asn
 50

<210> SEQ ID NO 73
 <211> LENGTH: 53
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 73

Ser Ser Arg Ser Ser Ser Arg Ser Arg Asn Ser Ser Arg Asn Ser Thr
 1 5 10 15

-continued

Pro Gly Ser Ser Arg Gly Thr Ser Pro Ala Arg Met Ala Gly Asn Gly
 20 25 30

Gly Asp Ala Ala Leu Ala Leu Leu Leu Asp Arg Leu Asn Gln Leu
 35 40 45

Glu Ser Lys Met Ser
 50

<210> SEQ ID NO 74
 <211> LENGTH: 54
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 74

Leu Asn Gln Leu Glu Ser Lys Met Ser Gly Lys Gly Gln Gln Gln Gln
 1 5 10 15

Gly Gln Thr Val Thr Lys Lys Ser Ala Ala Glu Ala Ser Lys Lys Pro
 20 25 30

Arg Gln Lys Arg Thr Ala Thr Lys Ala Tyr Asn Val Thr Gln Ala Phe
 35 40 45

Gly Arg Arg Gly Pro Glu
 50

<210> SEQ ID NO 75
 <211> LENGTH: 55
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 75

Thr Gln Ala Phe Gly Arg Arg Gly Pro Glu Gln Thr Gln Gly Asn Phe
 1 5 10 15

Gly Asp Gln Glu Leu Ile Arg Gln Gly Thr Asp Tyr Lys His Trp Pro
 20 25 30

Gln Ile Ala Gln Phe Ala Pro Ser Ala Ser Ala Phe Phe Gly Met Ser
 35 40 45

Arg Ile Gly Met Glu Val Thr
 50 55

<210> SEQ ID NO 76
 <211> LENGTH: 54
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 76

Gly Met Ser Arg Ile Gly Met Glu Val Thr Pro Ser Gly Thr Trp Leu
 1 5 10 15

Thr Tyr Thr Gly Ala Ile Lys Leu Asp Asp Lys Asp Pro Asn Phe Lys
 20 25 30

Asp Gln Val Ile Leu Leu Asn Lys His Ile Asp Ala Tyr Lys Thr Phe
 35 40 45

Pro Pro Thr Glu Pro Lys
 50

<210> SEQ ID NO 77
 <211> LENGTH: 113
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 77

-continued

Ser Trp Phe Thr Ala Leu Thr Gln His Gly Lys Glu Asp Leu Lys Phe
 1 5 10 15
 Pro Arg Gly Gln Gly Val Pro Ile Asn Thr Asn Ser Ser Pro Asp Asp
 20 25 30
 Gln Ile Gly Tyr Tyr Arg Arg Ala Thr Arg Arg Ile Arg Gly Gly Asp
 35 40 45
 Gly Lys Met Lys Asp Leu Ser Pro Arg Trp Tyr Phe Tyr Tyr Leu Gly
 50 55 60
 Thr Gly Pro Glu Ala Gly Leu Pro Tyr Gly Ala Asn Lys Asp Gly Ile
 65 70 75 80
 Ile Trp Val Ala Thr Glu Gly Ala Leu Asn Thr Pro Lys Asp His Ile
 85 90 95
 Gly Thr Arg Asn Pro Ala Asn Asn Ala Ala Ile Val Leu Gln Leu Pro
 100 105 110

Gly

<210> SEQ ID NO 78
 <211> LENGTH: 113
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 78

Asn Asn Ala Ala Ile Val Leu Gln Leu Pro Gln Gly Thr Thr Leu Pro
 1 5 10 15
 Lys Gly Phe Tyr Ala Glu Gly Ser Arg Gly Gly Ser Gln Ala Ser Ser
 20 25 30
 Arg Ser Ser Ser Arg Ser Arg Asn Ser Ser Arg Asn Ser Thr Pro Gly
 35 40 45
 Ser Ser Arg Gly Thr Ser Pro Ala Arg Met Ala Gly Asn Gly Gly Asp
 50 55 60
 Ala Ala Leu Ala Leu Leu Leu Asp Arg Leu Asn Gln Leu Glu Ser
 65 70 75 80
 Lys Met Ser Gly Lys Gly Gln Gln Gln Gln Gly Gln Thr Val Thr Lys
 85 90 95
 Lys Ser Ala Ala Glu Ala Ser Lys Lys Pro Arg Gln Lys Arg Thr Ala
 100 105 110

Thr

<210> SEQ ID NO 79
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 79

Ser Lys Lys Pro Arg Gln Lys Arg Thr Ala Thr Lys Ala Tyr Asn Val
 1 5 10 15
 Thr Gln Ala Phe Gly Arg Arg Gly Pro Glu Gln Thr Gln Gly Asn Phe
 20 25 30
 Gly Asp Gln Glu Leu Ile Arg Gln Gly Thr Asp Tyr Lys His Trp Pro
 35 40 45
 Gln Ile Ala Gln Phe Ala Pro Ser Ala Ser Ala Phe Phe Gly Met Ser
 50 55 60
 Arg Ile Gly Met Glu Val Thr Pro Ser Gly Thr Trp Leu Thr Tyr Thr

-continued

65	70	75	80
Gly Ala Ile Lys Leu Asp Asp Lys Asp Pro Asn Phe Lys Asp Gln Val	85	90	95
Ile Leu Leu Asn Lys His Ile Asp Ala Tyr Lys Thr Phe Pro Pro Thr	100	105	110
Glu Pro Lys	115		

<210> SEQ ID NO 80
 <211> LENGTH: 319
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 80

Ser Trp Phe Thr Ala Leu Thr Gln His Gly Lys Glu Asp Leu Lys Phe	1	5	10	15
Pro Arg Gly Gln Gly Val Pro Ile Asn Thr Asn Ser Ser Pro Asp Asp	20	25	30	
Gln Ile Gly Tyr Tyr Arg Arg Ala Thr Arg Arg Ile Arg Gly Gly Asp	35	40	45	
Gly Lys Met Lys Asp Leu Ser Pro Arg Trp Tyr Phe Tyr Tyr Leu Gly	50	55	60	
Thr Gly Pro Glu Ala Gly Leu Pro Tyr Gly Ala Asn Lys Asp Gly Ile	65	70	75	80
Ile Trp Val Ala Thr Glu Gly Ala Leu Asn Thr Pro Lys Asp His Ile	85	90	95	
Gly Thr Arg Asn Pro Ala Asn Asn Ala Ala Ile Val Leu Gln Leu Pro	100	105	110	
Gln Gly Thr Thr Leu Pro Lys Gly Phe Tyr Ala Glu Gly Ser Arg Gly	115	120	125	
Gly Ser Gln Ala Ser Ser Arg Ser Ser Arg Ser Arg Asn Ser Ser	130	135	140	
Arg Asn Ser Thr Pro Gly Ser Ser Arg Gly Thr Ser Pro Ala Arg Met	145	150	155	160
Ala Gly Asn Gly Gly Asp Ala Ala Leu Ala Leu Leu Leu Leu Asp Arg	165	170	175	
Leu Asn Gln Leu Glu Ser Lys Met Ser Gly Lys Gly Gln Gln Gln Gln	180	185	190	
Gly Gln Thr Val Thr Lys Lys Ser Ala Ala Glu Ala Ser Lys Lys Pro	195	200	205	
Arg Gln Lys Arg Thr Ala Thr Lys Ala Tyr Asn Val Thr Gln Ala Phe	210	215	220	
Gly Arg Arg Gly Pro Glu Gln Thr Gln Gly Asn Phe Gly Asp Gln Glu	225	230	235	240
Leu Ile Arg Gln Gly Thr Asp Tyr Lys His Trp Pro Gln Ile Ala Gln	245	250	255	
Phe Ala Pro Ser Ala Ser Ala Phe Phe Gly Met Ser Arg Ile Gly Met	260	265	270	
Glu Val Thr Pro Ser Gly Thr Trp Leu Thr Tyr Thr Gly Ala Ile Lys	275	280	285	
Leu Asp Asp Lys Asp Pro Asn Phe Lys Asp Gln Val Ile Leu Leu Asn	290	295	300	

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Lys His Ile Asp Ala Tyr Lys Thr Phe Pro Pro Thr Glu Pro Lys
305 310 315

<210> SEQ ID NO 81
 <211> LENGTH: 55
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 81

Cys Ala Ser Tyr Gln Thr Gln Thr Asn Ser Pro Arg Arg Ala Arg Ser
1 5 10 15
 Val Ala Ser Gln Ser Ile Ile Ala Tyr Thr Met Ser Leu Gly Ala Glu
20 25 30
 Asn Ser Val Ala Tyr Ser Asn Asn Ser Ile Ala Ile Pro Thr Asn Phe
35 40 45
 Thr Ile Ser Val Thr Thr Glu
50 55

<210> SEQ ID NO 82
 <211> LENGTH: 54
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 82

Thr Asn Phe Thr Ile Ser Val Thr Thr Glu Ile Leu Pro Val Ser Met
1 5 10 15
 Thr Lys Thr Ser Val Asp Cys Thr Met Tyr Ile Cys Gly Asp Ser Thr
20 25 30
 Glu Cys Ser Asn Leu Leu Leu Gln Tyr Gly Ser Phe Cys Thr Gln Leu
35 40 45
 Asn Arg Ala Leu Thr Gly
50

<210> SEQ ID NO 83
 <211> LENGTH: 55
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 83

Cys Thr Gln Leu Asn Arg Ala Leu Thr Gly Ile Ala Val Glu Gln Asp
1 5 10 15
 Lys Asn Thr Gln Glu Val Phe Ala Gln Val Lys Gln Ile Tyr Lys Thr
20 25 30
 Pro Pro Ile Lys Asp Phe Gly Gly Phe Asn Phe Ser Gln Ile Leu Pro
35 40 45
 Asp Pro Ser Lys Pro Ser Lys
50 55

<210> SEQ ID NO 84
 <211> LENGTH: 56
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 84

Ile Leu Pro Asp Pro Ser Lys Pro Ser Lys Arg Ser Phe Ile Glu Asp
1 5 10 15
 Leu Leu Phe Asn Lys Val Thr Leu Ala Asp Ala Gly Phe Ile Lys Gln
20 25 30

-continued

Tyr Gly Asp Cys Leu Gly Asp Ile Ala Ala Arg Asp Leu Ile Cys Ala
35 40 45

Gln Lys Phe Asn Gly Leu Thr Val
50 55

<210> SEQ ID NO 85
<211> LENGTH: 55
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 85

Cys Ala Gln Lys Phe Asn Gly Leu Thr Val Leu Pro Pro Leu Leu Thr
1 5 10 15

Asp Glu Met Ile Ala Gln Tyr Thr Ser Ala Leu Leu Ala Gly Thr Ile
20 25 30

Thr Ser Gly Trp Thr Phe Gly Ala Gly Ala Ala Leu Gln Ile Pro Phe
35 40 45

Ala Met Gln Met Ala Tyr Arg
50 55

<210> SEQ ID NO 86
<211> LENGTH: 55
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 86

Pro Phe Ala Met Gln Met Ala Tyr Arg Phe Asn Gly Ile Gly Val Thr
1 5 10 15

Gln Asn Val Leu Tyr Glu Asn Gln Lys Leu Ile Ala Asn Gln Phe Asn
20 25 30

Ser Ala Ile Gly Lys Ile Gln Asp Ser Leu Ser Ser Thr Ala Ser Ala
35 40 45

Leu Gly Lys Leu Gln Asp Val
50 55

<210> SEQ ID NO 87
<211> LENGTH: 55
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 87

Ala Ser Ala Leu Gly Lys Leu Gln Asp Val Val Asn Gln Asn Ala Gln
1 5 10 15

Ala Leu Asn Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile
20 25 30

Ser Ser Val Leu Asn Asp Ile Leu Ser Arg Leu Asp Lys Val Glu Ala
35 40 45

Glu Val Gln Ile Asp Arg Leu
50 55

<210> SEQ ID NO 88
<211> LENGTH: 55
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 88

Val Glu Ala Glu Val Gln Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln

-continued

1	5	10	15
Ser Leu Gln Thr Tyr Val Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile			
	20	25	30
Arg Ala Ser Ala Asn Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu			
	35	40	45
Gly Gln Ser Lys Arg Val Asp			
	50	55	

<210> SEQ ID NO 89
 <211> LENGTH: 56
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 89

Cys Val Leu Gly Gln Ser Lys Arg Val Asp Phe Cys Gly Lys Gly Tyr			
1	5	10	15
His Leu Met Ser Phe Pro Gln Ser Ala Pro His Gly Val Val Phe Leu			
	20	25	30
His Val Thr Tyr Val Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro			
	35	40	45
Ala Ile Cys His Asp Gly Lys Ala			
	50	55	

<210> SEQ ID NO 90
 <211> LENGTH: 55
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 90

Ala Pro Ala Ile Cys His Asp Gly Lys Ala His Phe Pro Arg Glu Gly			
1	5	10	15
Val Phe Val Ser Asn Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe			
	20	25	30
Tyr Glu Pro Gln Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn			
	35	40	45
Cys Asp Val Val Ile Gly Ile			
	50	55	

<210> SEQ ID NO 91
 <211> LENGTH: 55
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 91

Ser Gly Asn Cys Asp Val Val Ile Gly Ile Val Asn Asn Thr Val Tyr			
1	5	10	15
Asp Pro Leu Gln Pro Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys			
	20	25	30
Tyr Phe Lys Asn His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser			
	35	40	45
Gly Ile Asn Ala Ser Val Val			
	50	55	

<210> SEQ ID NO 92
 <211> LENGTH: 55
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

-continued

<400> SEQUENCE: 92

Asp Ile Ser Gly Ile Asn Ala Ser Val Val Asn Ile Gln Lys Glu Ile
1 5 10 15
Asp Arg Leu Asn Glu Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp
20 25 30
Leu Gln Glu Leu Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr
35 40 45
Ile Trp Leu Gly Phe Ile Ala
50 55

<210> SEQ ID NO 93

<211> LENGTH: 56

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 93

Pro Trp Tyr Ile Trp Leu Gly Phe Ile Ala Gly Leu Ile Ala Ile Val
1 5 10 15
Met Val Thr Ile Met Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu
20 25 30
Lys Gly Cys Cys Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp
35 40 45
Ser Glu Pro Val Leu Lys Gly Val
50 55

<210> SEQ ID NO 94

<211> LENGTH: 128

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 94

Cys Ala Ser Tyr Gln Thr Gln Thr Asn Ser Pro Arg Arg Ala Arg Ser
1 5 10 15
Val Ala Ser Gln Ser Ile Ile Ala Tyr Thr Met Ser Leu Gly Ala Glu
20 25 30
Asn Ser Val Ala Tyr Ser Asn Asn Ser Ile Ala Ile Pro Thr Asn Phe
35 40 45
Thr Ile Ser Val Thr Thr Glu Ile Leu Pro Val Ser Met Thr Lys Thr
50 55 60
Ser Val Asp Cys Thr Met Tyr Ile Cys Gly Asp Ser Thr Glu Cys Ser
65 70 75 80
Asn Leu Leu Leu Gln Tyr Gly Ser Phe Cys Thr Gln Leu Asn Arg Ala
85 90 95
Leu Thr Gly Ile Ala Val Glu Gln Asp Lys Asn Thr Gln Glu Val Phe
100 105 110
Ala Gln Val Lys Gln Ile Tyr Lys Thr Pro Pro Ile Lys Asp Phe Gly
115 120 125

<210> SEQ ID NO 95

<211> LENGTH: 129

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 95

Ile Tyr Lys Thr Pro Pro Ile Lys Asp Phe Gly Gly Phe Asn Phe Ser

-continued

1	5	10	15
Gln Ile Leu Pro Asp Pro Ser Lys Pro Ser Lys Arg Ser Phe Ile Glu	20	25	30
Asp Leu Leu Phe Asn Lys Val Thr Leu Ala Asp Ala Gly Phe Ile Lys	35	40	45
Gln Tyr Gly Asp Cys Leu Gly Asp Ile Ala Ala Arg Asp Leu Ile Cys	50	55	60
Ala Gln Lys Phe Asn Gly Leu Thr Val Leu Pro Pro Leu Leu Thr Asp	65	70	75
Glu Met Ile Ala Gln Tyr Thr Ser Ala Leu Leu Ala Gly Thr Ile Thr	85	90	95
Ser Gly Trp Thr Phe Gly Ala Gly Ala Ala Leu Gln Ile Pro Phe Ala	100	105	110
Met Gln Met Ala Tyr Arg Phe Asn Gly Ile Gly Val Thr Gln Asn Val	115	120	125

Leu

<210> SEQ ID NO 96
 <211> LENGTH: 128
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 96

Phe Asn Gly Ile Gly Val Thr Gln Asn Val Leu Tyr Glu Asn Gln Lys	1	5	10	15
Leu Ile Ala Asn Gln Phe Asn Ser Ala Ile Gly Lys Ile Gln Asp Ser	20	25	30	
Leu Ser Ser Thr Ala Ser Ala Leu Gly Lys Leu Gln Asp Val Val Asn	35	40	45	
Gln Asn Ala Gln Ala Leu Asn Thr Leu Val Lys Gln Leu Ser Ser Asn	50	55	60	
Phe Gly Ala Ile Ser Ser Val Leu Asn Asp Ile Leu Ser Arg Leu Asp	65	70	75	80
Lys Val Glu Ala Glu Val Gln Ile Asp Arg Leu Ile Thr Gly Arg Leu	85	90	95	
Gln Ser Leu Gln Thr Tyr Val Thr Gln Gln Leu Ile Arg Ala Ala Glu	100	105	110	
Ile Arg Ala Ser Ala Asn Leu Ala Ala Thr Lys Met Ser Glu Cys Val	115	120	125	

<210> SEQ ID NO 97
 <211> LENGTH: 130
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 97

Asn Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys	1	5	10	15
Arg Val Asp Phe Cys Gly Lys Gly Tyr His Leu Met Ser Phe Pro Gln	20	25	30	
Ala Pro His Gly Val Val Phe Leu His Val Thr Tyr Val Pro Ala Gln	35	40	45	
Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His Asp Gly Lys Ala	50	55	60	

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His Phe Pro Arg Glu Gly Val Phe Val Ser Asn Gly Thr His Trp Phe
65 70 75 80

Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln Ile Ile Thr Thr Asp Asn
85 90 95

Thr Phe Val Ser Gly Asn Cys Asp Val Val Ile Gly Ile Val Asn Asn
100 105 110

Thr Val Tyr Asp Pro Leu Gln Pro Glu Leu Ile Asp Ser Phe Lys Glu
115 120 125

Glu Leu
130

<210> SEQ ID NO 98
<211> LENGTH: 127
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 98

Gln Pro Glu Leu Ile Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe
1 5 10 15

Lys Asn His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile
20 25 30

Asn Ala Ser Val Val Asn Ile Gln Lys Glu Asp Arg Leu Asn Glu Val
35 40 45

Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu Gly Lys
50 55 60

Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu Gly Phe Ile
65 70 75 80

Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met Leu Cys Cys Met
85 90 95

Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys Ser Cys Gly Ser Cys
100 105 110

Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro Val Leu Lys Gly Val
115 120 125

<210> SEQ ID NO 99
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 99

Phe Leu Tyr Leu Tyr Ala Leu Val Tyr Phe Leu Gln Ser Ile Asn Phe
1 5 10 15

Val Arg Ile Ile Met Arg Leu Trp Leu Cys Trp Lys
20 25

<210> SEQ ID NO 100
<211> LENGTH: 25
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 100

Ile Ile Met Arg Leu Trp Leu Cys Trp Lys Cys Arg Ser Lys Asn Pro
1 5 10 15

Leu Leu Tyr Asp Ala Asn Tyr Phe Leu
20 25

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<210> SEQ ID NO 101
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 101

Pro Leu Leu Tyr Asp Ala Asn Tyr Phe Leu Cys Trp His Thr Asn Cys
1 5 10 15

Tyr Asp Tyr Cys Ile Pro Tyr Asn Ser Val Thr Ser
20 25

<210> SEQ ID NO 102
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 102

Tyr Cys Ile Pro Tyr Asn Ser Val Thr Ser Ser Ile Val Ile Thr Ser
1 5 10 15

Gly Asp Gly Thr Thr Ser Pro Ile Ser Glu His Asp
20 25

<210> SEQ ID NO 103
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 103

Gly Thr Thr Ser Pro Ile Ser Glu His Asp Tyr Gln Ile Gly Gly Tyr
1 5 10 15

Thr Glu Lys Trp Glu Ser Gly Val Lys Asp Cys Val
20 25

<210> SEQ ID NO 104
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 104

Lys Trp Glu Ser Gly Val Lys Asp Cys Val Val Leu His Ser Tyr Phe
1 5 10 15

Thr Ser Asp Tyr Tyr Gln Leu Tyr Ser Thr Gln Leu
20 25

<210> SEQ ID NO 105
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 105

Asp Tyr Tyr Gln Leu Tyr Ser Thr Gln Leu Ser Thr Asp Thr Gly Val
1 5 10 15

Glu His Val Thr Phe Phe Ile Tyr Asn Lys Ile Val Asp
20 25

<210> SEQ ID NO 106
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

-continued

<400> SEQUENCE: 106

Thr Phe Phe Ile Tyr Asn Lys Ile Val Asp Glu Pro Glu Glu His Val
1 5 10 15
Gln Ile His Thr Ile Asp Gly Ser Ser Gly Val Val Asn
20 25

<210> SEQ ID NO 107

<211> LENGTH: 28

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 107

Thr Ile Asp Gly Ser Ser Gly Val Val Asn Pro Val Met Glu Pro Ile
1 5 10 15
Tyr Asp Glu Pro Thr Thr Thr Thr Ser Val Pro Leu
20 25

<210> SEQ ID NO 108

<211> LENGTH: 50

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 108

Phe Leu Tyr Leu Tyr Ala Leu Val Tyr Phe Leu Gln Ser Ile Asn Phe
1 5 10 15
Val Arg Ile Ile Met Arg Leu Trp Leu Cys Trp Lys Cys Arg Ser Lys
20 25 30
Asn Pro Leu Leu Tyr Asp Ala Asn Tyr Phe Leu Cys Trp His Thr Asn
35 40 45
Cys Tyr
50

<210> SEQ ID NO 109

<211> LENGTH: 50

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 109

Tyr Phe Leu Cys Trp His Thr Asn Cys Tyr Asp Tyr Cys Ile Pro Tyr
1 5 10 15
Asn Ser Val Thr Ser Ser Ile Val Ile Thr Ser Gly Asp Gly Thr Thr
20 25 30
Ser Pro Ile Ser Glu His Asp Tyr Gln Ile Gly Gly Tyr Thr Glu Lys
35 40 45
Trp Glu
50

<210> SEQ ID NO 110

<211> LENGTH: 50

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 110

Gln Ile Gly Gly Tyr Thr Glu Lys Trp Glu Ser Gly Val Lys Asp Cys
1 5 10 15
Val Val Leu His Ser Tyr Phe Thr Ser Asp Tyr Tyr Gln Leu Tyr Ser
20 25 30

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Thr Gln Leu Ser Thr Asp Thr Gly Val Glu His Val Thr Phe Phe Ile
35 40 45

Tyr Asn
50

<210> SEQ ID NO 111
 <211> LENGTH: 51
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 111

Val Glu His Val Thr Phe Phe Ile Tyr Asn Lys Ile Val Asp Glu Pro
1 5 10 15

Glu Glu His Val Gln Ile His Thr Ile Asp Gly Ser Ser Gly Val Val
20 25 30

Asn Pro Val Met Glu Pro Ile Tyr Asp Glu Pro Thr Thr Thr Ser
35 40 45

Val Pro Leu
50

<210> SEQ ID NO 112
 <211> LENGTH: 91
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 112

Phe Leu Tyr Leu Tyr Ala Leu Val Tyr Phe Leu Gln Ser Ile Asn Phe
1 5 10 15

Val Arg Ile Ile Met Arg Leu Trp Leu Cys Trp Lys Cys Arg Ser Lys
20 25 30

Asn Pro Leu Leu Tyr Asp Ala Asn Tyr Phe Leu Cys Trp His Thr Asn
35 40 45

Cys Tyr Asp Tyr Cys Ile Pro Tyr Asn Ser Val Thr Ser Ser Ile Val
50 55 60

Ile Thr Ser Gly Asp Gly Thr Thr Ser Pro Ile Ser Glu His Asp Tyr
65 70 75 80

Gln Ile Gly Gly Tyr Thr Glu Lys Trp Glu Ser
85 90

<210> SEQ ID NO 113
 <211> LENGTH: 91
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 113

Gln Ile Gly Gly Tyr Thr Glu Lys Trp Glu Ser Gly Val Lys Asp Cys
1 5 10 15

Val Val Leu His Ser Tyr Phe Thr Ser Asp Tyr Tyr Gln Leu Tyr Ser
20 25 30

Thr Gln Leu Ser Thr Asp Thr Gly Val Glu His Val Thr Phe Phe Ile
35 40 45

Tyr Asn Lys Ile Val Asp Glu Pro Glu Glu His Val Gln Ile His Thr
50 55 60

Ile Asp Gly Ser Ser Gly Val Val Asn Pro Val Met Glu Pro Ile Tyr
65 70 75 80

Asp Glu Pro Thr Thr Thr Thr Ser Val Pro Leu

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85	90
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<210> SEQ ID NO 114
 <211> LENGTH: 171
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 114

Phe	Leu	Tyr	Leu	Tyr	Ala	Leu	Val	Tyr	Phe	Leu	Gln	Ser	Ile	Asn	Phe
1				5					10					15	
Val	Arg	Ile	Ile	Met	Arg	Leu	Trp	Leu	Cys	Trp	Lys	Cys	Arg	Ser	Lys
		20					25						30		
Asn	Pro	Leu	Leu	Tyr	Asp	Ala	Asn	Tyr	Phe	Leu	Cys	Trp	His	Thr	Asn
		35					40					45			
Cys	Tyr	Asp	Tyr	Cys	Ile	Pro	Tyr	Asn	Ser	Val	Thr	Ser	Ser	Ile	Val
	50					55					60				
Ile	Thr	Ser	Gly	Asp	Gly	Thr	Thr	Ser	Pro	Ile	Ser	Glu	His	Asp	Tyr
65				70					75					80	
Gln	Ile	Gly	Gly	Tyr	Thr	Glu	Lys	Trp	Glu	Ser	Gly	Val	Lys	Asp	Cys
		85						90					95		
Val	Val	Leu	His	Ser	Tyr	Phe	Thr	Ser	Asp	Tyr	Tyr	Gln	Leu	Tyr	Ser
		100						105					110		
Thr	Gln	Leu	Ser	Thr	Asp	Thr	Gly	Val	Glu	His	Val	Thr	Phe	Phe	Ile
	115						120					125			
Tyr	Asn	Lys	Ile	Val	Asp	Glu	Pro	Glu	Glu	His	Val	Gln	Ile	His	Thr
	130					135					140				
Ile	Asp	Gly	Ser	Ser	Gly	Val	Val	Asn	Pro	Val	Met	Glu	Pro	Ile	Tyr
145					150				155					160	
Asp	Glu	Pro	Thr	Thr	Thr	Thr	Ser	Val	Pro	Leu					
			165						170						

<210> SEQ ID NO 115
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 115

Met	Lys	Ile	Ile	Leu	Phe	Leu	Ala	Leu	Ile	Thr	Leu	Ala	Thr	Cys	Glu
1				5					10					15	
Leu	Tyr	His	Tyr	Gln	Glu	Cys	Val	Arg	Gly	Thr	Thr				
		20						25							

<210> SEQ ID NO 116
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 116

Tyr	Gln	Glu	Cys	Val	Arg	Gly	Thr	Thr	Val	Leu	Leu	Lys	Glu	Pro	Cys
1				5					10					15	
Ser	Ser	Gly	Thr	Tyr	Glu	Gly	Asn	Ser	Pro	Phe	His				
		20					25								

<210> SEQ ID NO 117
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

-continued

<400> SEQUENCE: 117

Gly Thr Tyr Glu Gly Asn Ser Pro Phe His Pro Leu Ala Asp Asn Lys
1 5 10 15

Phe Ala Leu Thr Cys Phe Ser Thr Gln Phe Ala Phe
20 25

<210> SEQ ID NO 118

<211> LENGTH: 29

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 118

Leu Thr Cys Phe Ser Thr Gln Phe Ala Phe Ala Cys Pro Asp Gly Val
1 5 10 15

Lys His Val Tyr Gln Leu Arg Ala Arg Ser Val Ser Pro
20 25

<210> SEQ ID NO 119

<211> LENGTH: 29

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 119

Tyr Gln Leu Arg Ala Arg Ser Val Ser Pro Lys Leu Phe Ile Arg Gln
1 5 10 15

Glu Glu Val Gln Glu Leu Tyr Ser Pro Ile Phe Leu Ile
20 25

<210> SEQ ID NO 120

<211> LENGTH: 28

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 120

Gln Glu Leu Tyr Ser Pro Ile Phe Leu Ile Val Ala Ala Ile Val Phe
1 5 10 15

Ile Thr Leu Cys Phe Thr Leu Lys Arg Lys Thr Glu
20 25

<210> SEQ ID NO 121

<211> LENGTH: 48

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 121

Met Lys Ile Ile Leu Phe Leu Ala Leu Ile Thr Leu Glu Ala Thr Cys
1 5 10 15

Glu Leu Tyr His Tyr Gln Glu Cys Val Arg Gly Thr Thr Val Leu Leu
20 25 30

Lys Glu Pro Cys Ser Ser Gly Thr Tyr Glu Gly Asn Ser Pro Phe His
35 40 45

<210> SEQ ID NO 122

<211> LENGTH: 47

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 122

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Gly Thr Tyr Glu Gly Asn Ser Pro Phe His Pro Leu Ala Asp Asn Lys
 1 5 10 15
 Phe Ala Leu Thr Cys Phe Ser Thr Gln Phe Ala Phe Ala Cys Pro Asp
 20 25 30
 Gly Val Lys His Val Tyr Gln Leu Arg Ala Arg Ser Val Ser Pro
 35 40 45

<210> SEQ ID NO 123
 <211> LENGTH: 47
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 123

Tyr Gln Leu Arg Ala Arg Ser Val Ser Pro Lys Leu Phe Ile Arg Gln
 1 5 10 15
 Glu Glu Val Gln Glu Leu Tyr Ser Pro Ile Phe Leu Ile Val Ala Ala
 20 25 30
 Ile Val Phe Ile Thr Leu Cys Phe Thr Leu Lys Arg Lys Thr Glu
 35 40 45

<210> SEQ ID NO 124
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 124

Met Lys Ile Ile Leu Phe Leu Ala Leu Ile Thr Leu Ala Thr Cys Glu
 1 5 10 15
 Leu Tyr His Tyr Gln Glu Cys Val Arg Gly Thr Thr Val Leu Leu Lys
 20 25 30
 Glu Pro Cys Ser Ser Gly Thr Tyr Glu Gly Asn Ser Pro Phe His Pro
 35 40 45
 Leu Ala Asp Asn Lys Phe Ala Leu Thr Cys Phe Ser Thr Gln Phe Ala
 50 55 60
 Phe Ala Cys Pro Asp Gly Val Lys His Val Tyr Gln Leu Arg Ala Arg
 65 70 75 80
 Ser Val Ser Pro Lys Leu Phe Ile Arg Gln Glu Glu Val Gln Glu Leu
 85 90 95
 Tyr Ser Pro Ile Phe Leu Ile Val Ala Ala Ile Val Phe Ile Thr Leu
 100 105 110
 Cys Phe Thr Leu Lys Arg Lys Thr Glu
 115 120

<210> SEQ ID NO 125
 <211> LENGTH: 30
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 125

Ser Ala Val Lys Arg Thr Ile Lys Gly Thr His His Trp Leu Leu Leu
 1 5 10 15
 Thr Ile Leu Thr Ser Leu Leu Val Leu Val Gln Ser Thr Gln
 20 25 30

<210> SEQ ID NO 126
 <211> LENGTH: 30
 <212> TYPE: PRT

-continued

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 126

Ser Leu Leu Val Leu Val Gln Ser Thr Gln Trp Ser Leu Phe Phe Phe
1 5 10 15

Leu Tyr Glu Asn Ala Phe Leu Pro Phe Ala Met Gly Ile Ile
20 25 30

<210> SEQ ID NO 127

<211> LENGTH: 30

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 127

Ala Phe Leu Pro Phe Ala Met Gly Ile Ile Ala Met Ser Ala Phe Ala
1 5 10 15

Met Met Phe Val Lys His Lys His Ala Phe Leu Cys Leu Phe
20 25 30

<210> SEQ ID NO 128

<211> LENGTH: 30

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 128

Lys His Lys His Ala Phe Leu Cys Leu Phe Leu Leu Pro Ser Leu Ala
1 5 10 15

Thr Val Ala Tyr Phe Asn Met Val Tyr Met Pro Ala Ser Trp
20 25 30

<210> SEQ ID NO 129

<211> LENGTH: 30

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 129

Phe Asn Met Val Tyr Met Pro Ala Ser Trp Val Met Arg Ile Met Thr
1 5 10 15

Trp Leu Asp Met Val Asp Thr Ser Leu Ser Gly Phe Lys Leu
20 25 30

<210> SEQ ID NO 130

<211> LENGTH: 30

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 130

Val Asp Thr Ser Leu Ser Gly Phe Lys Leu Lys Asp Cys Val Met Tyr
1 5 10 15

Ala Ser Ala Val Val Leu Leu Ile Leu Met Thr Ala Arg Thr
20 25 30

<210> SEQ ID NO 131

<211> LENGTH: 30

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 131

Val Leu Leu Ile Leu Met Thr Ala Arg Thr Val Tyr Asp Asp Gly Ala
1 5 10 15

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Arg Arg Val Trp Thr Leu Met Asn Val Leu Thr Leu Val Tyr
20 25 30

<210> SEQ ID NO 132
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 132

Thr Leu Met Asn Val Leu Thr Leu Val Tyr Lys Val Tyr Tyr Gly Asn
1 5 10 15

Ala Leu Asp Gln Ala Ile Ser Met Trp Ala Leu Ile Ile Ser
20 25 30

<210> SEQ ID NO 133
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 133

Ala Ile Ser Met Trp Ala Leu Ile Ile Ser Val Thr Ser Asn Tyr Ser
1 5 10 15

Gly Val Val Thr Thr Val Met Phe Leu Ala Arg Gly Ile Val
20 25 30

<210> SEQ ID NO 134
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 134

Thr Val Met Phe Leu Ala Arg Gly Ile Val Phe Met Cys Val Glu Tyr
1 5 10 15

Cys Pro Ile Phe Phe Ile Thr Gly Asn Thr Leu Gln Cys Ile
20 25 30

<210> SEQ ID NO 135
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 135

Phe Ile Thr Gly Asn Thr Leu Gln Cys Ile Met Leu Val Tyr Cys Phe
1 5 10 15

Leu Gly Tyr Phe Cys Thr Cys Tyr Phe Gly Leu Phe Cys Leu
20 25 30

<210> SEQ ID NO 136
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 136

Cys Thr Cys Tyr Phe Gly Leu Phe Cys Leu Leu Asn Arg Tyr Phe Arg
1 5 10 15

Leu Thr Leu Gly Val Tyr Asp Tyr Leu Val Ser Thr Gln Glu
20 25 30

<210> SEQ ID NO 137

-continued

<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 137

Val Tyr Asp Tyr Leu Val Ser Thr Gln Glu Phe Arg Tyr Met Asn Ser
1 5 10 15

Gln Gly Leu Leu Pro Pro Lys Asn Ser Ile Asp Ala Phe Lys
20 25 30

<210> SEQ ID NO 138
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 138

Pro Pro Lys Asn Ser Ile Asp Ala Phe Lys Leu Asn Ile Lys Leu Leu
1 5 10 15

Gly Val Gly Gly Lys Pro Cys Ile Lys Val Ala Thr Val Gln
20 25 30

<210> SEQ ID NO 139
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 139

Ser Lys Met Ser Asp Val Lys Cys Thr Ser Val Val Leu Leu Ser Val
1 5 10 15

Leu Gln Gln Leu Arg Val Glu Ser Ser Ser Lys Leu
20 25

<210> SEQ ID NO 140
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 140

Gln Leu Arg Val Glu Ser Ser Ser Lys Leu Trp Ala Gln Cys Val Gln
1 5 10 15

Leu His Asn Asp Ile Leu Leu Ala Lys Asp Thr Thr
20 25

<210> SEQ ID NO 141
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 141

Asn Asp Ile Leu Leu Ala Lys Asp Thr Thr Glu Ala Phe Glu Lys Met
1 5 10 15

Val Ser Leu Leu Ser Val Leu Leu Met Gln Gly
20 25

<210> SEQ ID NO 142
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 142

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Leu Leu Ser Val Leu Leu Ser Met Gln Gly Ala Val Asp Ile Asn Lys
1 5 10 15

Leu Cys Glu Glu Met Leu Asp Asn Arg Ala Thr Leu Gln
20 25

<210> SEQ ID NO 143
<211> LENGTH: 50
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 143

Ser Ala Val Lys Arg Thr Ile Lys Gly Thr His His Trp Leu Leu Leu
1 5 10 15

Thr Ile Leu Thr Ser Leu Leu Val Leu Val Gln Ser Thr Gln Trp Ser
20 25 30

Leu Phe Phe Phe Leu Tyr Glu Asn Ala Phe Leu Pro Phe Ala Met Gly
35 40 45

Ile Ile
50

<210> SEQ ID NO 144
<211> LENGTH: 50
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 144

Ala Phe Leu Pro Phe Ala Met Gly Ile Ile Ala Met Ser Ala Phe Ala
1 5 10 15

Met Met Phe Val Lys His Lys His Ala Phe Leu Cys Leu Phe Leu Leu
20 25 30

Pro Ser Leu Ala Thr Val Ala Tyr Phe Asn Met Val Tyr Met Pro Ala
35 40 45

Ser Trp
50

<210> SEQ ID NO 145
<211> LENGTH: 50
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 145

Phe Asn Met Val Tyr Met Pro Ala Ser Trp Val Met Arg Ile Met Thr
1 5 10 15

Trp Leu Asp Met Val Asp Thr Ser Leu Ser Gly Phe Lys Leu Lys Asp
20 25 30

Cys Val Met Tyr Ala Ser Ala Val Val Leu Leu Ile Leu Met Thr Ala
35 40 45

Arg Thr
50

<210> SEQ ID NO 146
<211> LENGTH: 50
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 146

Val Leu Leu Ile Leu Met Thr Ala Arg Thr Val Tyr Asp Asp Gly Ala
1 5 10 15

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Arg Arg Val Trp Thr Leu Met Asn Val Leu Thr Leu Val Tyr Lys Val
 20 25 30

Tyr Tyr Gly Asn Ala Leu Asp Gln Ala Ile Ser Met Trp Ala Leu Ile
 35 40 45

Ile Ser
 50

<210> SEQ ID NO 147
 <211> LENGTH: 50
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 147

Ala Ile Ser Met Trp Ala Leu Ile Ile Ser Val Thr Ser Asn Tyr Ser
 1 5 10 15

Gly Val Val Thr Thr Val Met Phe Leu Ala Arg Gly Ile Val Phe Met
 20 25 30

Cys Val Glu Tyr Cys Pro Ile Phe Phe Ile Thr Gly Asn Thr Leu Gln
 35 40 45

Cys Ile
 50

<210> SEQ ID NO 148
 <211> LENGTH: 50
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 148

Phe Ile Thr Gly Asn Thr Leu Gln Cys Ile Met Leu Val Tyr Cys Phe
 1 5 10 15

Leu Gly Tyr Phe Cys Thr Cys Tyr Phe Gly Leu Phe Cys Leu Leu Asn
 20 25 30

Arg Tyr Phe Arg Leu Thr Leu Gly Val Tyr Asp Tyr Leu Val Ser Thr
 35 40 45

Gln Glu
 50

<210> SEQ ID NO 149
 <211> LENGTH: 50
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 149

Val Tyr Asp Tyr Leu Val Ser Thr Gln Glu Phe Arg Tyr Met Asn Ser
 1 5 10 15

Gln Gly Leu Leu Pro Pro Lys Asn Ser Ile Asp Ala Phe Lys Leu Asn
 20 25 30

Ile Lys Leu Leu Gly Val Gly Gly Lys Pro Cys Ile Lys Val Ala Thr
 35 40 45

Val Gln
 50

<210> SEQ ID NO 150
 <211> LENGTH: 46
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

-continued

<400> SEQUENCE: 150

Ser Lys Met Ser Asp Val Lys Cys Thr Ser Val Val Leu Leu Ser Val
1 5 10 15
Leu Gln Gln Leu Arg Val Glu Ser Ser Ser Lys Leu Trp Ala Gln Cys
20 25 30
Val Gln Leu His Asn Asp Ile Leu Leu Ala Lys Asp Thr Thr
35 40 45

<210> SEQ ID NO 151

<211> LENGTH: 47

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 151

Asn Asp Ile Leu Leu Ala Lys Asp Thr Thr Glu Ala Phe Glu Lys Met
1 5 10 15
Val Ser Leu Leu Ser Val Leu Leu Ser Met Gln Gly Ala Val Asp Ile
20 25 30
Asn Lys Leu Cys Glu Glu Met Leu Asp Asn Arg Ala Thr Leu Gln
35 40 45

<210> SEQ ID NO 152

<211> LENGTH: 104

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 152

Ser Ala Val Lys Arg Thr Ile Lys Gly Thr His His Trp Leu Leu Leu
1 5 10 15
Thr Ile Leu Thr Ser Leu Leu Val Leu Val Gln Ser Thr Gln Trp Ser
20 25 30
Leu Phe Phe Phe Leu Tyr Glu Asn Ala Phe Leu Pro Phe Ala Met Gly
35 40 45
Ile Ile Ala Met Ser Ala Phe Ala Met Met Phe Val Lys His Lys His
50 55 60
Ala Phe Leu Cys Leu Phe Leu Leu Pro Ser Leu Ala Thr Val Ala Tyr
65 70 75 80
Phe Asn Met Val Tyr Met Pro Ala Ser Trp Val Met Arg Ile Met Thr
85 90 95
Trp Leu Asp Met Val Asp Thr Ser
100

<210> SEQ ID NO 153

<211> LENGTH: 104

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 153

Ile Met Thr Trp Leu Asp Met Val Asp Thr Ser Leu Ser Gly Phe Lys
1 5 10 15
Leu Lys Asp Cys Val Met Tyr Ala Ser Ala Val Val Leu Leu Ile Leu
20 25 30
Met Thr Ala Arg Thr Val Tyr Asp Asp Gly Ala Arg Arg Val Trp Thr
35 40 45
Leu Met Asn Val Leu Thr Leu Val Tyr Lys Val Tyr Tyr Gly Asn Ala
50 55 60

-continued

Leu Asp Gln Ala Ile Ser Met Trp Ala Leu Ile Ile Ser Val Thr Ser
 65 70 75 80

Asn Tyr Ser Gly Val Val Thr Thr Val Met Phe Leu Ala Arg Gly Ile
 85 90 95

Val Phe Met Cys Val Glu Tyr Cys
 100

<210> SEQ ID NO 154
 <211> LENGTH: 104
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 154

Arg Gly Ile Val Phe Met Cys Val Glu Tyr Cys Pro Ile Phe Phe Ile
 1 5 10 15

Thr Gly Asn Thr Leu Gln Cys Ile Met Leu Val Tyr Cys Phe Leu Gly
 20 25 30

Tyr Phe Cys Thr Cys Tyr Phe Gly Leu Phe Cys Leu Leu Asn Arg Tyr
 35 40 45

Phe Arg Leu Thr Leu Gly Val Tyr Asp Tyr Leu Val Ser Thr Gln Glu
 50 55 60

Phe Arg Tyr Met Asn Ser Gln Gly Leu Leu Pro Pro Lys Asn Ser Ile
 65 70 75 80

Asp Ala Phe Lys Leu Asn Ile Lys Leu Leu Gly Val Gly Gly Lys Pro
 85 90 95

Cys Ile Lys Val Ala Thr Val Gln
 100

<210> SEQ ID NO 155
 <211> LENGTH: 290
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 155

Ser Ala Val Lys Arg Thr Ile Lys Gly Thr His His Trp Leu Leu Leu
 1 5 10 15

Thr Ile Leu Thr Ser Leu Leu Val Leu Val Gln Ser Thr Gln Trp Ser
 20 25 30

Leu Phe Phe Phe Leu Tyr Glu Asn Ala Phe Leu Pro Phe Ala Met Gly
 35 40 45

Ile Ile Ala Met Ser Ala Phe Ala Met Met Phe Val Lys His Lys His
 50 55 60

Ala Phe Leu Cys Leu Phe Leu Leu Pro Ser Leu Ala Thr Val Ala Tyr
 65 70 75 80

Phe Asn Met Val Tyr Met Pro Ala Ser Trp Val Met Arg Ile Met Thr
 85 90 95

Trp Leu Asp Met Val Asp Thr Ser Leu Ser Gly Phe Lys Leu Lys Asp
 100 105 110

Cys Val Met Tyr Ala Ser Ala Val Val Leu Leu Ile Leu Met Thr Ala
 115 120 125

Arg Thr Val Tyr Asp Asp Gly Ala Arg Arg Val Trp Thr Leu Met Asn
 130 135 140

Val Leu Thr Leu Val Tyr Lys Val Tyr Tyr Gly Asn Ala Leu Asp Gln
 145 150 155 160

-continued

Glu Ile Leu Val Thr Tyr Asn Cys Cys Asp Asp Asp Tyr Phe Asn Lys

-continued

1	5	10	15
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Lys Asp Trp Tyr Asp Phe Val Glu Asn Pro Asp Ile
 20 25

<210> SEQ ID NO 159
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

 <400> SEQUENCE: 159

Trp Tyr Asp Phe Val Glu Asn Pro Asp Ile Leu Arg Val Tyr Ala Asn
1 5 10 15

Leu Gly Glu Arg Val Arg Gln Ala Leu Leu Lys Thr
 20 25

<210> SEQ ID NO 160
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

 <400> SEQUENCE: 160

Glu Arg Val Arg Gln Ala Leu Leu Lys Thr Val Gln Phe Cys Asp Ala
1 5 10 15

Met Arg Asn Ala Gly Ile Val Gly Val Leu Thr Leu
 20 25

<210> SEQ ID NO 161
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

 <400> SEQUENCE: 161

Asn Ala Gly Ile Val Gly Val Leu Thr Leu Asp Asn Gln Asp Leu Asn
1 5 10 15

Gly Asn Trp Tyr Asp Phe Gly Asp Phe Ile Gln Thr
 20 25

<210> SEQ ID NO 162
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

 <400> SEQUENCE: 162

Trp Tyr Asp Phe Gly Asp Phe Ile Gln Thr Thr Pro Gly Ser Gly Val
1 5 10 15

Pro Val Val Asp Ser Tyr Tyr Ser Leu Leu Met Pro
 20 25

<210> SEQ ID NO 163
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

 <400> SEQUENCE: 163

Val Asp Ser Tyr Tyr Ser Leu Leu Met Pro Ile Leu Thr Leu Thr Arg
1 5 10 15

Ala Leu Thr Ala Glu Ser His Val Asp Thr Asp Leu
 20 25

-continued

<210> SEQ ID NO 164
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 164

Thr Ala Glu Ser His Val Asp Thr Asp Leu Thr Lys Pro Tyr Ile Lys
1 5 10 15

Trp Asp Leu Leu Lys Tyr Asp Phe Thr Glu Glu Arg Leu
20 25

<210> SEQ ID NO 165
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 165

Leu Lys Tyr Asp Phe Thr Glu Glu Arg Leu Lys Leu Phe Asp Arg Tyr
1 5 10 15

Phe Lys Tyr Trp Asp Gln Thr Tyr His Pro Asn Cys Val
20 25

<210> SEQ ID NO 166
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 166

Trp Asp Gln Thr Tyr His Pro Asn Cys Val Asn Cys Leu Asp Asp Arg
1 5 10 15

Cys Ile Leu His Cys Ala Asn Phe Asn Val Leu Phe
20 25

<210> SEQ ID NO 167
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 167

Leu His Cys Ala Asn Phe Asn Val Leu Phe Ser Thr Val Phe Pro Pro
1 5 10 15

Thr Ser Phe Gly Pro Leu Val Arg Lys Ile Phe Val Asp
20 25

<210> SEQ ID NO 168
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 168

Gly Pro Leu Val Arg Lys Ile Phe Val Asp Gly Val Pro Phe Val Val
1 5 10 15

Ser Thr Gly Tyr His Phe Arg Glu Leu Gly Val Val His
20 25

<210> SEQ ID NO 169
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 169

-continued

Tyr His Phe Arg Glu Leu Gly Val Val His Asn Gln Asp Val Asn Leu
1 5 10 15

His Ser Ser Arg Leu Ser Phe Lys Glu Leu Leu Val Tyr Ala
20 25 30

<210> SEQ ID NO 170
<211> LENGTH: 50
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 170

Asp Leu Val Tyr Ala Leu Arg His Phe Asp Glu Gly Asn Cys Asp Thr
1 5 10 15

Leu Lys Glu Ile Leu Val Thr Tyr Asn Cys Cys Asp Asp Tyr Phe
20 25 30

Asn Lys Lys Asp Trp Tyr Asp Phe Val Glu Asn Pro Asp Ile Leu Arg
35 40 45

Val Tyr
50

<210> SEQ ID NO 171
<211> LENGTH: 50
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 171

Val Glu Asn Pro Asp Ile Leu Arg Val Tyr Ala Asn Leu Gly Glu Arg
1 5 10 15

Val Arg Gln Ala Leu Leu Lys Thr Val Gln Phe Cys Asp Ala Met Arg
20 25 30

Asn Ala Gly Ile Val Gly Val Leu Thr Leu Asp Asn Gln Asp Leu Asn
35 40 45

Gly Asn
50

<210> SEQ ID NO 172
<211> LENGTH: 50
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 172

Thr Leu Asp Asn Gln Asp Leu Asn Gly Asn Trp Tyr Asp Phe Gly Asp
1 5 10 15

Phe Ile Gln Thr Thr Pro Gly Ser Gly Val Pro Val Val Asp Ser Tyr
20 25 30

Tyr Ser Leu Leu Met Pro Ile Leu Thr Leu Thr Arg Ala Leu Thr Ala
35 40 45

Glu Ser
50

<210> SEQ ID NO 173
<211> LENGTH: 50
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 173

Thr Leu Thr Arg Ala Leu Thr Ala Glu Ser His Val Asp Thr Asp Leu

-continued

1	5	10	15
Thr Lys Pro Tyr Ile Lys Trp Asp Leu Leu Lys Tyr Asp Phe Thr Glu			
	20	25	30
Glu Arg Leu Lys Leu Phe Asp Arg Tyr Phe Lys Tyr Trp Asp Gln Thr			
	35	40	45
Tyr His			
50			

<210> SEQ ID NO 174
 <211> LENGTH: 50
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 174

Tyr Phe Lys Tyr Trp Asp Gln Thr Tyr His Pro Asn Cys Val Asn Cys			
1	5	10	15
Leu Asp Asp Arg Cys Ile Leu His Cys Ala Asn Phe Asn Val Leu Phe			
	20	25	30
Ser Thr Val Phe Pro Pro Thr Ser Phe Gly Pro Leu Val Arg Lys Ile			
	35	40	45
Phe Val			
50			

<210> SEQ ID NO 175
 <211> LENGTH: 50
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 175

Phe Gly Pro Leu Val Arg Lys Ile Phe Val Asp Gly Val Pro Phe Val			
1	5	10	15
Val Ser Thr Gly Tyr His Phe Arg Glu Leu Gly Val Val His Asn Gln			
	20	25	30
Asp Val Asn Leu His Ser Ser Arg Leu Ser Phe Lys Glu Leu Leu Val			
	35	40	45
Tyr Ala			
50			

<210> SEQ ID NO 176
 <211> LENGTH: 130
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 176

Asp Leu Val Tyr Ala Leu Arg His Phe Asp Glu Gly Asn Cys Asp Thr			
1	5	10	15
Leu Lys Glu Ile Leu Val Thr Tyr Asn Cys Cys Asp Asp Asp Tyr Phe			
	20	25	30
Asn Lys Lys Asp Trp Tyr Asp Phe Val Glu Asn Pro Asp Ile Leu Arg			
	35	40	45
Val Tyr Ala Asn Leu Gly Glu Arg Val Arg Gln Ala Leu Leu Lys Thr			
	50	55	60
Val Gln Phe Cys Asp Ala Met Arg Asn Ala Gly Ile Val Gly Val Leu			
65	70	75	80
Thr Leu Asp Asn Gln Asp Leu Asn Gly Asn Trp Tyr Asp Phe Gly Asp			
	85	90	95

-continued

Phe Ile Gln Thr Thr Pro Gly Ser Gly Val Pro Val Val Asp Ser Tyr
100 105 110

Tyr Ser Leu Leu Met Pro Ile Leu Thr Leu Thr Arg Ala Leu Thr Ala
115 120 125

Glu Ser
130

<210> SEQ ID NO 177

<211> LENGTH: 130

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 177

Thr Leu Thr Arg Ala Leu Thr Ala Glu Ser His Val Asp Thr Asp Leu
1 5 10 15

Thr Lys Pro Tyr Ile Lys Trp Asp Leu Leu Lys Tyr Asp Phe Thr Glu
20 25 30

Glu Arg Leu Lys Leu Phe Asp Arg Tyr Phe Lys Tyr Trp Asp Gln Thr
35 40 45

Tyr His Pro Asn Cys Val Asn Cys Leu Asp Asp Arg Cys Ile Leu His
50 55 60

Cys Ala Asn Phe Asn Val Leu Phe Ser Thr Val Phe Pro Pro Thr Ser
65 70 75 80

Phe Gly Pro Leu Val Arg Lys Ile Phe Val Asp Gly Val Pro Phe Val
85 90 95

Val Ser Thr Gly Tyr His Phe Arg Glu Leu Gly Val Val His Asn Gln
100 105 110

Asp Val Asn Leu His Ser Ser Arg Leu Ser Phe Lys Glu Leu Leu Val
115 120 125

Tyr Ala
130

<210> SEQ ID NO 178

<211> LENGTH: 250

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 178

Asp Leu Val Tyr Ala Leu Arg His Phe Asp Glu Gly Asn Cys Asp Thr
1 5 10 15

Leu Lys Glu Ile Leu Val Thr Tyr Asn Cys Cys Asp Asp Asp Tyr Phe
20 25 30

Asn Lys Lys Asp Trp Tyr Asp Phe Val Glu Asn Pro Asp Ile Leu Arg
35 40 45

Val Tyr Ala Asn Leu Gly Glu Arg Val Arg Gln Ala Leu Leu Lys Thr
50 55 60

Val Gln Phe Cys Asp Ala Met Arg Asn Ala Gly Ile Val Gly Val Leu
65 70 75 80

Thr Leu Asp Asn Gln Asp Leu Asn Gly Asn Trp Tyr Asp Phe Gly Asp
85 90 95

Phe Ile Gln Thr Thr Pro Gly Ser Gly Val Pro Val Val Asp Ser Tyr
100 105 110

Tyr Ser Leu Leu Met Pro Ile Leu Thr Leu Thr Arg Ala Leu Thr Ala
115 120 125

-continued

Glu Ser His Val Asp Thr Asp Leu Thr Lys Pro Tyr Ile Lys Trp Asp
 130 135 140
 Leu Leu Lys Tyr Asp Phe Thr Glu Glu Arg Leu Lys Leu Phe Asp Arg
 145 150 155 160
 Tyr Phe Lys Tyr Trp Asp Gln Thr Tyr His Pro Asn Cys Val Asn Cys
 165 170 175
 Leu Asp Asp Arg Cys Ile Leu His Cys Ala Asn Phe Asn Val Leu Phe
 180 185 190
 Ser Thr Val Phe Pro Pro Thr Ser Phe Gly Pro Leu Val Arg Lys Ile
 195 200 205
 Phe Val Asp Gly Val Pro Phe Val Val Ser Thr Gly Tyr His Phe Arg
 210 215 220
 Glu Leu Gly Val Val His Asn Gln Asp Val Asn Leu His Ser Ser Arg
 225 230 235 240
 Leu Ser Phe Lys Glu Leu Leu Val Tyr Ala
 245 250

<210> SEQ ID NO 179
 <211> LENGTH: 29
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 179

Ser Tyr Glu Asp Gln Asp Ala Leu Phe Ala Tyr Thr Lys Arg Asn Val
 1 5 10 15
 Ile Pro Thr Ile Thr Gln Met Asn Leu Lys Tyr Ala Ile
 20 25

<210> SEQ ID NO 180
 <211> LENGTH: 29
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 180

Thr Ile Thr Gln Met Asn Leu Lys Tyr Ala Ile Ser Ala Lys Asn Arg
 1 5 10 15
 Ala Arg Thr Val Ala Gly Val Ser Ile Cys Ser Thr Met
 20 25

<210> SEQ ID NO 181
 <211> LENGTH: 29
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 181

Val Ala Gly Val Ser Ile Cys Ser Thr Met Asn Arg Gln Phe His Gln
 1 5 10 15
 Lys Leu Leu Lys Ser Ile Ala Ala Thr Arg Gly Ala Thr
 20 25

<210> SEQ ID NO 182
 <211> LENGTH: 29
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 182

Leu Lys Ser Ile Ala Ala Thr Arg Gly Ala Thr Val Val Ile Gly Thr

-continued

1 5 10 15
 Ser Lys Phe Tyr Gly Gly Trp His Asn Met Leu Lys Thr
 20 25

<210> SEQ ID NO 183
 <211> LENGTH: 30
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 183

Tyr Gly Gly Trp His Asn Met Leu Lys Thr Val Tyr Ser Asp Val Glu
 1 5 10 15

Asn Pro His Leu Met Gly Trp Asp Tyr Pro Lys Cys Asp Arg
 20 25 30

<210> SEQ ID NO 184
 <211> LENGTH: 29
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 184

Met Gly Trp Asp Tyr Pro Lys Cys Asp Arg Ala Met Pro Asn Met Leu
 1 5 10 15

Arg Ile Met Ala Ser Leu Val Leu Ala Arg Lys His Thr
 20 25

<210> SEQ ID NO 185
 <211> LENGTH: 29
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 185

Ala Ser Leu Val Leu Ala Arg Lys His Thr Thr Cys Cys Ser Leu Ser
 1 5 10 15

His Arg Phe Tyr Arg Leu Ala Asn Glu Cys Ala Gln Val
 20 25

<210> SEQ ID NO 186
 <211> LENGTH: 31
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 186

Tyr Arg Leu Ala Asn Glu Cys Ala Gln Val Leu Ser Glu Met Val Met
 1 5 10 15

Cys Gly Gly Ser Leu Tyr Val Lys Pro Gly Gly Thr Ser Ser Gly
 20 25 30

<210> SEQ ID NO 187
 <211> LENGTH: 31
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 187

Tyr Val Lys Pro Gly Gly Thr Ser Ser Gly Asp Ala Thr Thr Ala Tyr
 1 5 10 15

Ala Asn Ser Val Phe Asn Ile Cys Gln Ala Val Thr Ala Asn Val
 20 25 30

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<210> SEQ ID NO 188
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 188

Asn Ile Cys Gln Ala Val Thr Ala Asn Val Asn Ala Leu Leu Ser Thr
1 5 10 15

Asp Gly Asn Lys Ile Ala Asp Lys Tyr Val Arg Asn Leu Gln
20 25 30

<210> SEQ ID NO 189
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 189

Ile Ala Asp Lys Tyr Val Arg Asn Leu Gln His Arg Leu Tyr Glu Cys
1 5 10 15

Leu Tyr Arg Asn Arg Asp Val Asp Thr Asp Phe Val Asn Glu
20 25 30

<210> SEQ ID NO 190
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 190

Arg Asp Val Asp Thr Asp Phe Val Asn Glu Phe Tyr Ala Tyr Leu Arg
1 5 10 15

Lys His Phe Ser Met Met Ile Leu Ser Asp Asp Ala Val Val
20 25 30

<210> SEQ ID NO 191
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 191

Met Met Ile Leu Ser Asp Asp Ala Val Val Cys Phe Asn Ser Thr Tyr
1 5 10 15

Ala Ser Gln Gly Leu Val Ala Ser Ile Lys Asn Phe Lys Ser
20 25 30

<210> SEQ ID NO 192
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 192

Leu Val Ala Ser Ile Lys Asn Phe Lys Ser Val Leu Tyr Tyr Gln Asn
1 5 10 15

Asn Val Phe Met Ser Glu Ala Lys Cys Trp Thr Glu Thr Asp
20 25 30

<210> SEQ ID NO 193
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 193

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Ser Glu Ala Lys Cys Trp Thr Glu Thr Asp Leu Thr Lys Gly Pro His
1 5 10 15

Glu Phe Cys Ser Gln His Thr Met Leu Val Lys Gln Gly Asp
20 25 30

<210> SEQ ID NO 194
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 194

Gln His Thr Met Leu Val Lys Gln Gly Asp Asp Tyr Val Tyr Leu Pro
1 5 10 15

Tyr Pro Asp Pro Ser Arg Ile Leu Gly Ala Gly Cys Phe Val
20 25 30

<210> SEQ ID NO 195
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 195

Ser Arg Ile Leu Gly Ala Gly Cys Phe Val Asp Asp Ile Val Lys Thr
1 5 10 15

Asp Gly Thr Leu Met Ile Glu Arg Phe Val Ser Leu Ala
20 25

<210> SEQ ID NO 196
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 196

Leu Met Ile Glu Arg Phe Val Ser Leu Ala Ile Asp Ala Tyr Pro Leu
1 5 10 15

Thr Lys His Pro Asn Gln Glu Tyr Ala Asp Val Phe His
20 25

<210> SEQ ID NO 197
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 197

Pro Asn Gln Glu Tyr Ala Asp Val Phe His Leu Tyr Leu Gln Tyr Ile
1 5 10 15

Arg Lys Leu His Asp Glu Leu Thr Gly His Met Leu Asp
20 25

<210> SEQ ID NO 198
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 198

His Asp Glu Leu Thr Gly His Met Leu Asp Met Tyr Ser Val Met Leu
1 5 10 15

Thr Asn Asp Asn Thr Ser Arg Tyr Trp Glu Pro Glu Phe
20 25

-continued

<210> SEQ ID NO 199
<211> LENGTH: 49
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 199

Ser Tyr Glu Asp Gln Asp Ala Leu Phe Ala Tyr Thr Lys Arg Asn Val
1 5 10 15
Ile Pro Thr Ile Thr Gln Met Asn Leu Lys Tyr Ala Ile Ser Ala Lys
20 25 30
Asn Arg Ala Arg Thr Val Ala Gly Val Ser Ile Cys Ser Thr Met Thr
35 40 45

Asn

<210> SEQ ID NO 200
<211> LENGTH: 49
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 200

Gly Val Ser Ile Cys Ser Thr Met Thr Asn Arg Gln Phe His Gln Lys
1 5 10 15
Leu Leu Lys Ser Ile Ala Ala Thr Arg Gly Ala Thr Val Val Ile Gly
20 25 30
Thr Ser Lys Phe Tyr Gly Gly Trp His Asn Met Leu Lys Thr Val Tyr
35 40 45

Ser

<210> SEQ ID NO 201
<211> LENGTH: 49
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 201

Trp His Asn Met Leu Lys Thr Val Tyr Ser Asp Val Glu Asn Pro His
1 5 10 15
Leu Met Gly Trp Asp Tyr Pro Lys Cys Asp Arg Ala Met Pro Asn Met
20 25 30
Leu Arg Ile Met Ala Ser Leu Val Leu Ala Arg Lys His Thr Thr Cys
35 40 45

Cys

<210> SEQ ID NO 202
<211> LENGTH: 49
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 202

Val Leu Ala Arg Lys His Thr Thr Cys Cys Ser Leu Ser His Arg Phe
1 5 10 15
Tyr Arg Leu Ala Asn Glu Cys Ala Gln Val Leu Ser Glu Met Val Met
20 25 30
Cys Gly Gly Ser Leu Tyr Val Lys Pro Gly Gly Thr Ser Ser Gly Asp
35 40 45

Ala

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<210> SEQ ID NO 203
<211> LENGTH: 49
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 203

Lys Pro Gly Gly Thr Ser Ser Gly Asp Ala Thr Thr Ala Tyr Ala Asn
1 5 10 15
Ser Val Phe Asn Ile Cys Gln Ala Val Thr Ala Asn Val Asn Ala Leu
20 25 30
Leu Ser Thr Asp Gly Asn Lys Ile Ala Asp Lys Tyr Val Arg Asn Leu
35 40 45
Gln

<210> SEQ ID NO 204
<211> LENGTH: 49
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 204

Ile Ala Asp Lys Tyr Val Arg Asn Leu Gln His Arg Leu Tyr Glu Cys
1 5 10 15
Leu Tyr Arg Asn Arg Asp Val Asp Thr Asp Phe Val Asn Glu Phe Tyr
20 25 30
Ala Tyr Leu Arg Lys His Phe Ser Met Met Ile Leu Ser Asp Asp Ala
35 40 45
Val

<210> SEQ ID NO 205
<211> LENGTH: 49
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 205

Ser Met Met Ile Leu Ser Asp Asp Ala Val Val Cys Phe Asn Ser Thr
1 5 10 15
Tyr Ala Ser Gln Gly Leu Val Ala Ser Ile Lys Asn Phe Lys Ser Val
20 25 30
Leu Tyr Tyr Gln Asn Asn Val Phe Met Ser Glu Ala Lys Cys Trp Thr
35 40 45
Glu

<210> SEQ ID NO 206
<211> LENGTH: 49
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 206

Phe Met Ser Glu Ala Lys Cys Trp Thr Glu Thr Asp Leu Thr Lys Gly
1 5 10 15
Pro His Glu Phe Cys Ser Gln His Thr Met Leu Val Lys Gln Gly Asp
20 25 30
Asp Tyr Val Tyr Leu Pro Tyr Pro Asp Pro Ser Arg Ile Leu Gly Ala
35 40 45
Gly

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<210> SEQ ID NO 207
<211> LENGTH: 49
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 207

Pro Asp Pro Ser Arg Ile Leu Gly Ala Gly Cys Phe Val Asp Asp Ile
1 5 10 15
Val Lys Thr Asp Gly Thr Leu Met Ile Glu Arg Phe Val Ser Leu Ala
20 25 30
Ile Asp Ala Tyr Pro Leu Thr Lys His Pro Asn Gln Glu Tyr Ala Asp
35 40 45
Val

<210> SEQ ID NO 208
<211> LENGTH: 50
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 208

Lys His Pro Asn Gln Glu Tyr Ala Asp Val Phe His Leu Tyr Leu Gln
1 5 10 15
Tyr Ile Arg Lys Leu His Asp Glu Leu Thr Gly His Met Leu Asp Met
20 25 30
Tyr Ser Val Met Leu Thr Asn Asp Asn Thr Ser Arg Tyr Trp Glu Pro
35 40 45
Glu Phe
50

<210> SEQ ID NO 209
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 209

Ser Tyr Glu Asp Gln Asp Ala Leu Phe Ala Tyr Thr Lys Arg Asn Val
1 5 10 15
Ile Pro Thr Ile Thr Gln Met Asn Leu Lys Tyr Ala Ile Ser Ala Lys
20 25 30
Asn Arg Ala Arg Thr Val Ala Gly Val Ser Ile Cys Ser Thr Met Thr
35 40 45
Asn Arg Gln Phe His Gln Lys Leu Leu Lys Ser Ile Ala Ala Thr Arg
50 55 60
Gly Ala Thr Val Val Ile Gly Thr Ser Lys Phe Tyr Gly Gly Trp His
65 70 75 80
Asn Met Leu Lys Thr Val Tyr Ser Asp Val Glu Asn Pro His Leu Met
85 90 95
Gly Trp Asp Tyr Pro Lys Cys Asp Arg Ala Met Pro Asn Met
100 105 110

<210> SEQ ID NO 210
<211> LENGTH: 109
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 210

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Tyr Pro Lys Cys Asp Arg Ala Met Pro Asn Met Leu Arg Ile Met Ala
 1 5 10 15
 Ser Leu Val Leu Ala Arg Lys His Thr Thr Cys Cys Ser Leu Ser His
 20 25 30
 Arg Phe Tyr Arg Leu Ala Asn Glu Cys Ala Gln Val Leu Ser Glu Met
 35 40 45
 Val Met Cys Gly Gly Ser Leu Tyr Val Lys Pro Gly Gly Thr Ser Ser
 50 55 60
 Gly Asp Ala Thr Thr Ala Tyr Ala Asn Ser Val Phe Asn Ile Cys Gln
 65 70 75 80
 Ala Val Thr Ala Asn Val Asn Ala Leu Leu Ser Thr Asp Gly Asn Lys
 85 90 95
 Ile Ala Asp Lys Tyr Val Arg Asn Leu Gln His Arg Leu
 100 105

<210> SEQ ID NO 211
 <211> LENGTH: 110
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 211

Asp Lys Tyr Val Arg Asn Leu Gln His Arg Leu Tyr Glu Cys Leu Tyr
 1 5 10 15
 Arg Asn Arg Asp Val Asp Thr Asp Phe Val Asn Glu Phe Tyr Ala Tyr
 20 25 30
 Leu Arg Lys His Phe Ser Met Met Ile Leu Ser Asp Asp Ala Val Val
 35 40 45
 Cys Phe Asn Ser Thr Tyr Ala Ser Gln Gly Leu Val Ala Ser Ile Lys
 50 55 60
 Asn Phe Lys Ser Val Leu Tyr Tyr Gln Asn Asn Val Phe Met Ser Glu
 65 70 75 80
 Ala Lys Cys Trp Thr Glu Thr Asp Leu Thr Lys Gly Pro His Glu Phe
 85 90 95
 Cys Ser Gln His Thr Met Leu Val Lys Gln Gly Asp Asp Tyr
 100 105 110

<210> SEQ ID NO 212
 <211> LENGTH: 105
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 212

His Thr Met Leu Val Lys Gln Gly Asp Asp Tyr Val Tyr Leu Pro Tyr
 1 5 10 15
 Pro Asp Pro Ser Arg Ile Leu Gly Ala Gly Cys Phe Val Asp Asp Ile
 20 25 30
 Val Lys Thr Asp Gly Thr Leu Met Ile Glu Arg Phe Val Ser Leu Ala
 35 40 45
 Ile Asp Ala Tyr Pro Leu Thr Lys His Pro Asn Gln Glu Tyr Ala Asp
 50 55 60
 Val Phe His Leu Tyr Leu Gln Tyr Ile Arg Lys Leu His Asp Glu Leu
 65 70 75 80
 Thr Gly His Met Leu Asp Met Tyr Ser Val Met Leu Thr Asn Asp Asn
 85 90 95

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Thr Ser Arg Tyr Trp Glu Pro Glu Phe
100 105

<210> SEQ ID NO 213
 <211> LENGTH: 401
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 213

Ser Tyr Glu Asp Gln Asp Ala Leu Phe Ala Tyr Thr Lys Arg Asn Val
1 5 10 15

Ile Pro Thr Ile Thr Gln Met Asn Leu Lys Tyr Ala Ile Ser Ala Lys
20 25 30

Asn Arg Ala Arg Thr Val Ala Gly Val Ser Ile Cys Ser Thr Met Thr
35 40 45

Asn Arg Gln Phe His Gln Lys Leu Leu Lys Ser Ile Ala Ala Thr Arg
50 55 60

Gly Ala Thr Val Val Ile Gly Thr Ser Lys Phe Tyr Gly Gly Trp His
65 70 75 80

Asn Met Leu Lys Thr Val Tyr Ser Asp Val Glu Asn Pro His Leu Met
85 90 95

Gly Trp Asp Tyr Pro Lys Cys Asp Arg Ala Met Pro Asn Met Leu Arg
100 105 110

Ile Met Ala Ser Leu Val Leu Ala Arg Lys His Thr Thr Cys Cys Ser
115 120 125

Leu Ser His Arg Phe Tyr Arg Leu Ala Asn Glu Cys Ala Gln Val Leu
130 135 140

Ser Glu Met Val Met Cys Gly Gly Ser Leu Tyr Val Lys Pro Gly Gly
145 150 155 160

Thr Ser Ser Gly Asp Ala Thr Thr Ala Tyr Ala Asn Ser Val Phe Asn
165 170 175

Ile Cys Gln Ala Val Thr Ala Asn Val Asn Ala Leu Leu Ser Thr Asp
180 185 190

Gly Asn Lys Ile Ala Asp Lys Tyr Val Arg Asn Leu Gln His Arg Leu
195 200 205

Tyr Glu Cys Leu Tyr Arg Asn Arg Asp Val Asp Thr Asp Phe Val Asn
210 215 220

Glu Phe Tyr Ala Tyr Leu Arg Lys His Phe Ser Met Met Ile Leu Ser
225 230 235 240

Asp Asp Ala Val Val Cys Phe Asn Ser Thr Tyr Ala Ser Gln Gly Leu
245 250 255

Val Ala Ser Ile Lys Asn Phe Lys Ser Val Leu Tyr Tyr Gln Asn Asn
260 265 270

Val Phe Met Ser Glu Ala Lys Cys Trp Thr Glu Thr Asp Leu Thr Lys
275 280 285

Gly Pro His Glu Phe Cys Ser Gln His Thr Met Leu Val Lys Gln Gly
290 295 300

Asp Asp Tyr Val Tyr Leu Pro Tyr Pro Asp Pro Ser Arg Ile Leu Gly
305 310 315 320

Ala Gly Cys Phe Val Asp Asp Ile Val Lys Thr Asp Gly Thr Leu Met
325 330 335

Ile Glu Arg Phe Val Ser Leu Ala Ile Asp Ala Tyr Pro Leu Thr Lys

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340	345	350
His Pro Asn Gln Glu Tyr Ala Asp Val Phe His Leu Tyr Leu Gln Tyr		
355	360	365
Ile Arg Lys Leu His Asp Glu Leu Thr Gly His Met Leu Asp Met Tyr		
370	375	380
Ser Val Met Leu Thr Asn Asp Asn Thr Ser Arg Tyr Trp Glu Pro Glu		
385	390	395
		400

Phe

<210> SEQ ID NO 214
 <211> LENGTH: 29
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 214

Thr Ala Asp Glu Ala Ile Lys Phe Leu Lys Glu Trp Asn Phe Ser Leu
1 5 10 15
Gly Ile Ile Leu Leu Phe Ile Thr Ile Ile Leu Gln Phe
20 25

<210> SEQ ID NO 215
 <211> LENGTH: 29
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 215

Leu Leu Phe Ile Thr Ile Ile Ile Leu Gln Phe Gly Tyr Thr Ser Arg
1 5 10 15
Ser Met Phe Val Tyr Val Ile Lys Met Ile Ile Leu Trp
20 25

<210> SEQ ID NO 216
 <211> LENGTH: 29
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 216

Val Tyr Val Ile Lys Met Ile Ile Leu Trp Leu Met Trp Pro Leu Thr
1 5 10 15
Ile Ile Leu Thr Ile Phe Asn Cys Val Tyr Ala Leu Asn
20 25

<210> SEQ ID NO 217
 <211> LENGTH: 29
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 217

Thr Ile Phe Asn Cys Val Tyr Ala Leu Asn Asn Val Tyr Leu Gly Leu
1 5 10 15
Ser Ile Val Phe Thr Ile Val Ala Ile Ile Met Trp Ile
20 25

<210> SEQ ID NO 218
 <211> LENGTH: 29
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 218

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Phe Thr Ile Val Ala Ile Ile Met Trp Ile Val Tyr Phe Val Asn Ser
1 5 10 15

Ile Arg Leu Phe Ile Arg Thr Gly Ser Phe Trp Ser Phe
20 25

<210> SEQ ID NO 219
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 219

Phe Ile Arg Thr Gly Ser Phe Trp Ser Phe Asn Pro Glu Thr Asn Asn
1 5 10 15

Leu Met Cys Ile Asp Met Lys Gly Thr Met Tyr Val Arg
20 25

<210> SEQ ID NO 220
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 220

Ile Asp Met Lys Gly Thr Met Tyr Val Arg Pro Ile Ile Glu Asp Tyr
1 5 10 15

His Thr Leu Thr Val Thr Ile Ile Arg Gly His Leu
20 25

<210> SEQ ID NO 221
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 221

Leu Thr Val Thr Ile Ile Arg Gly His Leu Tyr Ile Gln Gly Ile Lys
1 5 10 15

Leu Gly Thr Gly Tyr Ser Leu Ala Asp Leu Pro Ala
20 25

<210> SEQ ID NO 222
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 222

Thr Gly Tyr Ser Leu Ala Asp Leu Pro Ala Tyr Met Thr Val Ala Lys
1 5 10 15

Val Thr His Leu Cys Thr Tyr Lys Arg Gly Phe Leu
20 25

<210> SEQ ID NO 223
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 223

His Leu Cys Thr Tyr Lys Arg Gly Phe Leu Asp Arg Ile Ser Asp Thr
1 5 10 15

Ser Gly Phe Ala Val Tyr Val Lys Ser Lys Val Gly
20 25

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<210> SEQ ID NO 224
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 224

Phe Ala Val Tyr Val Lys Ser Lys Val Gly Asn Tyr Arg Leu Pro Ser
1 5 10 15

Thr Gln Lys Gly Ser Gly Met Asp Thr Ala Leu Leu
20 25

<210> SEQ ID NO 225
<211> LENGTH: 23
<212> TYPE: PRT
<213> ORGANISM: Human coronavirus HKU11

<400> SEQUENCE: 225

Asn Asn Ala Phe Leu Ala Phe Ser Ile Val Phe Thr Ile Ile Ser Ile
1 5 10 15

Val Ile Trp Ile Leu Tyr Phe
20

<210> SEQ ID NO 226
<211> LENGTH: 27
<212> TYPE: PRT
<213> ORGANISM: Human coronavirus HKU11

<400> SEQUENCE: 226

Lys Val Gln Val Leu Cys Thr Tyr Lys Arg Ala Phe Leu Asp Lys Leu
1 5 10 15

Asp Val Asn Ser Gly Phe Ala Val Phe Val Lys
20 25

<210> SEQ ID NO 227
<211> LENGTH: 44
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 227

Thr Ala Asp Glu Ala Ile Lys Phe Leu Lys Glu Trp Asn Phe Ser Leu
1 5 10 15

Gly Ile Ile Leu Leu Phe Ile Thr Ile Ile Leu Gln Phe Gly Tyr Thr
20 25 30

Ser Arg Ser Met Phe Val Tyr Val Ile Lys Met Ile
35 40

<210> SEQ ID NO 228
<211> LENGTH: 44
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 228

Ser Met Phe Val Tyr Val Ile Lys Met Ile Ile Leu Trp Leu Met Trp
1 5 10 15

Pro Leu Thr Ile Ile Leu Thr Ile Phe Asn Cys Val Tyr Ala Leu Asn
20 25 30

Asn Val Tyr Leu Gly Leu Ser Ile Val Phe Thr Ile
35 40

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<210> SEQ ID NO 229

<211> LENGTH: 44

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 229

Tyr Leu Gly Leu Ser Ile Val Phe Thr Ile Val Ala Ile Ile Met Trp
1 5 10 15

Ile Val Tyr Phe Val Asn Ser Ile Arg Leu Phe Ile Arg Thr Gly Ser
20 25 30

Phe Trp Ser Phe Asn Pro Glu Thr Asn Asn Leu Met
35 40

<210> SEQ ID NO 230

<211> LENGTH: 44

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 230

Ser Phe Asn Pro Glu Thr Asn Asn Leu Met Cys Ile Asp Met Lys Gly
1 5 10 15

Thr Met Phe Val Arg Pro Ile Ile Glu Asp Tyr His Thr Leu Thr Val
20 25 30

Thr Ile Ile Arg Gly His Leu Tyr Ile Gln Gly Ile
35 40

<210> SEQ ID NO 231

<211> LENGTH: 45

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 231

Ile Arg Gly His Leu Tyr Ile Gln Gly Ile Lys Leu Gly Thr Gly Tyr
1 5 10 15

Ser Leu Ala Asp Leu Pro Ala Tyr Met Thr Val Ala Lys Val Thr Tyr
20 25 30

Leu Cys Thr Tyr Lys Arg Gly Phe Leu Asp Lys Ile Ser
35 40 45

<210> SEQ ID NO 232

<211> LENGTH: 42

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 232

Tyr Lys Arg Gly Phe Leu Asp Lys Ile Ser Asp Thr Ser Gly Phe Ala
1 5 10 15

Val Tyr Val Lys Ser Lys Val Gly Asn Tyr Arg Leu Pro Ser Thr Gln
20 25 30

Lys Gly Ser Gly Met Asp Thr Ala Leu Leu
35 40

<210> SEQ ID NO 233

<211> LENGTH: 50

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

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polypeptide

<400> SEQUENCE: 233

Asn Asn Ala Phe Leu Ala Phe Ser Ile Val Phe Thr Ile Ile Ser Ile
 1 5 10 15

Val Ile Trp Ile Leu Tyr Phe Lys Val Gln Val Leu Cys Thr Tyr Lys
 20 25 30

Arg Ala Phe Leu Asp Lys Leu Asp Val Asn Ser Gly Phe Ala Val Phe
 35 40 45

Val Lys
 50

<210> SEQ ID NO 234
 <211> LENGTH: 138
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 234

Thr Ala Asp Glu Ala Ile Lys Phe Leu Lys Glu Trp Asn Phe Ser Leu
 1 5 10 15

Gly Ile Ile Leu Leu Phe Ile Thr Ile Ile Leu Gln Phe Gly Tyr Thr
 20 25 30

Ser Arg Ser Met Phe Val Tyr Val Ile Lys Met Ile Ile Leu Trp Leu
 35 40 45

Met Trp Pro Leu Thr Ile Ile Leu Thr Ile Phe Asn Cys Val Tyr Ala
 50 55 60

Leu Asn Asn Val Tyr Leu Gly Leu Ser Ile Val Phe Thr Ile Val Ala
 65 70 75 80

Ile Ile Met Trp Ile Val Tyr Phe Val Asn Ser Ile Arg Leu Phe Ile
 85 90 95

Arg Thr Gly Ser Phe Trp Ser Phe Asn Pro Glu Thr Asn Asn Leu Met
 100 105 110

Cys Ile Asp Met Lys Gly Thr Met Tyr Val Arg Pro Ile Ile Glu Asp
 115 120 125

Tyr His Thr Leu Thr Val Thr Ile Ile Arg
 130 135

<210> SEQ ID NO 235
 <211> LENGTH: 136
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 235

Asp Tyr His Thr Leu Thr Val Thr Ile Ile Arg Gly His Leu Tyr Ile
 1 5 10 15

Gln Gly Ile Lys Leu Gly Thr Gly Tyr Ser Leu Ala Asp Leu Pro Ala
 20 25 30

Tyr Met Thr Val Ala Lys Val Thr His Leu Cys Thr Tyr Lys Arg Gly
 35 40 45

Phe Leu Asp Arg Ile Ser Asp Thr Ser Gly Phe Ala Val Tyr Val Lys
 50 55 60

Ser Lys Val Gly Asn Tyr Arg Leu Pro Ser Thr Gln Lys Gly Ser Gly
 65 70 75 80

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Met Asp Thr Ala Leu Leu Asn Asn Ala Phe Leu Ala Phe Ser Ile Val
85 90 95

Phe Thr Ile Ile Ser Ile Val Ile Trp Ile Leu Tyr Phe Lys Val Gln
100 105 110

Val Leu Cys Thr Tyr Lys Arg Ala Phe Leu Asp Lys Leu Asp Val Asn
115 120 125

Ser Gly Phe Ala Val Phe Val Lys
130 135

<210> SEQ ID NO 236

<211> LENGTH: 263

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 236

Thr Ala Asp Glu Ala Ile Lys Phe Leu Lys Glu Trp Asn Phe Ser Leu
1 5 10 15

Gly Ile Ile Leu Leu Phe Ile Thr Ile Ile Leu Gln Phe Gly Tyr Thr
20 25 30

Ser Arg Ser Met Phe Val Tyr Val Ile Lys Met Ile Ile Leu Trp Leu
35 40 45

Met Trp Pro Leu Thr Ile Ile Leu Thr Ile Phe Asn Cys Val Tyr Ala
50 55 60

Leu Asn Asn Val Tyr Leu Gly Leu Ser Ile Val Phe Thr Ile Val Ala
65 70 75 80

Ile Ile Met Trp Ile Val Tyr Phe Val Asn Ser Ile Arg Leu Phe Ile
85 90 95

Arg Thr Gly Ser Phe Trp Ser Phe Asn Pro Glu Thr Asn Asn Leu Met
100 105 110

Cys Ile Asp Met Lys Gly Thr Met Tyr Val Arg Pro Ile Ile Glu Asp
115 120 125

Tyr His Thr Leu Thr Val Thr Ile Ile Arg Gly His Leu Tyr Ile Gln
130 135 140

Gly Ile Lys Leu Gly Thr Gly Tyr Ser Leu Ala Asp Leu Pro Ala Tyr
145 150 155 160

Met Thr Val Ala Lys Val Thr His Leu Cys Thr Tyr Lys Arg Gly Phe
165 170 175

Leu Asp Arg Ile Ser Asp Thr Ser Gly Phe Ala Val Tyr Val Lys Ser
180 185 190

Lys Val Gly Asn Tyr Arg Leu Pro Ser Thr Gln Lys Gly Ser Gly Met
195 200 205

Asp Thr Ala Leu Leu Asn Asn Ala Phe Leu Ala Phe Ser Ile Val Phe
210 215 220

Thr Ile Ile Ser Ile Val Ile Trp Ile Leu Tyr Phe Lys Val Gln Val
225 230 235 240

Leu Cys Thr Tyr Lys Arg Ala Phe Leu Asp Lys Leu Asp Val Asn Ser
245 250 255

Gly Phe Ala Val Phe Val Lys
260

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<210> SEQ ID NO 237

<211> LENGTH: 25

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 237

Ser Trp Phe Ser Gly Ile Thr Gln Phe Gln Lys Gly Lys Glu Phe Glu
1 5 10 15

Phe Val Glu Gly Gln Gly Val Pro Ile
20 25

<210> SEQ ID NO 238

<211> LENGTH: 33

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 238

Gly Tyr Trp Tyr Arg His Asn Arg Arg Ser Phe Lys Thr Ala Asp Gly
1 5 10 15

Asn Gln Arg Gln Leu Leu Pro Arg Trp Tyr Phe Tyr Tyr Leu Gly Thr
20 25 30

Gly

<210> SEQ ID NO 239

<211> LENGTH: 33

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 239

Arg Trp Tyr Phe Tyr Tyr Leu Gly Thr Gly Pro His Ala Lys Asp Gln
1 5 10 15

Tyr Gly Asp Ile Asp Gly Val Tyr Trp Val Ala Ser Asn Gln Ala Val
20 25 30

Asp

<210> SEQ ID NO 240

<211> LENGTH: 32

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 240

Tyr Trp Val Ala Ser Asn Gln Ala Asp Val Asn Thr Pro Ala Asp Ile
1 5 10 15

Val Asp Arg Asp Pro Ser Ser Asp Glu Ala Ile Pro Thr Arg Phe Pro
20 25 30

<210> SEQ ID NO 241

<211> LENGTH: 33

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 241

Ser Asp Glu Ala Ile Pro Thr Arg Phe Pro Pro Gly Thr Val Leu Pro
1 5 10 15

Gln Gly Tyr Tyr Ile Glu Gly Ser Gly Arg Ser Ala Pro Asn Ser Arg
20 25 30

Ser

-continued

<210> SEQ ID NO 242
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 242

Ser Gly Arg Ser Ala Pro Asn Ser Arg Ser Thr Ser Arg Thr Ser Ser
1 5 10 15

Arg Ala Ser Ser Ala Gly Ser Arg Ser Arg Ala Asn Ser Gly Asn Arg
20 25 30

Thr

<210> SEQ ID NO 243
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 243

Arg Ser Arg Ala Asn Ser Gly Asn Arg Thr Pro Thr Ser Gly Val Thr
1 5 10 15

Pro Asp Met Ala Asp Gln Ile Ala Ser Leu Val Leu Ala Lys Leu Gly
20 25 30

<210> SEQ ID NO 244
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 244

Ile Ala Ser Leu Val Leu Ala Lys Leu Gly Lys Asp Ala Thr Lys Pro
1 5 10 15

Gln Gln Val Thr Lys His Thr Ala Lys Glu Val Arg Gln Lys
20 25 30

<210> SEQ ID NO 245
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 245

Pro Arg Gln Lys Arg Ser Pro Asn Lys Gln Cys Thr Val Gln Gln Cys
1 5 10 15

Phe Gly Lys Arg Gly Pro Asn Gln Asn Phe Gly Gly Gly Glu Met Leu
20 25 30

Lys

<210> SEQ ID NO 246
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 246

Gln Asn Phe Gly Gly Gly Glu Met Leu Lys Leu Gly Thr Ser Asp Pro
1 5 10 15

Gln Phe Pro Ile Leu Ala Glu Leu Ala Pro Thr Ala Gly Ala Phe Phe
20 25 30

Phe

-continued

<210> SEQ ID NO 247
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 247

Leu Ala Pro Thr Ala Gly Ala Phe Phe Phe Gly Ser Arg Leu Glu Leu
1 5 10 15

Ala Lys Val Gln Asn Leu Ser Gly Asn Glu Leu Arg Tyr Asn Gly Ala
20 25 30

Ile Arg

<210> SEQ ID NO 248
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 248

Glu Leu Arg Tyr Asn Gly Ala Ile Arg Phe Asp Ser Thr Leu Ser Gly
1 5 10 15

Phe Glu Thr Ile Met Lys Val Leu Asn Glu Asn Leu Asn Ala Tyr Gln
20 25 30

<210> SEQ ID NO 249
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Human coronavirus HKU11

<400> SEQUENCE: 249

Pro Tyr Ala Asn Ala Ser Tyr Gly Glu Ser Leu Glu Gly Val Phe Trp
1 5 10 15

Val Ala Asn His Gln Ala Asp Thr Ser Thr
20 25

<210> SEQ ID NO 250
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: Human coronavirus HKU11

<400> SEQUENCE: 250

Val Ala Asn His Gln Ala Asp Thr Ser Thr Pro Ser Asp Val Ser Ser
1 5 10 15

Arg Asp Pro Thr Thr Gln Glu Ala
20

<210> SEQ ID NO 251
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: Human coronavirus HKU11

<400> SEQUENCE: 251

Arg Pro Gly Ser Arg Ser Gln Ser Arg Gly Pro Asn Asn Arg Ser Leu
1 5 10 15

Ser Arg Ser Asn Ser Asn Phe Arg His Ser Asp Ser Ile Val Lys Pro
20 25 30

<210> SEQ ID NO 252

-continued

<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Human coronavirus HKU11

<400> SEQUENCE: 252

Ser Lys Leu Asp Leu Val Lys Arg Asp Ser Glu Ala Asp Ser Pro Val
1 5 10 15

Lys

<210> SEQ ID NO 253
<211> LENGTH: 46
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 253

Ser Trp Phe Ser Gly Ile Thr Gln Phe Gln Lys Gly Lys Glu Phe Glu
1 5 10 15

Phe Val Glu Gly Gln Gly Val Pro Ile Gly Tyr Trp Tyr Arg His Asn
20 25 30

Arg Arg Ser Phe Lys Thr Ala Asp Gly Asn Gln Arg Gln Leu
35 40 45

<210> SEQ ID NO 254
<211> LENGTH: 46
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 254

Lys Thr Ala Asp Gly Asn Gln Arg Gln Leu Leu Pro Arg Trp Tyr Phe
1 5 10 15

Tyr Tyr Leu Gly Thr Gly Pro His Ala Lys Asp Gln Tyr Gly Thr Asp
20 25 30

Ile Asp Gly Val Tyr Trp Val Ala Ser Asn Gln Ala Asp Val
35 40 45

<210> SEQ ID NO 255
<211> LENGTH: 46
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 255

Tyr Trp Val Ala Ser Asn Gln Ala Asp Val Asn Thr Pro Ala Asp Ile
1 5 10 15

Val Asp Arg Asp Pro Ser Ser Asp Glu Ala Ile Pro Thr Arg Phe Pro
20 25 30

Pro Gly Thr Val Leu Pro Gln Gly Tyr Tyr Ile Glu Gly Ser
35 40 45

<210> SEQ ID NO 256
<211> LENGTH: 46
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 256

Leu Pro Gln Gly Tyr Tyr Ile Glu Gly Ser Gly Arg Ser Ala Pro Asn
1 5 10 15

-continued

Ser	Arg	Ser	Thr	Ser	Arg	Thr	Ser	Arg	Thr	Ser	Ser	Arg	Ala	Ser	Ser
			20							25			30		

Ala	Gly	Ser	Arg	Ser	Arg	Ala	Asn	Ser	Gly	Asn	Arg	Thr	Pro
		35					40					45	

<210> SEQ ID NO 257
 <211> LENGTH: 46
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 257

Ser	Arg	Ala	Asn	Ser	Gly	Asn	Arg	Thr	Pro	Thr	Ser	Gly	Val	Thr	Pro
1			5					10					15		

Asp	Met	Ala	Asp	Gln	Ile	Ala	Ser	Leu	Val	Leu	Ala	Lys	Leu	Gly	Lys
		20					25					30			

Asp	Ala	Thr	Lys	Pro	Gln	Gln	Val	Thr	Lys	His	Thr	Ala	Lys
		35				40						45	

<210> SEQ ID NO 258
 <211> LENGTH: 46
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 258

Pro	Gln	Gln	Val	Thr	Lys	His	Thr	Ala	Lys	Glu	Val	Arg	Gln	Lys	Pro
1			5					10					15		

Arg	Gln	Lys	Arg	Ser	Pro	Asn	Lys	Gln	Cys	Thr	Val	Gln	Gln	Cys	Phe
		20					25					30			

Gly	Lys	Arg	Gly	Pro	Asn	Gln	Asn	Phe	Gly	Gly	Gly	Glu	Met
		35				40						45	

<210> SEQ ID NO 259
 <211> LENGTH: 46
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 259

Pro	Asn	Gln	Asn	Phe	Gly	Gly	Gly	Glu	Met	Leu	Lys	Leu	Gly	Thr	Ser
1			5					10					15		

Asp	Pro	Gln	Phe	Pro	Ile	Leu	Ala	Glu	Leu	Ala	Pro	Thr	Ala	Gly	Ala
		20					25					30			

Phe	Phe	Phe	Gly	Ser	Arg	Leu	Glu	Leu	Ala	Lys	Val	Gln	Asn
		35				40						45	

<210> SEQ ID NO 260
 <211> LENGTH: 46
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 260

Ser	Arg	Leu	Glu	Leu	Ala	Lys	Val	Gln	Asn	Leu	Ser	Gly	Asn	Glu	Leu
1			5					10					15		

Arg	Tyr	Asn	Gly	Ala	Ile	Arg	Phe	Asp	Ser	Thr	Leu	Ser	Gly	Phe	Glu
		20					25					30			

-continued

Thr Ile Met Lys Val Leu Asn Glu Asn Leu Asn Ala Tyr Gln
35 40 45

<210> SEQ ID NO 261
 <211> LENGTH: 40
 <212> TYPE: PRT
 <213> ORGANISM: Human coronavirus HKU11

<400> SEQUENCE: 261

Pro Tyr Ala Asn Ala Ser Tyr Gly Glu Ser Leu Glu Gly Val Phe Trp
1 5 10 15
 Val Ala Asn His Gln Ala Asp Thr Ser Thr Pro Ser Asp Val Ser Ser
20 25 30
 Arg Asp Pro Thr Thr Gln Glu Ala
35 40

<210> SEQ ID NO 262
 <211> LENGTH: 49
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 262

Arg Pro Gly Ser Arg Ser Gln Ser Arg Gly Pro Asn Asn Arg Ser Leu
1 5 10 15
 Ser Arg Ser Asn Ser Asn Phe Arg His Ser Asp Ser Ile Val Lys Pro
20 25 30
 Ser Lys Leu Asp Leu Val Lys Arg Asp Ser Glu Ala Asp Ser Pro Val
35 40 45
 Lys

<210> SEQ ID NO 263
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 263

Ser Trp Phe Ser Gly Ile Thr Gln Phe Gln Lys Gly Lys Glu Phe Glu
1 5 10 15
 Phe Val Glu Gly Gln Gly Val Pro Ile Gly Tyr Trp Tyr Arg His Asn
20 25 30
 Arg Arg Ser Phe Lys Thr Ala Asp Gly Asn Gln Arg Gln Leu Leu Pro
35 40 45
 Arg Trp Tyr Phe Tyr Tyr Leu Gly Thr Gly Pro His Ala Lys Asp Gln
50 55 60
 Tyr Gly Thr Asp Ile Asp Gly Val Tyr Trp Val Ala Ser Asn Gln Ala
65 70 75 80
 Asp Val Asn Thr Pro Ala Asp Ile Val Asp Arg Asp Pro Ser Ser Asp
85 90 95
 Glu Ala Ile Pro Thr Arg Phe Pro Pro Gly Thr Val Leu Pro Gln
100 105 110

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<210> SEQ ID NO 264
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 264

Thr Arg Phe Pro Pro Gly Thr Val Leu Pro Gln Gly Tyr Tyr Ile Glu
1 5 10 15
Gly Ser Gly Arg Ser Ala Pro Asn Ser Arg Ser Thr Ser Arg Thr Ser
20 25 30
Ser Arg Ala Ser Ser Ala Gly Ser Arg Ser Arg Ala Asn Ser Gly Asn
35 40 45
Arg Thr Pro Thr Ser Gly Val Thr Pro Asp Met Ala Asp Gln Ile Ala
50 55 60
Ser Leu Val Leu Ala Lys Leu Gly Lys Asp Ala Thr Lys Pro Gln Gln
65 70 75 80
Val Thr Lys His Thr Ala Lys Glu Val Arg Gln Lys Pro Arg Gln Lys
85 90 95
Arg Ser Pro Asn Lys Gln Cys Thr Val Gln
100 105

<210> SEQ ID NO 265
<211> LENGTH: 99
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 265

Lys Arg Ser Pro Asn Lys Gln Cys Thr Val Gln Gln Cys Phe Gly Lys
1 5 10 15
Arg Gly Pro Asn Gln Asn Phe Gly Gly Gly Glu Met Leu Lys Gly Thr
20 25 30
Ser Asp Pro Gln Phe Pro Ile Leu Ala Glu Leu Ala Pro Thr Ala Gly
35 40 45
Ala Phe Phe Phe Gly Ser Arg Leu Glu Leu Ala Lys Val Gln Asn Leu
50 55 60
Ser Gly Asn Glu Leu Arg Tyr Asn Gly Ala Ile Arg Phe Asp Ser Thr
65 70 75 80
Leu Ser Gly Phe Glu Thr Ile Met Lys Val Leu Asn Glu Asn Leu Asn
85 90 95
Ala Tyr Gln

<210> SEQ ID NO 266
<211> LENGTH: 89
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 266

Pro Tyr Ala Asn Ala Ser Tyr Gly Glu Ser Leu Glu Gly Val Phe Trp
1 5 10 15
Val Ala Asn His Gln Ala Asp Thr Ser Thr Pro Ser Asp Val Ser Ser
20 25 30

-continued

Arg	Asp	Pro	Thr	Thr	Gln	Glu	Ala	Arg	Pro	Gly	Ser	Arg	Ser	Gln	Ser
	35						40					45			
Arg	Gly	Pro	Asn	Asn	Arg	Ser	Leu	Ser	Arg	Ser	Asn	Ser	Asn	Phe	Arg
	50					55					60				
His	Ser	Asp	Ser	Ile	Val	Lys	Pro	Ser	Lys	Leu	Asp	Leu	Val	Lys	Arg
65					70					75					80
Asp	Ser	Glu	Ala	Asp	Ser	Pro	Val	Lys							
				85											

<210> SEQ ID NO 267

<211> LENGTH: 383

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 267

Ser	Trp	Phe	Ser	Gly	Ile	Thr	Gln	Phe	Gln	Lys	Gly	Lys	Glu	Phe	Glu
1			5						10					15	
Phe	Val	Glu	Gly	Gln	Gly	Val	Pro	Ile	Gly	Tyr	Trp	Tyr	Arg	His	Asn
	20						25						30		
Arg	Arg	Ser	Phe	Lys	Thr	Ala	Asp	Gly	Asn	Gln	Arg	Gln	Leu	Leu	Pro
	35						40					45			
Arg	Trp	Tyr	Phe	Tyr	Tyr	Leu	Gly	Thr	Gly	Pro	His	Ala	Lys	Asp	Gln
	50					55					60				
Tyr	Gly	Asp	Ile	Asp	Gly	Val	Tyr	Trp	Val	Ala	Ser	Asn	Gln	Ala	Asp
65				70					75						80
Val	Asn	Thr	Pro	Ala	Asp	Ile	Val	Asp	Arg	Asp	Pro	Ser	Ser	Asp	Glu
			85						90					95	
Ala	Ile	Pro	Thr	Arg	Phe	Pro	Pro	Gly	Thr	Val	Leu	Pro	Gln	Gly	Tyr
	100							105					110		
Tyr	Ile	Glu	Gly	Ser	Gly	Arg	Ser	Ala	Pro	Asn	Ser	Arg	Ser	Thr	Ser
	115					120						125			
Arg	Thr	Ser	Ser	Arg	Ala	Ser	Ser	Ala	Gly	Ser	Arg	Ser	Arg	Ala	Asn
	130					135					140				
Ser	Gly	Asn	Arg	Thr	Pro	Thr	Ser	Gly	Val	Thr	Pro	Asp	Met	Ala	Asp
145					150					155					160
Gln	Ile	Ala	Ser	Leu	Val	Leu	Ala	Lys	Leu	Gly	Lys	Asp	Ala	Thr	Lys
			165						170					175	
Pro	Gln	Gln	Val	Thr	Lys	His	Thr	Ala	Lys	Glu	Val	Arg	Gln	Lys	Pro
	180							185					190		
Arg	Gln	Lys	Arg	Ser	Pro	Asn	Lys	Gln	Cys	Thr	Val	Gln	Gln	Cys	Phe
	195					200						205			
Gly	Lys	Arg	Gly	Pro	Asn	Gln	Asn	Phe	Gly	Gly	Gly	Glu	Met	Leu	Lys
	210				215						220				
Leu	Gly	Thr	Ser	Asp	Pro	Gln	Phe	Pro	Ile	Leu	Ala	Glu	Leu	Ala	Pro
225					230					235					240
Thr	Ala	Gly	Ala	Phe	Phe	Phe	Gly	Ser	Arg	Leu	Glu	Leu	Ala	Lys	Val
			245						250					255	
Gln	Asn	Leu	Ser	Gly	Asn	Glu	Leu	Arg	Tyr	Asn	Gly	Ala	Ile	Arg	Phe
	260							265					270		
Asp	Ser	Thr	Leu	Ser	Gly	Phe	Glu	Thr	Ile	Met	Lys	Val	Leu	Asn	Glu
	275						280					285			

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Asn Leu Asn Ala Tyr Gln Pro Tyr Ala Asn Ala Ser Tyr Gly Glu Ser
 290 295 300
 Leu Glu Gly Val Phe Trp Val Ala Asn His Gln Ala Asp Thr Ser Thr
 305 310 315 320
 Pro Ser Asp Val Ser Ser Arg Asp Pro Thr Thr Gln Glu Ala Arg Pro
 325 330 335
 Gly Ser Arg Ser Gln Ser Arg Gly Pro Asn Asn Arg Ser Leu Ser Arg
 340 345 350
 Ser Asn Ser Asn Phe Arg His Ser Asp Ser Ile Val Lys Pro Ser Lys
 355 360 365
 Leu Asp Leu Val Lys Arg Asp Ser Glu Ala Asp Ser Pro Val Lys
 370 375 380

<210> SEQ ID NO 268
 <211> LENGTH: 32
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 268

Ser Lys Ala Ser Ser Arg Ser Ala Ile Glu Asp Leu Leu Phe Asp Lys
 1 5 10 15
 Val Lys Leu Ser Asp Val Gly Phe Val Glu Ala Tyr Asn Asn Cys Thr
 20 25 30

<210> SEQ ID NO 269
 <211> LENGTH: 32
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 269

Gly Phe Val Glu Ala Tyr Asn Asn Cys Thr Gly Gly Ala Glu Ile Arg
 1 5 10 15
 Asp Leu Ile Cys Val Gln Ser Tyr Lys Gly Ile Lys Val Leu Pro Pro
 20 25 30

<210> SEQ ID NO 270
 <211> LENGTH: 32
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 270

Ser Tyr Lys Gly Ile Lys Val Leu Pro Pro Leu Leu Ser Glu Asn Gln
 1 5 10 15
 Ile Ser Gly Tyr Thr Leu Ala Ala Thr Ser Ala Ser Leu Phe Pro Pro
 20 25 30

<210> SEQ ID NO 271
 <211> LENGTH: 32
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 271

Ala Ala Thr Ser Ala Ser Leu Phe Pro Pro Trp Thr Ala Ala Ala Gly
 1 5 10 15
 Val Pro Phe Tyr Leu Asn Val Gln Tyr Arg Ile Asn Gly Leu Gly Val
 20 25 30

-continued

<210> SEQ ID NO 272
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 272

Val Gln Tyr Arg Ile Asn Gly Leu Gly Val Thr Met Asp Val Leu Ser
1 5 10 15

Gln Asn Gln Lys Leu Ile Ala Asn Ala Phe Asn Asn Ala Leu Tyr Ala
20 25 30

<210> SEQ ID NO 273
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 273

Ala Asn Ala Phe Asn Asn Ala Leu Tyr Ala Ile Gln Glu Gly Phe Asp
1 5 10 15

Ala Thr Asn Ser Ala Leu Val Lys Ile Gln Ala Val Val Asn Ala Asn
20 25 30

<210> SEQ ID NO 274
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 274

Val Lys Ile Gln Ala Val Val Asn Ala Asn Ala Glu Ala Leu Asn Asn
1 5 10 15

Leu Leu Gln Gln Leu Ser Asn Arg Phe Gly Ala Ile Ser Ala Ser Leu
20 25 30

<210> SEQ ID NO 275
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 275

Asn Arg Phe Gly Ala Ile Ser Ala Ser Leu Gln Glu Ile Leu Ser Arg
1 5 10 15

Leu Asp Ala Leu Glu Ala Glu Ala Gln Ile Asp Arg Leu Ile Asn Gly
20 25 30

Arg

<210> SEQ ID NO 276
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 276

Ala Gln Ile Asp Arg Leu Ile Asn Gly Arg Leu Thr Ala Leu Asn Ala
1 5 10 15

Tyr Val Ser Gln Gln Leu Ser Asp Ser Thr Leu Val Lys Phe Ser Ala
20 25 30

Ala

<210> SEQ ID NO 277
<211> LENGTH: 33

-continued

<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 277

Asp Ser Thr Leu Val Lys Phe Ser Ala Ala Gln Ala Met Glu Lys Val
1 5 10 15

Asn Glu Cys Val Lys Ser Gln Ser Ser Arg Ile Asn Phe Cys Gly Asn
20 25 30

Gly

<210> SEQ ID NO 278
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 278

Gln Ser Ser Arg Ile Asn Phe Cys Gly Asn Gly Asn His Ile Ile Ser
1 5 10 15

Leu Val Gln Asn Ala Pro Tyr Gly Leu Tyr Phe Ile His Phe Ser Tyr
20 25 30

Val Pro

<210> SEQ ID NO 279
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 279

Pro Asn Leu Pro Asp Phe Lys Glu Glu Leu Asp Gln Trp Phe Lys Asn
1 5 10 15

Gln Thr Ser Val Ala Pro Asp Leu Ser Leu Asp Tyr Ile Asn Val Thr
20 25 30

<210> SEQ ID NO 280
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 280

Asp Leu Ser Leu Asp Tyr Ile Asn Val Thr Phe Leu Asp Leu Gln Val
1 5 10 15

Glu Met Asn Arg Leu Gln Glu Ala Ile Lys Val Leu Asn Gln Ser Tyr
20 25 30

<210> SEQ ID NO 281
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 281

Glu Ala Ile Lys Val Leu Asn Gln Ser Tyr Ile Asn Leu Lys Asp Ile
1 5 10 15

Gly Thr Tyr Glu Tyr Tyr Val Lys Trp Pro Trp Tyr Val Trp Leu
20 25 30

<210> SEQ ID NO 282
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Human coronavirus HKU11

-continued

<400> SEQUENCE: 282

Pro Lys Leu Ser Asp Phe Glu Ser Glu Leu Ser His Trp Phe Lys Asn
1 5 10 15

Gln Thr Ser Ile Ala Pro Asn Leu Thr Leu Asn Leu His Thr
20 25 30

<210> SEQ ID NO 283

<211> LENGTH: 32

<212> TYPE: PRT

<213> ORGANISM: Human coronavirus HKU11

<400> SEQUENCE: 283

Ala Pro Asn Leu Thr Leu Asn Leu His Thr Ile Asn Ala Thr Phe Leu
1 5 10 15

Asp Leu Tyr Tyr Glu Met Asn Leu Ile Gln Glu Ser Ile Lys Ser Leu
20 25 30

<210> SEQ ID NO 284

<211> LENGTH: 45

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 284

Ser Lys Ala Ser Ser Arg Ser Ala Ile Glu Asp Leu Leu Phe Asp Lys
1 5 10 15

Val Lys Leu Ser Asp Val Gly Phe Val Glu Ala Tyr Asn Asn Cys Thr
20 25 30

Gly Gly Ala Glu Ile Arg Asp Leu Ile Cys Val Gln Ser
35 40 45

<210> SEQ ID NO 285

<211> LENGTH: 46

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 285

Glu Ile Arg Asp Leu Ile Cys Val Gln Ser Tyr Lys Gly Ile Lys Val
1 5 10 15

Leu Pro Pro Leu Leu Ser Glu Asn Gln Ile Ser Gly Tyr Thr Leu Ala
20 25 30

Ala Thr Ser Ala Ser Leu Phe Pro Pro Trp Thr Ala Ala Ala
35 40 45

<210> SEQ ID NO 286

<211> LENGTH: 45

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 286

Ser Leu Phe Pro Pro Trp Thr Ala Ala Ala Gly Val Pro Phe Tyr Leu
1 5 10 15

Asn Val Gln Tyr Arg Ile Asn Gly Leu Gly Val Thr Met Asp Val Leu
20 25 30

Ser Gln Asn Gln Lys Leu Ile Ala Asn Ala Phe Asn Asn
35 40 45

<210> SEQ ID NO 287

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<211> LENGTH: 45
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 287

Gln Lys Leu Ile Ala Asn Ala Phe Asn Asn Ala Leu Tyr Ala Ile Gln
1 5 10 15

Glu Gly Phe Asp Ala Thr Asn Ser Ala Leu Val Lys Ile Gln Ala Val
20 25 30

Val Asn Ala Asn Ala Glu Ala Leu Asn Asn Leu Leu Gln
35 40 45

<210> SEQ ID NO 288
<211> LENGTH: 45
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 288

Asn Ala Glu Ala Leu Asn Asn Leu Leu Gln Gln Leu Ser Asn Arg Phe
1 5 10 15

Gly Ala Ile Ser Ala Ser Leu Gln Glu Ile Leu Ser Arg Leu Asp Ala
20 25 30

Leu Glu Ala Glu Ala Gln Ile Asp Arg Leu Ile Asn Gly
35 40 45

<210> SEQ ID NO 289
<211> LENGTH: 45
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 289

Glu Ala Gln Ile Asp Arg Leu Ile Asn Gly Arg Leu Thr Ala Leu Asn
1 5 10 15

Ala Tyr Val Ser Gln Gln Leu Ser Asp Ser Thr Leu Val Lys Phe Ser
20 25 30

Ala Ala Gln Ala Met Glu Lys Val Asn Glu Cys Lys Ser
35 40 45

<210> SEQ ID NO 290
<211> LENGTH: 45
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 290

Ala Met Glu Lys Val Asn Glu Cys Val Lys Ser Gln Ser Ser Arg Ile
1 5 10 15

Asn Phe Cys Gly Asn Gly Asn His Ile Ile Ser Leu Val Gln Asn Ala
20 25 30

Pro Tyr Gly Leu Tyr Phe Ile His Phe Ser Tyr Val Pro
35 40 45

<210> SEQ ID NO 291
<211> LENGTH: 42
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 291

Pro Asn Leu Pro Asp Phe Lys Glu Glu Leu Asp Gln Trp Phe Lys Asn
1 5 10 15

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Gln Thr Ser Val Ala Pro Asp Leu Ser Leu Asp Tyr Ile Asn Val Thr
20 25 30

Phe Leu Asp Leu Gln Val Glu Met Asn Arg
35 40

<210> SEQ ID NO 292

<211> LENGTH: 43

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 292

Phe Leu Asp Leu Gln Val Glu Met Asn Arg Leu Gln Glu Ala Ile Lys
1 5 10 15

Val Leu Asn Gln Ser Tyr Ile Asn Leu Lys Asp Ile Gly Thr Tyr Glu
20 25 30

Tyr Tyr Val Lys Trp Pro Trp Tyr Val Trp Leu
35 40

<210> SEQ ID NO 293

<211> LENGTH: 47

<212> TYPE: PRT

<213> ORGANISM: Human coronavirus HKU11

<400> SEQUENCE: 293

Pro Lys Leu Ser Asp Phe Glu Ser Glu Leu Ser His Trp Phe Lys Asn
1 5 10 15

Gln Thr Ser Ile Ala Pro Asn Leu Thr Leu Asn Leu His Thr Ile Asn
20 25 30

Ala Thr Phe Leu Met Asn Leu Ile Gln Glu Ser Ile Lys Ser Leu
35 40 45

<210> SEQ ID NO 294

<211> LENGTH: 136

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 294

Ser Lys Ala Ser Ser Arg Ser Ala Ile Glu Asp Leu Leu Phe Asp Lys
1 5 10 15

Val Lys Leu Ser Asp Val Gly Phe Val Glu Ala Tyr Asn Asn Cys Thr
20 25 30

Gly Gly Ala Glu Ile Arg Asp Leu Ile Cys Val Gln Ser Tyr Lys Gly
35 40 45

Ile Lys Val Leu Pro Pro Leu Leu Ser Glu Asn Gln Ile Ser Gly Tyr
50 55 60

Thr Leu Ala Ala Thr Ser Ala Ser Leu Phe Pro Pro Trp Thr Ala Ala
65 70 75 80

Ala Gly Val Pro Phe Tyr Leu Asn Val Gln Tyr Arg Ile Asn Gly Leu
85 90 95

Gly Val Thr Met Asp Val Leu Ser Gln Asn Gln Lys Leu Ile Ala Asn
100 105 110

Ala Phe Asn Asn Ala Leu Tyr Ala Ile Gln Glu Gly Phe Asp Ala Thr
115 120 125

Asn Ser Ala Leu Val Lys Ile Gln
130 135

-continued

<210> SEQ ID NO 295

<211> LENGTH: 132

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 295

Phe Asp Ala Thr Asn Ser Ala Leu Val Lys Ile Gln Ala Val Val Asn
1 5 10 15

Ala Asn Ala Glu Ala Leu Asn Asn Leu Leu Gln Gln Leu Ser Asn Arg
20 25 30

Phe Gly Ala Ile Ser Ala Ser Leu Gln Glu Ile Leu Ser Arg Leu Asp
35 40 45

Ala Leu Glu Ala Glu Ala Gln Ile Asp Arg Leu Ile Asn Gly Arg Leu
50 55 60

Thr Ala Leu Asn Ala Tyr Val Ser Gln Gln Leu Ser Asp Ser Thr Leu
65 70 75 80

Val Lys Phe Ser Ala Ala Gln Ala Met Glu Lys Val Asn Glu Cys Val
85 90 95

Lys Ser Gln Ser Ser Arg Ile Asn Phe Cys Gly Asn Gly Asn His Ile
100 105 110

Ile Ser Leu Val Gln Asn Ala Pro Tyr Gly Leu Tyr Phe Ile His Phe
115 120 125

Ser Tyr Val Pro
130

<210> SEQ ID NO 296

<211> LENGTH: 120

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 296

Pro Asn Leu Pro Asp Phe Lys Glu Glu Leu Asp Gln Trp Phe Lys Asn
1 5 10 15

Gln Thr Ser Val Ala Pro Asp Leu Ser Asp Tyr Ile Asn Val Phe Leu
20 25 30

Asp Leu Gln Val Glu Met Asn Arg Leu Gln Glu Ala Ile Lys Val Leu
35 40 45

Asn Gln Ser Tyr Ile Asn Leu Lys Asp Ile Gly Thr Tyr Glu Tyr Tyr
50 55 60

Val Lys Trp Pro Trp Tyr Val Trp Leu Pro Lys Leu Ser Asp Phe Glu
65 70 75 80

Ser Glu Leu Ser His Trp Phe Lys Asn Gln Thr Ser Ile Ala Pro Asn
85 90 95

Leu Thr Leu Asn Leu His Thr Ile Asn Ala Thr Phe Leu Met Asn Leu
100 105 110

Ile Gln Glu Ser Ile Lys Ser Leu
115 120

<210> SEQ ID NO 297

<211> LENGTH: 32

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

-continued

<400> SEQUENCE: 297

Phe Met Arg Phe Tyr Ile Ile Ile Ala Ser Phe Ile Lys Leu Phe Ser
1 5 10 15
Leu Phe Arg His Val Ala Tyr Gly Cys Ser Lys Ser Gly Cys Leu Phe
20 25 30

<210> SEQ ID NO 298

<211> LENGTH: 32

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 298

Tyr Gly Cys Ser Lys Ser Gly Cys Leu Phe Tyr Lys Arg Asn Arg Ser
1 5 10 15
Leu Arg Val Lys Cys Ser Thr Ile Val Gly Gly Met Ile Arg Tyr Tyr
20 25 30

<210> SEQ ID NO 299

<211> LENGTH: 30

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 299

Thr Ile Val Gly Gly Met Ile Arg Tyr Tyr Asp Val Met Ala Asn Gly
1 5 10 15
Gly Thr Gly Phe Cys Ser Lys His Gln Trp Asn Cys Ile Asp
20 25 30

<210> SEQ ID NO 300

<211> LENGTH: 31

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 300

Cys Ser Lys His Gln Trp Asn Cys Ile Asp Cys Asp Ser Tyr Lys Pro
1 5 10 15
Gly Asn Thr Phe Ile Thr Val Glu Ala Ala Leu Asp Leu Ser Lys
20 25 30

<210> SEQ ID NO 301

<211> LENGTH: 32

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 301

Thr Val Glu Ala Ala Leu Asp Leu Ser Lys Glu Leu Lys Arg Pro Ile
1 5 10 15
Gln Pro Thr Asp Asn Ala Ala Val Phe Tyr Ala Gln Ser Leu Phe Arg
20 25 30

<210> SEQ ID NO 302

<211> LENGTH: 31

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 302

Ala Val Phe Tyr Ala Gln Ser Leu Phe Arg Pro Ile Leu Met Val Asp

-continued

1	5	10	15
Lys	Asn	Leu	Ile
Thr	Thr	Ala	Asn
Thr	Gly	Thr	Ser
Val	Thr	Glu	
20	25	30	

<210> SEQ ID NO 303
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 303

Thr	Ala	Asn	Thr	Gly	Thr	Ser	Val	Thr	Glu	Thr	Met	Phe	Asp	Val	Tyr
1			5					10						15	
Val	Asp	Thr	Phe	Leu	Ser	Met	Phe	Asp	Val	Asp	Lys	Lys	Ser	Leu	
	20					25							30		

<210> SEQ ID NO 304
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 304

Ser	Met	Phe	Asp	Val	Asp	Lys	Lys	Ser	Leu	Asn	Ala	Leu	Ile	Ala	Thr
1				5					10					15	
Ala	Glu	Leu	Thr	Asp	Glu	Ser	Cys	Asn	Asn	Leu	Val	Pro	Thr	Tyr	Leu
	20						25							30	

<210> SEQ ID NO 305
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 305

Ser	Cys	Asn	Asn	Leu	Val	Pro	Thr	Tyr	Leu	Lys	Ser	Asp	Asn	Ile	Val
1				5					10					15	
Ala	Ala	Asp	Leu	Gly	Val	Leu	Ile	Gln	Asn	Ser	Ala	Lys	His	Val	
	20						25						30		

<210> SEQ ID NO 306
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 306

Val	Leu	Ile	Gln	Asn	Ser	Ala	Lys	His	Val	Gln	Gly	Asn	Val	Ala	Lys
1				5					10					15	
Ile	Ala	Gly	Val	Ser	Cys	Ile	Trp	Ser	Val	Asp	Ala	Phe			
	20						25								

<210> SEQ ID NO 307
<211> LENGTH: 42
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 307

Phe	Met	Arg	Phe	Tyr	Ile	Ile	Ile	Ala	Ser	Phe	Ile	Lys	Leu	Phe	Ser
1				5					10					15	
Leu	Phe	Arg	His	Val	Ala	Tyr	Gly	Cys	Ser	Lys	Ser	Gly	Cys	Leu	Phe

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	20		25		30
Cys Tyr Lys Arg Asn Arg Ser Leu Arg Val					
	35		40		

<210> SEQ ID NO 308
 <211> LENGTH: 42
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 308

Cys Tyr Arg Asn Arg Ser Leu Arg Val Lys Cys Ser Thr Ile Val Gly					
1		5		10	15
Gly Met Ile Arg Tyr Tyr Asp Val Met Ala Asn Gly Gly Thr Gly Phe					
	20		25		30
Cys Ser Lys His Gln Trp Asn Cys Ile Asp					
	35		40		

<210> SEQ ID NO 309
 <211> LENGTH: 41
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 309

Cys Ser Lys His Gln Trp Asn Cys Ile Asp Cys Asp Ser Tyr Lys Pro					
1		5		10	15
Gly Asn Thr Phe Ile Thr Val Glu Ala Ala Leu Asp Leu Ser Lys Glu					
	20		25		30
Leu Lys Arg Pro Ile Gln Pro Thr Asp					
	35		40		

<210> SEQ ID NO 310
 <211> LENGTH: 44
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 310

Asn Ala Ala Val Phe Tyr Ala Gln Ser Leu Phe Arg Pro Ile Leu Met					
1		5		10	15
Val Asp Lys Asn Leu Ile Thr Thr Ala Asn Thr Gly Thr Ser Val Thr					
	20		25		30
Glu Thr Met Phe Asp Val Tyr Val Asp Thr Phe Leu					
	35		40		

<210> SEQ ID NO 311
 <211> LENGTH: 50
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 311

Met Phe Asp Val Tyr Val Asp Thr Phe Leu Ser Met Phe Asp Val Asp					
1		5		10	15
Lys Lys Ser Leu Asn Ala Leu Ile Ala Thr Ala Glu Leu Thr Asp Glu					
	20		25		30
Ser Cys Asn Asn Leu Val Pro Thr Tyr Leu Lys Ser Asp Asn Ile Val					
	35		40		45

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Ala Ala
50

<210> SEQ ID NO 312
 <211> LENGTH: 42
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 312

Tyr Leu Lys Ser Asp Asn Ile Val Ala Ala Asp Leu Gly Val Leu Ile
 1 5 10 15
 Gln Asn Ser Ala Lys His Val Gln Gly Asn Val Ala Lys Ile Ala Gly
 20 25 30
 Val Ser Cys Ile Trp Ser Val Asp Ala Phe
 35 40

<210> SEQ ID NO 313
 <211> LENGTH: 106
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 313

Phe Met Arg Phe Tyr Ile Ile Ile Ala Ser Phe Ile Lys Leu Phe Ser
 1 5 10 15
 Leu Phe Arg His Val Ala Tyr Gly Cys Ser Lys Ser Gly Cys Leu Phe
 20 25 30
 Cys Tyr Lys Arg Asn Arg Ser Leu Arg Val Lys Cys Ser Thr Ile Val
 35 40 45
 Gly Gly Met Ile Arg Tyr Tyr Asp Val Met Ala Asn Gly Gly Thr Gly
 50 55 60
 Phe Cys Ser Lys His Gln Trp Asn Cys Ile Asp Cys Asp Ser Tyr Lys
 65 70 75 80
 Pro Gly Asn Thr Phe Ile Thr Val Glu Ala Ala Leu Asp Leu Ser Lys
 85 90 95
 Glu Leu Lys Arg Pro Ile Gln Pro Thr Asp
 100 105

<210> SEQ ID NO 314
 <211> LENGTH: 116
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 314

Asn Ala Ala Val Phe Tyr Ala Gln Ser Leu Phe Arg Pro Ile Leu Met
 1 5 10 15
 Val Asp Lys Asn Leu Ile Thr Thr Ala Asn Thr Gly Thr Ser Val Thr
 20 25 30
 Glu Thr Met Phe Asp Val Tyr Val Asp Thr Phe Leu Ser Met Phe Asp
 35 40 45
 Val Asp Lys Lys Ser Leu Asn Ala Leu Ile Ala Thr Ala Glu Leu Thr
 50 55 60
 Asp Glu Ser Cys Asn Asn Leu Val Pro Thr Tyr Leu Lys Ser Asp Asn
 65 70 75 80
 Ile Val Ala Ala Asp Leu Gly Val Leu Ile Gln Asn Ser Ala Lys His

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85	90	95
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Val Gln Gly Asn Val Ala Lys Ile Ala Gly Val Ser Cys Ile Trp Ser
 100 105 110

Val Asp Ala Phe
 115

<210> SEQ ID NO 315
 <211> LENGTH: 222
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 315

Phe Met Arg Phe Tyr Ile Ile Ile Ala Ser Phe Ile Lys Leu Phe Ser	
1 5 10 15	
Leu Phe Arg His Val Ala Tyr Gly Cys Ser Lys Ser Gly Cys Leu Phe	
20 25 30	
Cys Tyr Lys Arg Asn Arg Ser Leu Arg Val Lys Cys Ser Thr Ile Val	
35 40 45	
Gly Gly Met Ile Arg Tyr Tyr Asp Val Met Ala Asn Gly Gly Thr Gly	
50 55 60	
Phe Cys Ser Lys His Gln Trp Asn Cys Ile Asp Cys Asp Ser Tyr Lys	
65 70 75 80	
Pro Gly Asn Thr Phe Ile Thr Val Glu Ala Ala Leu Asp Leu Ser Lys	
85 90 95	
Glu Leu Lys Arg Pro Ile Gln Pro Thr Asp Asn Ala Ala Val Phe Tyr	
100 105 110	
Ala Gln Ser Leu Phe Arg Pro Ile Leu Met Val Asp Lys Asn Leu Ile	
115 120 125	
Thr Thr Ala Asn Thr Gly Thr Ser Val Thr Glu Thr Met Phe Asp Val	
130 135 140	
Tyr Val Asp Thr Phe Leu Ser Met Phe Asp Val Asp Lys Lys Ser Leu	
145 150 155 160	
Asn Ala Leu Ile Ala Thr Ala Glu Leu Thr Asp Glu Ser Cys Asn Asn	
165 170 175	
Leu Val Pro Thr Tyr Leu Lys Ser Asp Asn Ile Val Ala Ala Asp Leu	
180 185 190	
Gly Val Leu Ile Gln Asn Ser Ala Lys His Val Gln Gly Asn Val Ala	
195 200 205	
Lys Ile Ala Gly Val Ser Cys Ile Trp Ser Val Asp Ala Phe	
210 215 220	

<210> SEQ ID NO 316
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 316

Ser Ala Asp Gly Val Gln Cys Tyr Thr Pro His Ser Gln Ile Ser Tyr	
1 5 10 15	
Ser Asn Phe Tyr Ala Ser Gly Cys Val Leu Ser Ser	
20 25	

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<210> SEQ ID NO 317

<211> LENGTH: 26

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 317

Phe Tyr Ala Ser Gly Cys Val Leu Ser Ser Ala Cys Thr Met Phe Met
1 5 10 15

Ala Asp Gly Ser Pro Gln Pro Tyr Cys Tyr
20 25

<210> SEQ ID NO 318

<211> LENGTH: 26

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 318

Ser Leu Val Pro His Val Arg Tyr Asn Leu Ala Asn Ala Lys Gly Phe
1 5 10 15

Ile Arg Phe Pro Glu Val Leu Arg Glu Gly
20 25

<210> SEQ ID NO 319

<211> LENGTH: 26

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 319

Ile Arg Phe Pro Glu Val Leu Arg Glu Gly Leu Val Arg Ile Val Arg
1 5 10 15

Thr Arg Ser Met Ser Tyr Cys Arg Val Gly
20 25

<210> SEQ ID NO 320

<211> LENGTH: 25

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 320

Thr Arg Ser Met Ser Tyr Cys Arg Val Gly Leu Cys Glu Glu Ala Asp
1 5 10 15

Glu Gly Ile Cys Phe Asn Phe Asn Gly
20 25

<210> SEQ ID NO 321

<211> LENGTH: 25

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 321

Asp Glu Gly Ile Cys Phe Asn Phe Asn Gly Ser Trp Val Leu Asn Asn
1 5 10 15

Asp Tyr Tyr Arg Ser Leu Pro Gly Thr
20 25

<210> SEQ ID NO 322

<211> LENGTH: 25

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 322

-continued

Asn Asp Tyr Tyr Arg Ser Leu Pro Gly Thr Phe Cys Gly Arg Asp Val
1 5 10 15

Phe Asp Leu Ile Tyr Gln Leu Phe Lys
20 25

<210> SEQ ID NO 323
<211> LENGTH: 25
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 323

Val Phe Asp Leu Ile Tyr Gln Leu Phe Lys Gly Leu Ala Gln Pro Val
1 5 10 15

Asp Phe Leu Ala Leu Thr Ala Ser Ser
20 25

<210> SEQ ID NO 324
<211> LENGTH: 25
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 324

Val Asp Phe Leu Ala Leu Thr Ala Ser Ser Ile Ala Gly Ala Ile Leu
1 5 10 15

Ala Val Ile Val Val Leu Val Phe Tyr
20 25

<210> SEQ ID NO 325
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 325

Leu Ala Val Ile Val Val Leu Val Phe Tyr Tyr Leu Ile Lys Leu Lys
1 5 10 15

Arg Ala Phe Gly Asp Tyr Thr Ser Val Val
20 25

<210> SEQ ID NO 326
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 326

Asn Arg Tyr Leu Ser Leu Tyr Asn Lys Tyr Arg Tyr Tyr Ser Gly Lys
1 5 10 15

Met Asp Thr Ala Ala Tyr Arg Glu Ala Ala Cys Ser Gln Leu
20 25 30

<210> SEQ ID NO 327
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 327

Ala Tyr Arg Glu Ala Ala Cys Ser Gln Leu Ala Lys Ala Met Asp Thr
1 5 10 15

Phe Thr Asn Asn Asn Gly Ser Asp Val Leu Tyr Gln Pro Pro
20 25 30

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<210> SEQ ID NO 328

<211> LENGTH: 45

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 328

Ser Ala Asp Gly Val Gln Cys Tyr Thr Pro His Ser Gln Ile Ser Tyr
1 5 10 15

Ser Asn Phe Tyr Ala Ser Gly Cys Val Leu Ser Ser Ala Cys Thr Met
20 25 30

Phe Thr Met Ala Asp Gly Ser Pro Gln Pro Tyr Cys Tyr
35 40 45

<210> SEQ ID NO 329

<211> LENGTH: 46

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 329

Ser Leu Val Pro His Val Arg Tyr Asn Leu Ala Asn Ala Lys Gly Phe
1 5 10 15

Ile Arg Phe Pro Glu Val Leu Arg Glu Gly Leu Val Arg Ile Val Arg
20 25 30

Thr Arg Ser Met Ser Tyr Cys Arg Val Gly Leu Cys Glu Glu
35 40 45

<210> SEQ ID NO 330

<211> LENGTH: 44

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 330

Ser Tyr Cys Arg Val Gly Leu Cys Glu Glu Ala Asp Glu Gly Ile Cys
1 5 10 15

Phe Asn Phe Asn Gly Ser Trp Val Leu Asn Asn Asp Tyr Tyr Arg Ser
20 25 30

Leu Pro Gly Thr Phe Cys Gly Arg Asp Val Phe Asp
35 40

<210> SEQ ID NO 331

<211> LENGTH: 45

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 331

Gly Thr Phe Cys Gly Arg Asp Val Phe Asp Leu Ile Tyr Gln Leu Phe
1 5 10 15

Lys Gly Leu Ala Gln Pro Val Asp Phe Leu Ala Leu Thr Ala Ser Ser
20 25 30

Ile Ala Gly Ala Ile Leu Ala Val Ile Val Val Leu Val
35 40 45

<210> SEQ ID NO 332

<211> LENGTH: 44

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

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polypeptide

<400> SEQUENCE: 332

Ile Leu Ala Val Ile Val Val Leu Val Phe Tyr Tyr Leu Ile Lys Leu
1 5 10 15

Lys Arg Ala Phe Gly Asp Tyr Thr Ser Val Val Asn Arg Tyr Leu Ser
20 25 30

Leu Tyr Asn Lys Tyr Arg Tyr Tyr Ser Gly Lys Met
35 40

<210> SEQ ID NO 333
<211> LENGTH: 42
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 333

Lys Tyr Arg Tyr Tyr Ser Gly Lys Met Asp Thr Ala Ala Tyr Arg Glu
1 5 10 15

Ala Ala Cys Ser Gln Leu Ala Lys Ala Met Asp Thr Phe Thr Asn Asn
20 25 30

Asn Gly Ser Asp Val Leu Tyr Gln Pro Pro
35 40

<210> SEQ ID NO 334
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 334

Ser Ala Asp Gly Val Gln Cys Tyr Thr Pro His Ser Gln Ile Ser Tyr
1 5 10 15

Ser Asn Phe Tyr Ala Ser Gly Cys Val Leu Ser Ser Ala Cys Thr Met
20 25 30

Phe Thr Met Ala Asp Gly Ser Pro Gln Pro Tyr Cys Tyr Ser Leu Val
35 40 45

Pro His Val Arg Tyr Asn Leu Ala Asn Ala Lys Gly Phe Ile Arg Phe
50 55 60

Pro Glu Val Leu Arg Glu Gly Leu Val Arg Ile Val Arg Thr Arg Ser
65 70 75 80

Met Ser Tyr Cys Arg Val Gly Leu Cys Glu Glu Ala Asp Glu Gly Ile
85 90 95

Cys Phe Asn Phe Asn Gly Ser Trp Val Leu Asn Asn Asp Tyr Tyr Arg
100 105 110

Ser Leu Pro Gly Thr Phe Cys Gly
115 120

<210> SEQ ID NO 335
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 335

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Tyr Tyr Arg Ser Leu Pro Gly Thr Phe Cys Gly Arg Asp Val Phe Asp
1      5      10      15
Leu Ile Tyr Gln Leu Phe Lys Gly Leu Ala Gln Pro Val Asp Phe Leu
20      25      30
Ala Leu Thr Ala Ser Ser Ile Ala Gly Ala Ile Leu Ala Val Ile Val
35      40      45
Val Leu Val Phe Tyr Tyr Leu Ile Lys Leu Lys Arg Ala Phe Gly Asp
50      55      60
Tyr Thr Ser Val Val Asn Arg Tyr Leu Ser Leu Tyr Asn Lys Tyr Arg
65      70      75      80
Tyr Tyr Ser Gly Lys Met Asp Thr Ala Ala Tyr Arg Glu Ala Ala Cys
85      90      95
Ser Gln Leu Ala Lys Ala Met Asp Thr Phe Thr Asn Asn Asn Gly Ser
100     105     110
Asp Val Leu Tyr Gln Pro Pro
115

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<210> SEQ ID NO 336
<211> LENGTH: 228
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 336

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Ser Ala Asp Gly Val Gln Cys Tyr Thr Pro His Ser Gln Ile Ser Tyr
1      5      10      15
Ser Asn Phe Tyr Ala Ser Gly Cys Val Leu Ser Ser Ala Cys Thr Met
20      25      30
Phe Thr Met Ala Asp Gly Ser Pro Gln Pro Tyr Cys Tyr Ser Leu Val
35      40      45
Pro His Val Arg Tyr Asn Leu Ala Asn Ala Lys Gly Phe Ile Arg Phe
50      55      60
Pro Glu Val Leu Arg Glu Gly Leu Val Arg Ile Val Arg Thr Arg Ser
65      70      75      80
Met Ser Tyr Cys Arg Val Gly Leu Cys Glu Glu Ala Asp Glu Gly Ile
85      90      95
Cys Phe Asn Phe Asn Gly Ser Trp Val Leu Asn Asn Asp Tyr Tyr Arg
100     105     110
Ser Leu Pro Gly Thr Phe Cys Gly Arg Asp Val Phe Asp Leu Ile Tyr
115     120     125
Gln Leu Phe Lys Gly Leu Ala Gln Pro Val Asp Phe Leu Ala Leu Thr
130     135     140
Ala Ser Ser Ile Ala Gly Ala Ile Leu Ala Val Ile Val Val Leu Val
145     150     155     160
Phe Tyr Tyr Leu Ile Lys Leu Lys Arg Ala Phe Gly Asp Tyr Thr Ser
165     170     175
Val Val Asn Arg Tyr Leu Ser Leu Tyr Asn Lys Tyr Arg Tyr Tyr Ser
180     185     190
Gly Lys Met Asp Thr Ala Ala Tyr Arg Glu Ala Ala Cys Ser Gln Leu
195     200     205
Ala Lys Ala Met Asp Thr Phe Thr Asn Asn Asn Gly Ser Asp Val Leu
210     215     220

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Tyr Gln Pro Pro
225

<210> SEQ ID NO 337
<211> LENGTH: 21
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 337

Ser Leu Ala Met Leu Leu Val Lys His Lys His Leu Tyr Leu Thr Met
1 5 10 15

Tyr Ile Thr Pro Val
20

<210> SEQ ID NO 338
<211> LENGTH: 21
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 338

Leu Tyr Leu Thr Met Tyr Ile Thr Pro Val Leu Phe Thr Leu Leu Tyr
1 5 10 15

Asn Asn Tyr Leu Val
20

<210> SEQ ID NO 339
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 339

Ile Lys Ile Val Leu Leu Cys Tyr Leu Phe Ile Gly Tyr Ile Ile Ser
1 5 10 15

Cys Tyr Trp Gly Leu Phe Ser Leu Met Asn Ser Leu
20 25

<210> SEQ ID NO 340
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 340

Trp Gly Leu Phe Ser Leu Met Asn Ser Leu Phe Arg Met Pro Leu Gly
1 5 10 15

Val Tyr Asn Tyr Lys Ile Ser Val Gln Glu Leu Arg Tyr
20 25

<210> SEQ ID NO 341
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 341

Tyr Lys Ile Ser Val Gln Glu Leu Arg Tyr Met Asn Ala Asn Gly Leu
1 5 10 15

Arg Pro Pro Lys Asn Ser Phe Glu Ala Leu Met Leu
20 25

<210> SEQ ID NO 342

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<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 342

Pro Lys Asn Ser Phe Glu Ala Leu Met Leu Asn Phe Lys Leu Leu Gly
1 5 10 15

Ile Gly Gly Val Pro Ile Ile Glu Val Ser Gln Phe Gln
20 25

<210> SEQ ID NO 343
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 343

Ser Leu Ala Met Leu Leu Val Lys His Lys His Leu Tyr Leu Thr Met
1 5 10 15

Tyr Ile Thr Pro Val Leu Phe Thr Leu Leu Tyr Asn Asn Tyr Leu Val
20 25 30

<210> SEQ ID NO 344
<211> LENGTH: 47
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 344

Ile Lys Ile Val Leu Leu Cys Tyr Leu Phe Ile Gly Tyr Ile Ile Ser
1 5 10 15

Cys Tyr Trp Gly Leu Phe Ser Leu Met Asn Ser Leu Phe Arg Met Pro
20 25 30

Leu Gly Val Tyr Asn Tyr Lys Ile Ser Val Gln Glu Leu Arg Tyr
35 40 45

<210> SEQ ID NO 345
<211> LENGTH: 47
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 345

Tyr Lys Ile Ser Val Gln Glu Leu Arg Tyr Met Asn Ala Asn Gly Leu
1 5 10 15

Arg Pro Pro Lys Asn Ser Phe Glu Ala Leu Met Leu Asn Phe Lys Leu
20 25 30

Leu Gly Ile Gly Gly Val Pro Ile Ile Glu Val Ser Gln Phe Gln
35 40 45

<210> SEQ ID NO 346
<211> LENGTH: 116
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 346

Ser Leu Ala Met Leu Leu Val Lys His Lys His Leu Tyr Leu Thr Met
1 5 10 15

Tyr Ile Thr Pro Val Leu Phe Thr Leu Leu Tyr Asn Asn Tyr Leu Val
20 25 30

-continued

Ile Lys Ile Val Leu Leu Cys Tyr Leu Phe Ile Gly Tyr Ile Ile Ser
35 40 45
Cys Tyr Trp Gly Leu Phe Ser Leu Met Asn Ser Leu Phe Arg Met Pro
50 55 60
Leu Gly Val Tyr Asn Tyr Lys Ile Ser Val Gln Glu Leu Arg Tyr Met
65 70 75 80
Asn Ala Asn Gly Leu Arg Pro Pro Lys Asn Ser Phe Glu Ala Leu Met
85 90 95
Leu Asn Phe Lys Leu Leu Gly Ile Gly Gly Val Pro Ile Ile Glu Val
100 105 110
Ser Gln Phe Gln
115

<210> SEQ ID NO 347
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 347

Ser Lys Leu Thr Asp Val Lys Cys Ala Asn Val Val Leu Leu Asn Cys
1 5 10 15
Leu Gln His Leu His Val Ala Ser Asn Ser
20 25

<210> SEQ ID NO 348
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 348

Leu Gln His Leu His Val Ala Ser Asn Ser Lys Leu Trp His Tyr Cys
1 5 10 15
Ser Thr Leu His Asn Glu Ile Leu Ala Thr
20 25

<210> SEQ ID NO 349
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 349

Ser Thr Leu His Asn Glu Ile Leu Ala Thr Ser Asp Leu Ser Val Ala
1 5 10 15
Phe Glu Lys Leu Ala Gln Leu Leu Ile Val
20 25

<210> SEQ ID NO 350
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 350

Phe Glu Lys Leu Ala Gln Leu Leu Ile Val Leu Phe Ala Asn Pro Ala
1 5 10 15
Ala Val Asp Ser Lys Cys Leu Thr Ser Ile
20 25

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<210> SEQ ID NO 351
<211> LENGTH: 25
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 351

Ala Val Asp Ser Lys Cys Leu Thr Ser Ile Glu Glu Val Cys Asp Asp
1 5 10 15

Tyr Ala Lys Asp Asn Thr Val Leu Gln
20 25

<210> SEQ ID NO 352
<211> LENGTH: 39
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 352

Ser Lys Leu Thr Asp Val Lys Cys Ala Asn Val Val Leu Leu Asn Cys
1 5 10 15

Leu Gln His Leu His Val Ala Ser Asn Ser Lys Leu Trp His Tyr Cys
20 25 30

Ser Thr Leu His Asn Glu Ile
35

<210> SEQ ID NO 353
<211> LENGTH: 38
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 353

His Tyr Cys Ser Thr Leu His Asn Glu Ile Leu Ala Thr Ser Asp Leu
1 5 10 15

Ser Val Ala Phe Glu Lys Leu Ala Gln Leu Leu Ile Val Leu Phe Ala
20 25 30

Asn Pro Ala Ala Val Asp
35

<210> SEQ ID NO 354
<211> LENGTH: 89
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 354

Ser Lys Leu Thr Asp Val Lys Cys Ala Asn Val Val Leu Leu Asn Cys
1 5 10 15

Leu Gln His Leu His Val Ala Ser Asn Ser Lys Leu Trp His Tyr Cys
20 25 30

Ser Thr Leu His Asn Glu Ile Leu Ala Thr Ser Asp Leu Ser Val Ala
35 40 45

Phe Glu Lys Leu Ala Gln Leu Leu Ile Val Leu Phe Ala Asn Pro Ala
50 55 60

Ala Val Asp Ser Lys Cys Leu Thr Ser Ile Glu Glu Val Cys Asp Asp
65 70 75 80

Tyr Ala Lys Asp Asn Thr Val Leu Gln
85

<210> SEQ ID NO 355
<211> LENGTH: 31

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<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1
<400> SEQUENCE: 355
Tyr Phe Thr Lys Lys Asp Trp Tyr Asp Phe Val Glu Asn Pro Asp Ile
1 5 10 15
Ile Asn Val Tyr Lys Lys Leu Gly Pro Ile Phe Asn Arg Ala Leu
20 25 30
<210> SEQ ID NO 356
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1
<400> SEQUENCE: 356
Lys Leu Gly Pro Ile Phe Asn Arg Ala Leu Val Ser Ala Thr Glu Phe
1 5 10 15
Ala Asp Lys Leu Val Glu Val Gly Leu Val Gly Val Leu Thr
20 25 30
<210> SEQ ID NO 357
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1
<400> SEQUENCE: 357
Val Glu Val Gly Leu Val Gly Val Leu Thr Leu Asp Asn Gln Asp Leu
1 5 10 15
Asn Gly Lys Trp Tyr Asp Phe Gly Asp Tyr Val Ile Ala
20 25
<210> SEQ ID NO 358
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1
<400> SEQUENCE: 358
Trp Tyr Asp Phe Gly Asp Tyr Val Ile Ala Ala Pro Gly Cys Gly Val
1 5 10 15
Ala Ile Ala Asp Ser Tyr Tyr Ser Tyr Ile Met Pro Met Leu Thr
20 25 30
<210> SEQ ID NO 359
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1
<400> SEQUENCE: 359
Asp Leu Val Gln Tyr Asp Phe Thr Asp Tyr Lys Leu Glu Leu Phe Asn
1 5 10 15
Lys Tyr Phe Lys His Trp Ser Met Pro Tyr His Pro Asn Thr
20 25 30
<210> SEQ ID NO 360
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1
<400> SEQUENCE: 360
His Trp Ser Met Pro Tyr His Pro Asn Thr Val Asp Cys Gln Asp Asp

-continued

1 5 10 15
 Arg Cys Ile Ile His Cys Ala Asn Phe Asn Ile Leu Phe Ser
 20 25 30

<210> SEQ ID NO 361
 <211> LENGTH: 29
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1
 <400> SEQUENCE: 361

His Cys Ala Asn Phe Asn Ile Leu Phe Ser Met Val Leu Pro Asn Thr
 1 5 10 15

Cys Phe Gly Pro Leu Val Arg Gln Ile Phe Val Asp Gly
 20 25

<210> SEQ ID NO 362
 <211> LENGTH: 29
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1
 <400> SEQUENCE: 362

Pro Leu Val Arg Gln Ile Phe Val Asp Gly Val Pro Phe Val Val Ser
 1 5 10 15

Ile Gly Tyr His Tyr Lys Glu Leu Gly Ile Val Met Asn
 20 25

<210> SEQ ID NO 363
 <211> LENGTH: 29
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1
 <400> SEQUENCE: 363

His Tyr Lys Glu Leu Gly Ile Val Met Asn Met Asp Val Asp Thr His
 1 5 10 15

Arg Tyr Arg Leu Ser Leu Lys Asp Leu Leu Tyr Ala
 20 25

<210> SEQ ID NO 364
 <211> LENGTH: 50
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1
 <400> SEQUENCE: 364

Tyr Phe Thr Lys Lys Asp Trp Tyr Asp Phe Val Glu Asn Pro Asp Ile
 1 5 10 15

Ile Asn Val Tyr Lys Lys Leu Gly Pro Ile Phe Asn Arg Ala Leu Val
 20 25 30

Ser Ala Thr Glu Phe Ala Asp Lys Leu Val Glu Val Gly Leu Val Gly
 35 40 45

Val Leu
 50

<210> SEQ ID NO 365
 <211> LENGTH: 51
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1
 <400> SEQUENCE: 365

Leu Val Glu Val Gly Leu Val Gly Val Leu Thr Leu Asp Asn Gln Asp

-continued

1	5	10	15
Leu Asn Gly	Lys Trp Tyr Asp Phe Gly Asp Tyr Val Ile Ala Ala Pro		
	20	25	30
Gly Cys Gly	Val Ala Ile Ala Asp Ser Tyr Tyr Ser Tyr Ile Met Pro		
	35	40	45
Met Leu Thr			
50			

<210> SEQ ID NO 366
 <211> LENGTH: 42
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 366

Asp Leu Val Gln Tyr Asp Phe Thr Asp Tyr Lys Leu Glu Leu Phe Asn		
1	5	10 15
Lys Tyr Phe Lys His Trp Ser Met Pro Tyr His Pro Asn Thr Val Asp		
	20	25 30
Cys Gln Asp Asp Arg Cys Ile Ile His Cys		
	35	40

<210> SEQ ID NO 367
 <211> LENGTH: 42
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 367

Cys Gln Asp Asp Arg Cys Ile Ile His Cys Ala Asn Phe Asn Ile Leu		
1	5	10 15
Phe Ser Met Val Leu Pro Asn Thr Cys Phe Gly Pro Leu Val Arg Gln		
	20	25 30
Ile Phe Val Asp Gly Val Pro Phe Val Val		
	35	40

<210> SEQ ID NO 368
 <211> LENGTH: 43
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 368

Ile Phe Val Asp Gly Val Pro Phe Val Val Ser Ile Gly Tyr His Tyr		
1	5	10 15
Lys Glu Leu Gly Ile Val Met Asn Met Asp Val Asp Thr His Arg Tyr		
	20	25 30
Arg Leu Ser Leu Lys Asp Leu Leu Tyr Ala		
	35	40

<210> SEQ ID NO 369
 <211> LENGTH: 91
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 369

Tyr Phe Thr Lys Lys Asp Trp Tyr Asp Phe Val Glu Asn Pro Asp Ile		
1	5	10 15
Ile Asn Val Tyr Lys Lys Leu Gly Pro Ile Phe Asn Arg Ala Leu Val		
	20	25 30

-continued

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Ser Ala Thr Glu Phe Ala Asp Lys Leu Val Glu Val Gly Leu Val Gly
   35           40           45

Val Leu Thr Leu Asp Asn Gln Asp Leu Asn Gly Lys Trp Tyr Asp Phe
   50           55           60

Gly Asp Tyr Val Ile Ala Ala Pro Gly Cys Gly Val Ala Ile Ala Asp
   65           70           75           80

Ser Tyr Tyr Ser Tyr Ile Met Pro Met Leu Thr
           85           90

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<210> SEQ ID NO 370
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

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<400> SEQUENCE: 370

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Asp Leu Val Gln Tyr Asp Phe Thr Asp Tyr Lys Leu Glu Leu Phe Asn
 1           5           10           15

Lys Tyr Phe Lys His Trp Ser Met Pro Tyr His Pro Asn Thr Val Asp
 20           25           30

Cys Gln Asp Asp Arg Cys Ile Ile His Cys Ala Asn Phe Asn Ile Leu
 35           40           45

Phe Ser Met Val Leu Pro Asn Thr Cys Phe Gly Pro Leu Val Arg Gln
 50           55           60

Ile Phe Val Asp Gly Val Pro Phe Val Val Ser Ile Gly Tyr His Tyr
 65           70           75           80

Lys Glu Leu Gly Ile Val Met Asn Met Asp Val Asp Thr His Arg Tyr
 85           90           95

Arg Leu Ser Leu Lys Asp Leu Leu Leu Tyr Ala
 100           105

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<210> SEQ ID NO 371
<211> LENGTH: 198
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
    polypeptide

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<400> SEQUENCE: 371

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Tyr Phe Thr Lys Lys Asp Trp Tyr Asp Phe Val Glu Asn Pro Asp Ile
 1           5           10           15

Ile Asn Val Tyr Lys Lys Leu Gly Pro Ile Phe Asn Arg Ala Leu Val
 20           25           30

Ser Ala Thr Glu Phe Ala Asp Lys Leu Val Glu Val Gly Leu Val Gly
 35           40           45

Val Leu Thr Leu Asp Asn Gln Asp Leu Asn Gly Lys Trp Tyr Asp Phe
 50           55           60

Gly Asp Tyr Val Ile Ala Ala Pro Gly Cys Gly Val Ala Ile Ala Asp
 65           70           75           80

Ser Tyr Tyr Ser Tyr Ile Met Pro Met Leu Thr Asp Leu Val Gln Tyr
 85           90           95

Asp Phe Thr Asp Tyr Lys Leu Glu Leu Phe Asn Lys Tyr Phe Lys His
 100           105           110

Trp Ser Met Pro Tyr His Pro Asn Thr Val Asp Cys Gln Asp Asp Arg
 115           120           125

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-continued

Cys Ile Ile His Cys Ala Asn Phe Asn Ile Leu Phe Ser Met Val Leu
 130 135 140

Pro Asn Thr Cys Phe Gly Pro Leu Val Arg Gln Ile Phe Val Asp Gly
 145 150 155 160

Val Pro Phe Val Val Ser Ile Gly Tyr His Tyr Lys Glu Leu Gly Ile
 165 170 175

Val Met Asn Met Asp Val Asp Thr His Arg Tyr Arg Leu Ser Leu Lys
 180 185 190

Asp Leu Leu Leu Tyr Ala
 195

<210> SEQ ID NO 372
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 372

Ala Tyr Thr Lys Arg Asn Val Leu Pro Thr Leu Thr Gln Met Asn Leu
 1 5 10 15

Lys Tyr Ala Ile Ser Ala Lys Asn Arg Ala Arg Thr
 20 25

<210> SEQ ID NO 373
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 373

Ala Ile Ser Ala Lys Asn Arg Ala Arg Thr Val Ala Gly Val Ser Ile
 1 5 10 15

Leu Ser Thr Met Thr Gly Arg Met Phe His Gln Lys
 20 25

<210> SEQ ID NO 374
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 374

Thr Met Thr Gly Arg Met Phe His Gln Lys Cys Leu Lys Ser Ile Ala
 1 5 10 15

Ala Thr Arg Gly Val Pro Val Val Ile Gly Thr Thr
 20 25

<210> SEQ ID NO 375
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 375

Arg Gly Val Pro Val Val Ile Gly Thr Thr Lys Phe Tyr Gly Gly Trp
 1 5 10 15

Asp Asp Met Leu Arg Arg Leu Ile Lys Asp Val Asp
 20 25

<210> SEQ ID NO 376
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

-continued

<400> SEQUENCE: 376

Met Leu Arg Arg Leu Ile Lys Asp Val Asp Asn Pro Val Leu Met Gly
1 5 10 15

Trp Asp Tyr Pro Lys Cys Asp Arg Ala Met Pro Asn
20 25

<210> SEQ ID NO 377

<211> LENGTH: 29

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 377

Tyr Pro Lys Cys Asp Arg Ala Met Pro Asn Leu Leu Arg Ile Val Ser
1 5 10 15

Ser Leu Val Leu Ala Arg Lys His Glu Thr Cys Cys Ser
20 25

<210> SEQ ID NO 378

<211> LENGTH: 28

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 378

Leu Ala Arg Lys His Glu Thr Cys Cys Ser Gln Ser Asp Arg Phe Tyr
1 5 10 15

Arg Leu Ala Asn Glu Cys Ala Gln Val Leu Ser Glu
20 25

<210> SEQ ID NO 379

<211> LENGTH: 28

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 379

Ala Asn Glu Cys Ala Gln Val Leu Ser Glu Ile Val Met Cys Gly Gly
1 5 10 15

Cys Tyr Tyr Val Lys Pro Gly Gly Thr Ser Ser Gly
20 25

<210> SEQ ID NO 380

<211> LENGTH: 29

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 380

Tyr Val Lys Pro Gly Gly Thr Ser Ser Gly Asp Ala Thr Thr Ala Phe
1 5 10 15

Ala Asn Ser Val Phe Asn Ile Cys Gln Ala Val Ser Ala
20 25

<210> SEQ ID NO 381

<211> LENGTH: 28

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 381

Val Phe Asn Ile Cys Gln Ala Val Ser Ala Asn Val Cys Ala Leu Met
1 5 10 15

-continued

Ser Cys Asn Gly Asn Lys Ile Glu Asp Leu Ser Ile
20 25

<210> SEQ ID NO 382
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 382

Asn Gly Asn Lys Ile Glu Asp Leu Ser Ile Arg Ala Leu Gln Lys Arg
1 5 10 15
Leu Tyr Ser His Val Tyr Arg Ser Asp Lys Val Asp
20 25

<210> SEQ ID NO 383
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 383

Ser His Val Tyr Arg Ser Asp Lys Val Asp Ser Thr Phe Val Thr Glu
1 5 10 15
Tyr Tyr Glu Phe Leu Asn Lys His Phe Ser Met Met
20 25

<210> SEQ ID NO 384
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 384

Glu Phe Leu Asn Lys His Phe Ser Met Met Ile Leu Ser Asp Asp Gly
1 5 10 15
Val Val Cys Tyr Asn Ser Asp Tyr Ala Ser Lys Gly Tyr
20 25

<210> SEQ ID NO 385
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 385

Tyr Asn Ser Asp Tyr Ala Ser Lys Gly Tyr Ile Ala Asn Ile Ser Ala
1 5 10 15
Phe Gln Gln Val Leu Tyr Tyr Gln Asn Asn Val Phe Met
20 25

<210> SEQ ID NO 386
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 386

Val Leu Tyr Tyr Gln Asn Asn Val Phe Met Ser Glu Ser Lys Cys Trp
1 5 10 15
Val Glu His Asp Ile Asn Asn Gly Pro His Glu Phe
20 25

<210> SEQ ID NO 387
<211> LENGTH: 29

-continued

<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 387

His Asp Ile Asn Asn Gly Pro His Glu Phe Cys Ser Gln His Thr Met
1 5 10 15

Leu Val Lys Met Asp Gly Asp Asp Val Tyr Leu Pro Tyr
 20 25

<210> SEQ ID NO 388
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 388

Met Asp Gly Asp Asp Val Tyr Leu Pro Tyr Pro Asn Pro Ser Arg Ile
1 5 10 15

Leu Gly Ala Gly Cys Phe Val Asp Asp Leu Leu Lys Thr
 20 25

<210> SEQ ID NO 389
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 389

Gly Cys Phe Val Asp Asp Leu Leu Lys Thr Asp Ser Val Leu Leu Ile
1 5 10 15

Glu Arg Phe Val Ser Leu Ala Ile Asp Ala Tyr Pro Leu
 20 25

<210> SEQ ID NO 390
<211> LENGTH: 47
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 390

Ala Tyr Thr Lys Arg Asn Val Leu Pro Thr Leu Thr Gln Met Asn Leu
1 5 10 15

Lys Tyr Ala Ile Ser Ala Lys Asn Arg Ala Arg Thr Val Ala Gly Val
 20 25 30

Ser Ile Leu Ser Thr Met Thr Gly Arg Met Phe His Gln Lys Cys
 35 40 45

<210> SEQ ID NO 391
<211> LENGTH: 47
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 391

Met Thr Gly Arg Met Phe His Gln Lys Cys Leu Lys Ser Ile Ala Ala
1 5 10 15

Thr Arg Gly Val Pro Val Val Ile Gly Thr Thr Lys Phe Tyr Gly Gly
 20 25 30

Trp Asp Asp Met Leu Arg Arg Leu Ile Lys Asp Val Asp Asn Pro
 35 40 45

<210> SEQ ID NO 392
<211> LENGTH: 46

-continued

<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 392

Arg Arg Leu Ile Lys Asp Val Asp Asn Pro Val Leu Met Gly Trp Asp
1 5 10 15

Tyr Pro Lys Cys Asp Arg Ala Met Pro Asn Leu Leu Arg Ile Val Ser
 20 25 30

Ser Leu Val Leu Ala Arg Lys His Glu Thr Cys Cys Ser Gln
 35 40 45

<210> SEQ ID NO 393
<211> LENGTH: 46
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 393

Ala Arg Lys His Glu Thr Cys Cys Ser Gln Ser Asp Arg Phe Tyr Arg
1 5 10 15

Leu Ala Asn Glu Cys Ala Gln Val Leu Ser Glu Ile Val Met Cys Gly
 20 25 30

Gly Cys Tyr Tyr Val Lys Pro Gly Gly Thr Ser Ser Gly Asp
 35 40 45

<210> SEQ ID NO 394
<211> LENGTH: 46
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 394

Val Lys Pro Gly Gly Thr Ser Ser Gly Asp Ala Thr Thr Ala Phe Ala
1 5 10 15

Asn Ser Val Phe Asn Ile Cys Gln Ala Val Ser Ala Asn Val Cys Ala
 20 25 30

Leu Met Ser Cys Asn Gly Asn Lys Ile Glu Asp Leu Ser Ile
 35 40 45

<210> SEQ ID NO 395
<211> LENGTH: 46
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 395

Asn Gly Asn Lys Ile Glu Asp Leu Ser Ile Arg Ala Leu Gln Lys Arg
1 5 10 15

Leu Tyr Ser His Val Tyr Arg Ser Asp Lys Val Asp Ser Thr Phe Val
 20 25 30

Thr Glu Tyr Tyr Glu Phe Leu Asn Lys His Phe Ser Met Met
 35 40 45

<210> SEQ ID NO 396
<211> LENGTH: 47
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 396

Glu Phe Leu Asn Lys His Phe Ser Met Met Ile Leu Ser Asp Asp Gly
1 5 10 15

-continued

Val	Val	Cys	Tyr	Asn	Ser	Asp	Tyr	Ala	Ser	Lys	Gly	Tyr	Ile	Ala	Asn
			20					25					30		

Ile	Ser	Ala	Phe	Gln	Gln	Val	Leu	Tyr	Tyr	Gln	Asn	Asn	Val	Phe	
		35					40					45			

<210> SEQ ID NO 397

<211> LENGTH: 47

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 397

Gln	Val	Leu	Tyr	Tyr	Gln	Asn	Asn	Val	Phe	Met	Ser	Glu	Ser	Lys	Cys
1				5					10					15	

Trp	Val	Glu	His	Asp	Ile	Asn	Asn	Gly	Pro	His	Glu	Phe	Cys	Ser	Gln
		20						25					30		

His	Thr	Met	Leu	Val	Lys	Met	Asp	Gly	Asp	Asp	Val	Tyr	Leu	Pro	
		35					40					45			

<210> SEQ ID NO 398

<211> LENGTH: 49

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 398

Lys	Met	Asp	Gly	Asp	Asp	Val	Tyr	Leu	Pro	Tyr	Pro	Asn	Pro	Ser	Arg
1				5					10					15	

Ile	Leu	Gly	Ala	Gly	Cys	Phe	Val	Asp	Asp	Leu	Leu	Lys	Thr	Asp	Ser
		20						25					30		

Val	Leu	Leu	Ile	Glu	Arg	Phe	Val	Ser	Leu	Ala	Ile	Asp	Ala	Tyr	Pro
		35					40					45			

Leu

<210> SEQ ID NO 399

<211> LENGTH: 125

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 399

Ala	Tyr	Thr	Lys	Arg	Asn	Val	Leu	Pro	Thr	Leu	Thr	Gln	Met	Asn	Leu
1			5						10					15	

Lys	Tyr	Ala	Ile	Ser	Ala	Lys	Asn	Arg	Ala	Arg	Thr	Val	Ala	Gly	Val
		20					25						30		

Ser	Ile	Leu	Ser	Thr	Met	Thr	Gly	Arg	Met	Phe	His	Gln	Lys	Cys	Leu
		35					40					45			

Lys	Ser	Ile	Ala	Ala	Thr	Arg	Gly	Val	Pro	Val	Val	Ile	Gly	Thr	Thr
		50			55						60				

Lys	Phe	Tyr	Gly	Gly	Trp	Asp	Asp	Met	Leu	Arg	Arg	Leu	Ile	Lys	Asp
65					70				75					80	

Val	Asp	Asn	Pro	Val	Leu	Met	Gly	Trp	Asp	Tyr	Pro	Lys	Cys	Asp	Arg
			85					90						95	

Ala	Met	Pro	Asn	Leu	Leu	Arg	Ile	Val	Ser	Ser	Leu	Val	Leu	Ala	Arg
			100					105					110		

Lys	His	Glu	Thr	Cys	Cys	Ser	Gln	Ser	Asp	Arg	Phe	Tyr			
		115					120					125			

<210> SEQ ID NO 400

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<211> LENGTH: 125
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 400

Glu Thr Cys Cys Ser Gln Ser Asp Arg Phe Tyr Arg Leu Ala Asn Glu
1          5          10          15

Cys Ala Gln Val Leu Ser Glu Ile Val Met Cys Gly Gly Cys Tyr Tyr
20          25          30

Val Lys Pro Gly Gly Thr Ser Ser Gly Asp Ala Thr Thr Ala Phe Ala
35          40          45

Asn Ser Val Phe Asn Ile Cys Gln Ala Val Ser Ala Asn Val Cys Ala
50          55          60

Leu Met Ser Cys Asn Gly Asn Lys Ile Glu Asp Leu Ser Ile Arg Ala
65          70          75          80

Leu Gln Lys Arg Leu Tyr Ser His Val Tyr Arg Ser Asp Lys Val Asp
85          90          95

Ser Thr Phe Val Thr Glu Tyr Tyr Glu Phe Leu Asn Lys His Phe Ser
100         105         110

Met Met Ile Leu Ser Asp Asp Gly Val Val Cys Tyr Asn
115         120         125

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<210> SEQ ID NO 401
<211> LENGTH: 113
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 401

Ile Leu Ser Asp Asp Gly Val Val Cys Tyr Asn Ser Asp Tyr Ala Ser
1          5          10          15

Lys Gly Tyr Ile Ala Asn Ile Ser Ala Phe Gln Gln Val Leu Tyr Tyr
20          25          30

Gln Asn Asn Val Phe Met Ser Glu Ser Lys Cys Trp Val Glu His Asp
35          40          45

Ile Asn Asn Gly Pro His Glu Phe Cys Ser Gln His Thr Met Leu Val
50          55          60

Lys Met Asp Gly Asp Asp Val Tyr Leu Pro Tyr Pro Asn Pro Ser Arg
65          70          75          80

Ile Leu Gly Ala Gly Cys Phe Val Asp Asp Leu Leu Lys Thr Asp Ser
85          90          95

Val Leu Leu Ile Glu Arg Phe Val Ser Leu Ala Ile Asp Ala Tyr Pro
100         105         110

Leu

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<210> SEQ ID NO 402
<211> LENGTH: 341
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 402

Ala Tyr Thr Lys Arg Asn Val Leu Pro Thr Leu Thr Gln Met Asn Leu
1          5          10          15

Lys Tyr Ala Ile Ser Ala Lys Asn Arg Ala Arg Thr Val Ala Gly Val
20          25          30

Ser Ile Leu Ser Thr Met Thr Gly Arg Met Phe His Gln Lys Cys Leu

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35					40					45					
Lys	Ser	Ile	Ala	Ala	Thr	Arg	Gly	Val	Pro	Val	Val	Ile	Gly	Thr	Thr
50						55					60				
Lys	Phe	Tyr	Gly	Gly	Trp	Asp	Asp	Met	Leu	Arg	Arg	Leu	Ile	Lys	Asp
65					70					75					80
Val	Asp	Asn	Pro	Val	Leu	Met	Gly	Trp	Asp	Tyr	Pro	Lys	Cys	Asp	Arg
				85					90					95	
Ala	Met	Pro	Asn	Leu	Leu	Arg	Ile	Val	Ser	Ser	Leu	Val	Leu	Ala	Arg
			100					105					110		
Lys	His	Glu	Thr	Cys	Cys	Ser	Gln	Ser	Asp	Arg	Phe	Tyr	Arg	Leu	Ala
		115					120					125			
Asn	Glu	Cys	Ala	Gln	Val	Leu	Ser	Glu	Ile	Val	Met	Cys	Gly	Gly	Cys
	130					135					140				
Tyr	Tyr	Val	Lys	Pro	Gly	Gly	Thr	Ser	Ser	Gly	Asp	Ala	Thr	Thr	Ala
145					150					155					160
Phe	Ala	Asn	Ser	Val	Phe	Asn	Ile	Cys	Gln	Ala	Val	Ser	Ala	Asn	Val
				165					170						175
Cys	Ala	Leu	Met	Ser	Cys	Asn	Gly	Asn	Lys	Ile	Glu	Asp	Leu	Ser	Ile
		180						185					190		
Arg	Ala	Leu	Gln	Lys	Arg	Leu	Tyr	Ser	His	Val	Tyr	Arg	Ser	Asp	Lys
		195					200					205			
Val	Asp	Ser	Thr	Phe	Val	Thr	Glu	Tyr	Tyr	Glu	Phe	Leu	Asn	Lys	His
	210					215					220				
Phe	Ser	Met	Met	Ile	Leu	Ser	Asp	Asp	Gly	Val	Val	Cys	Tyr	Asn	Ser
225					230					235					240
Asp	Tyr	Ala	Ser	Lys	Gly	Tyr	Ile	Ala	Asn	Ile	Ser	Ala	Phe	Gln	Gln
				245					250					255	
Val	Leu	Tyr	Tyr	Gln	Asn	Asn	Val	Phe	Met	Ser	Glu	Ser	Lys	Cys	Trp
		260						265					270		
Val	Glu	His	Asp	Ile	Asn	Asn	Gly	Pro	His	Glu	Phe	Cys	Ser	Gln	His
		275					280					285			
Thr	Met	Leu	Val	Lys	Met	Asp	Gly	Asp	Asp	Val	Tyr	Leu	Pro	Tyr	Pro
	290					295					300				
Asn	Pro	Ser	Arg	Ile	Leu	Gly	Ala	Gly	Cys	Phe	Val	Asp	Asp	Leu	Leu
305					310					315					320
Lys	Thr	Asp	Ser	Val	Leu	Leu	Ile	Glu	Arg	Phe	Val	Ser	Leu	Ala	Ile
				325					330					335	
Asp	Ala	Tyr	Pro	Leu											
			340												

<210> SEQ ID NO 403

<211> LENGTH: 120

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Description of Unknown:
Bovine papillomavirus (BPV) sequence

<400> SEQUENCE: 403

Ser	Leu	Val	Ser	Thr	Asp	Asn	Gln	Ile	Phe	Asn	Arg	Pro	Tyr	Trp	Leu
1				5					10					15	
Phe	Arg	Ala	Gln	Gly	Met	Asn	Asn	Gly	Ile	Ala	Trp	Asn	Asn	Leu	Leu
		20						25				30			

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Phe Leu Thr Val Gly Asp Asn Thr Arg Gly Thr Asn Leu Thr Ile Ser
   35                               40                               45

Val Ala Ser Asp Gly Thr Pro Leu Thr Glu Tyr Asp Ser Ser Lys Phe
   50                               55                               60

Asn Val Tyr His Arg His Met Glu Glu Tyr Lys Leu Ala Phe Ile Leu
  65                               70                               75                               80

Glu Leu Cys Ser Val Glu Ile Thr Ala Gln Thr Val Ser His Leu Gln
   85                               90                               95

Gly Leu Met Pro Ser Val Leu Glu Asn Trp Glu Ile Gly Val Gln Pro
  100                               105                               110

Pro Thr Ser Ser Ile Leu Glu Asp
  115                               120

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<210> SEQ ID NO 404
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Description of Unknown:
      BPV isolate NY8385 sequence

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<400> SEQUENCE: 404

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Ser Leu Val Thr Ser Ser Asn Gln Leu Phe Asn Lys Pro Tyr Trp Ile
 1           5           10           15

Arg Arg Ala Gln Gly Asn Asn Asn Gly Ile Leu Trp Gly Asn Gln Cys
 20           25           30

Phe Val Thr Val Gly Asp Thr Thr Arg Gly Thr Asn Phe Asp Ile Ser
 35           40           45

Val Pro Thr Ala Gly Gly Thr Gln Tyr Lys Ala Thr Glu Phe Asn Gln
 50           55           60

Tyr Lys Arg His Val Glu Glu Tyr Glu Leu Gln Phe Val Leu Gln Ala
 65           70           75           80

Cys Val Val Arg Leu Thr Pro Glu Thr Val Ala His Leu Tyr Gln Met
 85           90           95

Asp Ser Arg Ile Leu Asp Asp Trp Asn Leu Gly Val Val Thr Ser Gln
100           105           110

Asn Gly Ala Leu Ala Glu
115

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<210> SEQ ID NO 405
<211> LENGTH: 212
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
<400> SEQUENCE: 405

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Thr Val Glu Glu Leu Lys Lys Leu Leu Glu Gln Trp Asn Leu Val Ile
 1           5           10           15

Gly Phe Leu Phe Leu Thr Trp Ile Cys Leu Leu Gln Phe Ala Tyr Ala
 20           25           30

Asn Arg Asn Arg Phe Leu Tyr Ile Ile Lys Leu Ile Phe Leu Trp Leu
 35           40           45

Leu Trp Pro Val Thr Leu Ala Cys Phe Val Leu Ala Ala Val Tyr Arg
 50           55           60

Ile Asn Trp Ile Thr Gly Gly Ile Ala Ile Ala Met Ala Cys Leu Val
 65           70           75           80

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Gly Leu Met Trp Leu Ser Tyr Phe Ile Ala Ser Phe Arg Leu Phe Ala
85 90 95

Arg Thr Arg Ser Met Trp Ser Phe Asn Pro Glu Thr Asn Ile Leu Leu
100 105 110

Asn Val Pro Leu His Gly Thr Ile Leu Thr Arg Pro Leu Leu Glu Ser
115 120 125

Glu Leu Val Ile Gly Ala Val Ile Leu Arg Gly His Leu Arg Ile Ala
130 135 140

Gly His His Leu Gly Arg Cys Asp Ile Lys Asp Leu Pro Lys Glu Ile
145 150 155 160

Thr Val Ala Thr Ser Arg Thr Leu Ser Tyr Tyr Lys Leu Gly Ala Ser
165 170 175

Gln Arg Val Ala Gly Asp Ser Gly Phe Ala Ala Tyr Ser Arg Tyr Arg
180 185 190

Ile Gly Asn Tyr Lys Leu Asn Thr Asp His Ser Ser Ser Ser Asp Asn
195 200 205

Ile Ala Leu Leu
210

<210> SEQ ID NO 406

<211> LENGTH: 213

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 406

Thr Ala Asp Glu Ala Ile Lys Phe Leu Lys Glu Trp Asn Phe Ser Leu
1 5 10 15

Gly Ile Ile Leu Leu Phe Ile Thr Ile Ile Leu Gln Phe Gly Tyr Thr
20 25 30

Ser Arg Ser Met Phe Val Tyr Val Ile Lys Met Ile Ile Leu Trp Leu
35 40 45

Met Trp Pro Leu Thr Ile Ile Leu Thr Ile Phe Asn Cys Val Tyr Ala
50 55 60

Leu Asn Asn Val Tyr Leu Gly Leu Ser Ile Val Phe Thr Ile Val Ala
65 70 75 80

Ile Ile Met Trp Ile Val Tyr Phe Val Asn Ser Ile Arg Leu Phe Ile
85 90 95

Arg Thr Gly Ser Phe Trp Ser Phe Asn Pro Glu Thr Asn Asn Leu Met
100 105 110

Cys Ile Asp Met Lys Gly Thr Met Tyr Val Arg Pro Ile Ile Glu Asp
115 120 125

Tyr His Thr Leu Thr Val Thr Ile Ile Arg Gly His Leu Tyr Ile Gln
130 135 140

Gly Ile Lys Leu Gly Thr Gly Tyr Ser Leu Ala Asp Leu Pro Ala Tyr
145 150 155 160

Met Thr Val Ala Lys Val Thr His Leu Cys Thr Tyr Lys Arg Gly Phe
165 170 175

Leu Asp Arg Ile Ser Asp Thr Ser Gly Phe Ala Val Tyr Val Lys Ser
180 185 190

Lys Val Gly Asn Tyr Arg Leu Pro Ser Thr Gln Lys Gly Ser Gly Met
195 200 205

Asp Thr Ala Leu Leu
210

-continued

<210> SEQ ID NO 407

<211> LENGTH: 214

<212> TYPE: PRT

<213> ORGANISM: Human coronavirus HKU1

<400> SEQUENCE: 407

Phe Thr Ser Asp Gln Ala Val Thr Phe Leu Lys Glu Trp Asn Phe Ser
1 5 10 15

Leu Gly Val Ile Leu Leu Phe Ile Thr Ile Ile Leu Gln Phe Gly Tyr
20 25 30

Thr Ser Arg Ser Met Phe Val Tyr Leu Ile Lys Met Ile Ile Leu Trp
35 40 45

Leu Met Trp Pro Leu Thr Ile Thr Leu Thr Ile Phe Asn Cys Phe Tyr
50 55 60

Ala Leu Asn Asn Ala Phe Leu Ala Phe Ser Ile Val Phe Thr Ile Ile
65 70 75 80

Ser Ile Val Ile Trp Ile Leu Tyr Phe Val Asn Ser Ile Arg Leu Phe
85 90 95

Ile Arg Thr Gly Ser Trp Trp Ser Phe Asn Pro Glu Thr Asn Asn Leu
100 105 110

Met Cys Ile Asp Met Lys Gly Lys Met Phe Val Arg Pro Val Ile Glu
115 120 125

Asp Tyr His Thr Leu Thr Ala Thr Val Ile Arg Gly His Leu Tyr Ile
130 135 140

Gln Gly Val Lys Leu Gly Thr Gly Tyr Thr Leu Ser Asp Leu Pro Val
145 150 155 160

Tyr Val Thr Val Ala Lys Val Gln Val Leu Cys Thr Tyr Lys Arg Ala
165 170 175

Phe Leu Asp Lys Leu Asp Val Asn Ser Gly Phe Ala Val Phe Val Lys
180 185 190

Ser Lys Val Gly Asn Tyr Arg Leu Pro Ser Ser Lys Pro Ser Gly Met
195 200 205

Asp Thr Ala Leu Leu Arg
210

<210> SEQ ID NO 408

<211> LENGTH: 215

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 408

Trp Thr Ala Asp Glu Ala Ile Lys Phe Leu Lys Glu Trp Asn Phe Ser
1 5 10 15

Leu Gly Ile Ile Leu Leu Phe Ile Thr Ile Ile Leu Gln Phe Gly Tyr
20 25 30

Thr Ser Arg Ser Met Phe Val Tyr Val Ile Lys Met Ile Ile Leu Trp
35 40 45

Leu Met Trp Pro Leu Thr Ile Ile Leu Thr Ile Phe Asn Cys Val Tyr
50 55 60

Ala Leu Asn Asn Val Tyr Leu Gly Leu Ser Ile Val Phe Thr Ile Val
65 70 75 80

Ala Ile Ile Met Trp Ile Val Tyr Phe Val Asn Ser Ile Arg Leu Phe
85 90 95

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Ile Arg Thr Gly Ser Phe Trp Ser Phe Asn Pro Glu Thr Asn Asn Leu
      100                      105                      110

Met Cys Ile Asp Met Lys Gly Thr Met Tyr Val Arg Pro Ile Ile Glu
      115                      120                      125

Asp Tyr His Thr Leu Thr Val Thr Ile Ile Arg Gly His Leu Tyr Ile
      130                      135                      140

Gln Gly Ile Lys Leu Gly Thr Gly Tyr Ser Leu Ala Asp Leu Pro Ala
      145                      150                      155                      160

Tyr Met Thr Val Ala Lys Val Thr His Leu Cys Thr Tyr Lys Arg Gly
      165                      170                      175

Phe Leu Asp Arg Ile Ser Asp Thr Ser Gly Phe Ala Val Tyr Val Lys
      180                      185                      190

Ser Lys Val Gly Asn Tyr Arg Leu Pro Ser Thr Gln Lys Gly Ser Gly
      195                      200                      205

Met Asp Thr Ala Leu Leu Arg
      210                      215

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<210> SEQ ID NO 409

<211> LENGTH: 269

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 409

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Gly Tyr Trp Tyr Arg His Asn Arg Arg Ser Phe Lys Thr Ala Asp Gly
1      5      10      15

Asn Gln Arg Gln Leu Leu Pro Arg Trp Tyr Phe Tyr Tyr Leu Gly Thr
20     25     30

Gly Pro His Ala Lys Asp Gln Tyr Gly Asp Ile Asp Gly Val Tyr Trp
35     40     45

Val Ala Ser Asn Gln Ala Asp Val Asn Thr Pro Ala Asp Ile Val Asp
50     55     60

Arg Asp Pro Ser Ser Asp Glu Ala Ile Pro Thr Arg Phe Pro Pro Gly
65     70     75     80

Thr Val Leu Pro Gln Gly Tyr Tyr Ile Glu Gly Ser Gly Arg Ser Ala
85     90     95

Pro Asn Ser Arg Ser Thr Ser Arg Thr Ser Ser Arg Ala Ser Ser Ala
100    105    110

Gly Ser Arg Ser Arg Ala Asn Ser Gly Asn Arg Thr Pro Thr Ser Gly
115    120    125

Val Thr Pro Asp Met Ala Asp Gln Ile Ala Ser Leu Val Leu Ala Lys
130    135    140

Leu Gly Lys Asp Ala Thr Lys Pro Gln Gln Val Thr Lys His Thr Ala
145    150    155    160

Lys Glu Val Arg Gln Lys Pro Arg Gln Lys Arg Ser Pro Asn Lys Gln
165    170    175

Cys Thr Val Gln Gln Cys Phe Gly Lys Arg Gly Pro Asn Gln Asn Phe
180    185    190

Gly Gly Gly Glu Met Leu Lys Leu Gly Thr Ser Asp Pro Gln Phe Pro
195    200    205

Ile Leu Ala Glu Leu Ala Pro Thr Ala Gly Ala Phe Phe Phe Gly Ser
210    215    220

Arg Leu Glu Leu Ala Lys Val Gln Asn Leu Ser Gly Asn Glu Leu Arg

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225	230	235	240
Tyr Asn Gly Ala Ile Arg Phe Asp Ser Thr Leu Ser Gly Phe Glu Thr			
	245	250	255
Ile Met Lys Val Leu Asn Glu Asn Leu Asn Ala Tyr Gln			
	260	265	

<210> SEQ ID NO 410
 <211> LENGTH: 273
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 410

Gly Tyr Tyr Arg Arg Ala Thr Arg Arg Ile Arg Gly Gly Asp Gly Lys			
1	5	10	15
Met Lys Asp Leu Ser Pro Arg Trp Tyr Phe Tyr Tyr Leu Gly Thr Gly			
	20	25	30
Pro Glu Ala Gly Leu Pro Tyr Gly Ala Asn Lys Asp Gly Ile Ile Trp			
	35	40	45
Val Ala Thr Glu Gly Ala Leu Asn Thr Pro Lys Asp His Ile Gly Thr			
	50	55	60
Arg Asn Pro Ala Asn Asn Ala Ala Ile Val Leu Gln Leu Pro Gln Gly			
	65	70	75
Thr Thr Leu Pro Lys Gly Phe Tyr Ala Glu Gly Ser Arg Gly Gly Ser			
	85	90	95
Gln Ala Ser Ser Arg Ser Ser Ser Arg Ser Arg Asn Ser Ser Arg Asn			
	100	105	110
Ser Thr Pro Gly Ser Ser Arg Gly Thr Ser Pro Ala Arg Met Ala Gly			
	115	120	125
Asn Gly Gly Asp Ala Ala Leu Ala Asp Arg Leu Asn Gln Leu Glu Ser			
	130	135	140
Lys Met Ser Gly Lys Gly Gln Gln Gln Gln Gly Gln Thr Val Thr Lys			
	145	150	155
Lys Ser Ala Ala Glu Ala Ser Lys Lys Pro Arg Gln Lys Arg Thr Ala			
	165	170	175
Thr Lys Ala Tyr Asn Val Thr Gln Ala Phe Gly Arg Arg Gly Pro Glu			
	180	185	190
Gln Thr Gln Gly Asn Phe Gly Asp Gln Glu Leu Ile Arg Gln Gly Thr			
	195	200	205
Asp Tyr Lys His Trp Pro Gln Ile Ala Gln Phe Ala Pro Ser Ala Ser			
	210	215	220
Ala Phe Phe Gly Met Ser Arg Ile Gly Met Glu Val Thr Pro Ser Gly			
	225	230	235
Thr Trp Leu Thr Tyr Thr Gly Ala Ile Lys Leu Asp Asp Lys Asp Pro			
	245	250	255
Asn Phe Lys Asp Gln Val Ile Leu Leu Asn Lys His Ile Asp Ala Tyr			
	260	265	270

Lys

<210> SEQ ID NO 411
 <211> LENGTH: 24
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2
 <400> SEQUENCE: 411

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Ser Trp Phe Thr Ala Leu Thr Gln His Gly Lys Glu Asp Leu Lys Phe
1 5 10 15

Pro Arg Gly Gln Gly Val Pro Ile
20

<210> SEQ ID NO 412
<211> LENGTH: 40
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 412

Pro His Ala Lys Asp Gln Tyr Gly Thr Asp Ile Asp Gly Val Tyr Trp
1 5 10 15

Val Ala Ser Asn Gln Ala Asp Val Asn Thr Pro Ala Asp Ile Val Asp
20 25 30

Arg Asp Pro Ser Ser Asp Glu Ala
35 40

<210> SEQ ID NO 413
<211> LENGTH: 40
<212> TYPE: PRT
<213> ORGANISM: Human coronavirus HKU1

<400> SEQUENCE: 413

Pro Tyr Ala Asn Ala Ser Tyr Gly Glu Ser Leu Glu Gly Val Phe Trp
1 5 10 15

Val Ala Asn His Gln Ala Asp Thr Ser Thr Pro Ser Asp Val Ser Ser
20 25 30

Arg Asp Pro Thr Thr Gln Glu Ala
35 40

<210> SEQ ID NO 414
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 414

Arg Ser Thr Ser Arg Thr Ser Ser Arg Ala Ser Ser Ala Gly Ser Arg
1 5 10 15

Ser Arg Ala Asn Ser Gly Asn Arg Thr Pro Thr Ser Gly Val Thr Pro
20 25 30

<210> SEQ ID NO 415
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: Human coronavirus HKU1

<400> SEQUENCE: 415

Arg Pro Gly Ser Arg Ser Gln Ser Arg Gly Pro Asn Asn Arg Ser Leu
1 5 10 15

Ser Arg Ser Asn Ser Asn Phe Arg His Ser Asp Ser Ile Val Lys Pro
20 25 30

<210> SEQ ID NO 416
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 416

-continued

Ser Arg Leu Glu Leu Ala Lys Val Gln Asn Leu Ser Gly Asn Pro
1 5 10 15

<210> SEQ ID NO 417
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Human coronavirus HKU1

<400> SEQUENCE: 417

Ser Lys Leu Asp Leu Val Lys Arg Asp Ser Glu Ala Asp Ser Pro
1 5 10 15

<210> SEQ ID NO 418
<211> LENGTH: 256
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 418

Ser Lys Ala Ser Ser Arg Ser Ala Ile Glu Asp Leu Leu Phe Asp Lys
1 5 10 15

Val Lys Leu Ser Asp Val Gly Phe Val Glu Ala Tyr Asn Asn Cys Thr
20 25 30

Gly Gly Ala Glu Ile Arg Asp Leu Ile Cys Val Gln Ser Tyr Lys Gly
35 40 45

Ile Lys Val Leu Pro Pro Leu Leu Ser Glu Asn Gln Ile Ser Gly Tyr
50 55 60

Thr Leu Ala Ala Thr Ser Ala Ser Leu Phe Pro Trp Thr Ala Ala
65 70 75 80

Ala Gly Val Pro Phe Tyr Leu Asn Val Gln Tyr Arg Ile Asn Gly Leu
85 90 95

Gly Val Thr Met Asp Val Leu Ser Gln Asn Gln Lys Leu Ile Ala Asn
100 105 110

Ala Phe Asn Asn Ala Leu Tyr Ala Ile Gln Glu Gly Phe Asp Ala Thr
115 120 125

Asn Ser Ala Leu Val Lys Ile Gln Ala Val Val Asn Ala Asn Ala Glu
130 135 140

Ala Leu Asn Asn Leu Leu Gln Gln Leu Ser Asn Arg Phe Gly Ala Ile
145 150 155 160

Ser Ala Ser Leu Gln Glu Ile Leu Ser Arg Leu Asp Ala Leu Glu Ala
165 170 175

Glu Ala Gln Ile Asp Arg Leu Ile Asn Gly Arg Leu Thr Ala Leu Asn
180 185 190

Ala Tyr Val Ser Gln Gln Leu Ser Asp Ser Thr Leu Val Lys Phe Ser
195 200 205

Ala Ala Gln Ala Met Glu Lys Val Asn Glu Cys Val Lys Ser Gln Ser
210 215 220

Ser Arg Ile Asn Phe Cys Gly Asn Gly Asn His Ile Ile Ser Leu Val
225 230 235 240

Gln Asn Ala Pro Tyr Gly Leu Tyr Phe Ile His Phe Ser Tyr Val Pro
245 250 255

<210> SEQ ID NO 419
<211> LENGTH: 260
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

-continued

<400> SEQUENCE: 419

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Ser Lys Pro Ser Lys Arg Ser Phe Ile Glu Asp Leu Leu Phe Asn Lys
1      5      10      15
Val Thr Leu Ala Asp Ala Gly Phe Ile Lys Gln Tyr Gly Asp Cys Leu
20     25     30
Gly Asp Ile Ala Ala Arg Asp Leu Ile Cys Ala Gln Lys Phe Asn Gly
35     40     45
Leu Thr Val Leu Pro Pro Leu Leu Thr Asp Glu Met Ile Ala Gln Tyr
50     55     60
Thr Ser Ala Leu Leu Ala Gly Thr Ile Thr Ser Gly Trp Thr Phe Gly
65     70     75     80
Ala Gly Ala Ala Leu Gln Ile Pro Phe Ala Met Gln Met Ala Tyr Arg
85     90     95
Phe Asn Gly Ile Gly Val Thr Gln Asn Val Leu Tyr Glu Asn Gln Lys
100    105    110
Leu Ile Ala Asn Gln Phe Asn Ser Ala Ile Gly Lys Ile Gln Asp Ser
115    120    125
Leu Ser Ser Thr Ala Ser Ala Leu Gly Lys Leu Gln Asp Val Val Asn
130    135    140
Gln Asn Ala Gln Ala Leu Asn Thr Leu Val Lys Gln Leu Ser Ser Asn
145    150    155    160
Phe Gly Ala Ile Ser Ser Val Leu Asn Asp Ile Leu Ser Arg Leu Asp
165    170    175
Lys Val Glu Ala Glu Val Gln Ile Asp Arg Leu Ile Thr Gly Arg Leu
180    185    190
Gln Ser Leu Gln Thr Tyr Val Thr Gln Gln Leu Ile Arg Ala Ala Glu
195    200    205
Ile Arg Ala Ser Ala Asn Leu Ala Ala Thr Lys Met Ser Glu Cys Val
210    215    220
Leu Gly Gln Ser Lys Arg Val Asp Phe Cys Gly Lys Gly Tyr His Leu
225    230    235    240
Met Ser Phe Pro Gln Ser Ala Pro His Gly Val Val Phe Leu His Val
245    250    255
Thr Tyr Val Pro
260

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<210> SEQ ID NO 420

<211> LENGTH: 75

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 420

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Pro Asn Leu Pro Asp Phe Lys Glu Glu Leu Asp Gln Trp Phe Lys Asn
1      5      10      15
Gln Thr Ser Val Ala Pro Asp Leu Ser Leu Asp Tyr Ile Asn Val Thr
20     25     30
Phe Leu Asp Leu Gln Val Glu Met Asn Arg Leu Gln Glu Ala Ile Lys
35     40     45
Val Leu Asn Gln Ser Tyr Ile Asn Leu Lys Asp Ile Gly Thr Tyr Glu
50     55     60
Tyr Tyr Val Lys Trp Pro Trp Tyr Val Trp Leu
65     70     75

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<210> SEQ ID NO 421
<211> LENGTH: 76
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 421

Pro Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn
1 5 10 15
His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn Ala
20 25 30
Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu Val Ala
35 40 45
Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu Gly Lys Tyr
50 55 60
Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu
65 70 75

<210> SEQ ID NO 422
<211> LENGTH: 73
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 422

Pro Asn Leu Pro Asp Phe Lys Glu Glu Leu Asp Gln Trp Phe Lys Asn
1 5 10 15
Gln Thr Ser Val Ala Pro Asp Leu Ser Leu Asp Tyr Ile Asn Val Thr
20 25 30
Phe Leu Asp Leu Gln Val Glu Met Asn Arg Leu Gln Glu Ala Ile Lys
35 40 45
Val Leu Asn Gln Ser Tyr Ile Asn Leu Lys Asp Ile Gly Thr Tyr Glu
50 55 60
Tyr Tyr Val Lys Trp Pro Trp Tyr Val
65 70

<210> SEQ ID NO 423
<211> LENGTH: 75
<212> TYPE: PRT
<213> ORGANISM: Human coronavirus HKU1

<400> SEQUENCE: 423

Pro Lys Leu Ser Asp Phe Glu Ser Glu Leu Ser His Trp Phe Lys Asn
1 5 10 15
Gln Thr Ser Ile Ala Pro Asn Leu Thr Leu Asn Leu His Thr Ile Asn
20 25 30
Ala Thr Phe Leu Asp Leu Tyr Tyr Glu Met Asn Leu Ile Gln Glu Ser
35 40 45
Ile Lys Ser Leu Asn Asn Ser Tyr Ile Asn Leu Lys Asp Ile Gly Thr
50 55 60
Tyr Glu Met Tyr Val Lys Trp Pro Trp Tyr Val
65 70 75

<210> SEQ ID NO 424
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

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<400> SEQUENCE: 424

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Met Val Arg Met Tyr Ile Phe Phe Ala Ser Phe Tyr Tyr Val Trp Lys
1           5           10           15
Ser Tyr Val His Val Val Asp Gly Cys Asn Ser Ser Thr Cys Met Met
           20           25           30
Cys Tyr Lys Arg Asn Arg Ala Thr Arg Val Glu Cys Thr Thr Ile Val
           35           40           45
Asn Gly Val Arg Arg Ser Phe Tyr Val Tyr Ala Asn Gly Gly Lys Gly
           50           55           60
Phe Cys Lys Leu His Asn Trp Asn Cys Val Asn Cys Asp Thr Phe Cys
65           70           75           80
Ala Gly Ser Thr Phe Ile Ser Asp Glu Val Ala Arg Asp Leu Ser Leu
           85           90           95
Gln Phe Lys Arg Pro Ile Asn Pro Thr Asp
           100           105

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<210> SEQ ID NO 425

<211> LENGTH: 106

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 425

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Phe Met Arg Phe Tyr Ile Ile Ile Ala Ser Phe Ile Lys Leu Phe Ser
1           5           10           15
Leu Phe Arg His Val Ala Tyr Gly Cys Ser Lys Ser Gly Cys Leu Phe
           20           25           30
Cys Tyr Lys Arg Asn Arg Ser Leu Arg Val Lys Cys Ser Thr Ile Val
           35           40           45
Gly Gly Met Ile Arg Tyr Tyr Asp Val Met Ala Asn Gly Gly Thr Gly
           50           55           60
Phe Cys Ser Lys His Gln Trp Asn Cys Ile Asp Cys Asp Ser Tyr Lys
65           70           75           80
Pro Gly Asn Thr Phe Ile Thr Val Glu Ala Ala Leu Asp Leu Ser Lys
           85           90           95
Glu Leu Lys Arg Pro Ile Gln Pro Thr Asp
           100           105

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<210> SEQ ID NO 426

<211> LENGTH: 61

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 426

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Asn Ala Ala Val Phe Tyr Ala Gln Ser Leu Phe Arg Pro Ile Leu Met
1           5           10           15
Val Asp Lys Asn Leu Ile Thr Thr Ala Asn Thr Gly Thr Ser Val Thr
           20           25           30
Glu Thr Met Phe Asp Val Tyr Val Asp Thr Phe Leu Ser Met Phe Asp
           35           40           45
Val Asp Lys Lys Ser Leu Asn Ala Leu Ile Ala Thr Ala
           50           55           60

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<210> SEQ ID NO 427

<211> LENGTH: 61

<212> TYPE: PRT

-continued

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 427

Ser Ala Ser Val Tyr Tyr Ser Gln Leu Met Cys Gln Pro Ile Leu Leu
1 5 10 15
Leu Asp Gln Ala Leu Val Ser Asp Val Gly Asp Ser Ala Glu Val Ala
20 25 30
Val Lys Met Phe Asp Ala Tyr Val Asn Thr Phe Ser Ser Thr Phe Asn
35 40 45
Val Pro Met Glu Lys Leu Lys Thr Leu Val Ala Thr Ala
50 55 60

<210> SEQ ID NO 428

<211> LENGTH: 55

<212> TYPE: PRT

<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 428

Glu Leu Thr Asp Glu Ser Cys Asn Asn Leu Val Pro Thr Tyr Leu Lys
1 5 10 15
Ser Asp Asn Ile Val Ala Ala Asp Leu Gly Val Leu Ile Gln Asn Ser
20 25 30
Ala Lys His Val Gln Gly Asn Val Ala Lys Ile Ala Gly Val Ser Cys
35 40 45
Ile Trp Ser Val Asp Ala Phe
50 55

<210> SEQ ID NO 429

<211> LENGTH: 55

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 429

Glu Val Thr Gly Asp Ser Cys Asn Asn Tyr Met Leu Thr Tyr Asn Lys
1 5 10 15
Val Glu Asn Met Thr Pro Arg Asp Leu Gly Ala Cys Ile Asp Cys Ser
20 25 30
Ala Arg His Ile Asn Ala Gln Val Ala Lys Ser His Asn Ile Ala Leu
35 40 45
Ile Trp Asn Val Lys Asp Phe
50 55

<210> SEQ ID NO 430

<211> LENGTH: 45

<212> TYPE: PRT

<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 430

Ser Ala Val Gly Asn Ile Cys Tyr Thr Pro Ser Lys Leu Ile Glu Tyr
1 5 10 15
Thr Asp Phe Ala Thr Ser Ala Cys Val Leu Ala Ala Glu Cys Thr Ile
20 25 30
Phe Lys Asp Ala Ser Gly Lys Pro Val Pro Tyr Cys Tyr
35 40 45

<210> SEQ ID NO 431

<211> LENGTH: 133

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<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 431

Ser Leu Val Pro His Val Arg Tyr Asn Leu Ala Asn Ala Lys Gly Phe
1      5      10      15
Ile Arg Phe Pro Glu Val Leu Arg Glu Gly Leu Val Arg Ile Val Arg
20      25      30
Thr Arg Ser Met Ser Tyr Cys Arg Val Gly Leu Cys Glu Glu Ala Asp
35      40      45
Glu Gly Ile Cys Phe Asn Phe Asn Gly Ser Trp Val Leu Asn Asn Asp
50      55      60
Tyr Tyr Arg Ser Leu Pro Gly Thr Phe Cys Gly Arg Asp Val Phe Asp
65      70      75      80
Leu Ile Tyr Gln Leu Phe Lys Gly Leu Ala Gln Pro Val Asp Phe Leu
85      90      95
Ala Leu Thr Ala Ser Ser Ile Ala Gly Ala Ile Leu Ala Val Ile Val
100     105     110
Val Leu Val Phe Tyr Tyr Leu Ile Lys Leu Lys Arg Ala Phe Gly Asp
115     120     125

Tyr Thr Ser Val Val
130

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<210> SEQ ID NO 432
<211> LENGTH: 132
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 432

Ser Leu Arg Pro Asp Thr Arg Tyr Val Leu Met Asp Gly Ser Ile Ile
1      5      10      15
Gln Phe Pro Asn Thr Tyr Leu Glu Gly Ser Val Arg Val Val Thr Thr
20      25      30
Phe Asp Ser Glu Tyr Cys Arg His Gly Thr Cys Glu Arg Ser Glu Ala
35      40      45
Gly Val Cys Val Ser Thr Ser Gly Arg Trp Val Leu Asn Asn Asp Tyr
50      55      60
Tyr Arg Ser Leu Pro Gly Val Phe Cys Gly Val Asp Ala Val Asn Leu
65      70      75      80
Leu Thr Asn Met Phe Thr Pro Leu Ile Gln Pro Ile Gly Ala Leu Asp
85      90      95
Ile Ser Ala Ser Ile Val Ala Gly Gly Ile Val Ala Ile Val Val Thr
100     105     110
Cys Leu Ala Tyr Tyr Phe Met Arg Phe Arg Arg Ala Phe Gly Glu Tyr
115     120     125

Ser His Val Val
130

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<210> SEQ ID NO 433
<211> LENGTH: 50
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 433

Asn Arg Tyr Leu Ser Leu Tyr Asn Lys Tyr Arg Tyr Tyr Ser Gly Lys

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1	5	10	15
Met Asp Thr	Ala Ala Tyr Arg	Glu Ala Ala Cys Ser	Gln Leu Ala Lys
	20	25	30
Ala Met Asp Thr Phe Thr Asn	Asn Asn Gly Ser Asp	Val Leu Tyr Gln	
	35	40	45
Pro Pro			
50			

<210> SEQ ID NO 434
 <211> LENGTH: 49
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 434

Asn Arg Tyr Leu Ala Leu Tyr Asn Lys Tyr Lys Tyr Phe Ser Gly Ala	
1	5 10 15
Met Asp Thr Thr Ser Tyr Arg Glu Ala Ala Cys Cys His Leu Ala Lys	
	20 25 30
Ala Leu Asn Asp Phe Ser Asn Ser Gly Ser Asp Val Leu Tyr Gln Pro	
	35 40 45
Pro	

<210> SEQ ID NO 435
 <211> LENGTH: 84
 <212> TYPE: PRT
 <213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 435

Ile Lys Ile Val Leu Leu Cys Tyr Leu Phe Ile Gly Tyr Ile Ile Ser	
1	5 10 15
Cys Tyr Trp Gly Leu Phe Ser Leu Met Asn Ser Leu Phe Arg Met Pro	
	20 25 30
Leu Gly Val Tyr Asn Tyr Lys Ile Ser Val Gln Glu Leu Arg Tyr Met	
	35 40 45
Asn Ala Asn Gly Leu Arg Pro Pro Lys Asn Ser Phe Glu Ala Leu Met	
	50 55 60
Leu Asn Phe Lys Leu Leu Gly Ile Gly Gly Val Pro Ile Ile Glu Val	
65	70 75 80
Ser Gln Phe Gln	

<210> SEQ ID NO 436
 <211> LENGTH: 84
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 436

Leu Gln Cys Ile Met Leu Val Tyr Cys Phe Leu Gly Tyr Phe Cys Thr	
1	5 10 15
Cys Tyr Phe Gly Leu Phe Cys Leu Leu Asn Arg Tyr Phe Arg Leu Thr	
	20 25 30
Leu Gly Val Tyr Asp Tyr Leu Val Ser Thr Gln Glu Phe Arg Tyr Met	
	35 40 45
Asn Ser Gln Gly Leu Leu Pro Pro Lys Asn Ser Ile Asp Ala Phe Lys	
	50 55 60
Leu Asn Ile Lys Leu Leu Gly Val Gly Gly Lys Pro Cys Ile Lys Val	

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65	70	75	80
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Ala Thr Val Gln

<210> SEQ ID NO 437
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 437

Ser Leu Ala Met Leu Leu Val Lys His Lys His Leu Tyr Leu Thr Met
1 5 10 15

Tyr Ile Thr Pro Val Leu Phe Thr Leu Leu Tyr Asn Asn Tyr Leu
20 25 30

<210> SEQ ID NO 438
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 438

Ala Phe Ala Met Met Phe Val Lys His Lys His Ala Phe Leu Cys Leu
1 5 10 15

Phe Leu Leu Pro Ser Leu Ala Thr Val Ala Tyr Phe Asn Met Val
20 25 30

<210> SEQ ID NO 439
<211> LENGTH: 67
<212> TYPE: PRT
<213> ORGANISM: Betacoronavirus 1

<400> SEQUENCE: 439

Ser Lys Leu Thr Asp Val Lys Cys Ala Asn Val Val Leu Leu Asn Cys
1 5 10 15

Leu Gln His Leu His Val Ala Ser Asn Ser Lys Leu Trp His Tyr Cys
20 25 30

Ser Thr Leu His Asn Glu Ile Leu Ala Thr Ser Asp Leu Ser Val Ala
35 40 45

Phe Glu Lys Leu Ala Gln Leu Leu Ile Val Leu Phe Ala Asn Pro Ala
50 55 60

Ala Val Asp
65

<210> SEQ ID NO 440
<211> LENGTH: 67
<212> TYPE: PRT
<213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 440

Ser Lys Met Ser Asp Val Lys Cys Thr Ser Val Val Leu Leu Ser Val
1 5 10 15

Leu Gln Gln Leu Arg Val Glu Ser Ser Ser Lys Leu Trp Ala Gln Cys
20 25 30

Val Gln Leu His Asn Asp Ile Leu Leu Ala Lys Asp Thr Thr Glu Ala
35 40 45

Phe Glu Lys Met Val Ser Leu Leu Ser Val Leu Leu Ser Met Gln Gly
50 55 60

Ala Val Asp

-continued

65

<210> SEQ ID NO 441

<211> LENGTH: 495

<212> TYPE: PRT

<213> ORGANISM: Deltapapillomavirus 4

<400> SEQUENCE: 441

Met Ala Leu Trp Gln Gln Gly Gln Lys Leu Tyr Leu Pro Pro Thr Pro
1 5 10 15

Val Ser Lys Val Leu Cys Ser Glu Thr Tyr Val Gln Arg Lys Ser Ile
20 25 30

Phe Tyr His Ala Glu Thr Glu Arg Leu Leu Thr Ile Gly His Pro Tyr
35 40 45

Tyr Pro Val Ser Ile Gly Ala Lys Thr Val Pro Lys Val Ser Ala Asn
50 55 60

Gln Tyr Arg Val Phe Lys Ile Gln Leu Pro Asp Pro Asn Gln Phe Ala
65 70 75 80

Leu Pro Asp Arg Thr Val His Asn Pro Ser Lys Glu Arg Leu Val Trp
85 90 95

Ala Val Ile Gly Val Gln Val Ser Arg Gly Gln Pro Leu Gly Gly Thr
100 105 110

Val Thr Gly His Pro Thr Phe Asn Ala Leu Leu Asp Ala Glu Asn Val
115 120 125

Asn Arg Lys Val Thr Thr Gln Thr Thr Asp Asp Arg Lys Gln Thr Gly
130 135 140

Leu Asp Ala Lys Gln Gln Gln Ile Leu Leu Leu Gly Cys Thr Pro Ala
145 150 155 160

Glu Gly Glu Tyr Trp Thr Thr Ala Arg Pro Cys Val Thr Asp Arg Leu
165 170 175

Glu Asn Gly Ala Cys Pro Pro Leu Glu Leu Lys Asn Lys His Ile Glu
180 185 190

Asp Gly Asp Met Met Glu Ile Gly Phe Gly Ala Ala Asn Phe Lys Glu
195 200 205

Ile Asn Ala Ser Lys Ser Asp Leu Pro Leu Asp Ile Gln Asn Glu Ile
210 215 220

Cys Leu Tyr Pro Asp Tyr Leu Lys Met Ala Glu Asp Ala Ala Gly Asn
225 230 235 240

Ser Met Phe Phe Phe Ala Arg Lys Glu Gln Val Tyr Val Arg His Ile
245 250 255

Trp Thr Arg Gly Gly Ser Glu Lys Glu Ala Pro Thr Thr Asp Phe Tyr
260 265 270

Leu Lys Asn Asn Lys Gly Asp Ala Thr Leu Lys Ile Pro Ser Val His
275 280 285

Phe Gly Ser Pro Ser Gly Ser Leu Val Ser Thr Asp Asn Gln Ile Phe
290 295 300

Asn Arg Pro Tyr Trp Leu Phe Arg Ala Gln Gly Met Asn Asn Gly Ile
305 310 315 320

Ala Trp Asn Asn Leu Leu Phe Leu Thr Val Gly Asp Asn Thr Arg Gly
325 330 335

Thr Asn Leu Thr Ile Ser Val Ala Ser Asp Gly Thr Pro Leu Thr Glu
340 345 350

-continued

Tyr Asp Ser Ser Lys Phe Asn Val Tyr His Arg His Met Glu Glu Tyr
 355 360 365
 Lys Leu Ala Phe Ile Leu Glu Leu Cys Ser Val Glu Ile Thr Ala Gln
 370 375 380
 Thr Val Ser His Leu Gln Gly Leu Met Pro Ser Val Leu Glu Asn Trp
 385 390 395 400
 Glu Ile Gly Val Gln Pro Pro Thr Ser Ser Ile Leu Glu Asp Thr Tyr
 405 410 415
 Arg Tyr Ile Glu Ser Pro Ala Thr Lys Cys Ala Ser Asn Val Ile Pro
 420 425 430
 Ala Lys Glu Asp Pro Tyr Ala Gly Phe Lys Phe Trp Asn Ile Asp Leu
 435 440 445
 Lys Glu Lys Leu Ser Leu Asp Leu Asp Gln Phe Pro Leu Gly Arg Arg
 450 455 460
 Phe Leu Ala Gln Gln Gly Ala Gly Cys Ser Thr Val Arg Lys Arg Arg
 465 470 475 480
 Ile Ser Gln Lys Thr Ser Ser Lys Pro Ala Lys Lys Lys Lys Lys
 485 490 495

<210> SEQ ID NO 442
 <211> LENGTH: 91
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 442

Tyr Phe Asn Lys Lys Asp Trp Tyr Asp Phe Val Glu Asn Pro Asp Ile
 1 5 10 15
 Leu Arg Val Tyr Ala Asn Leu Gly Glu Arg Val Arg Gln Ala Leu Leu
 20 25 30
 Lys Thr Val Gln Phe Cys Asp Ala Met Arg Asn Ala Gly Ile Val Gly
 35 40 45
 Val Leu Thr Leu Asp Asn Gln Asp Leu Asn Gly Asn Trp Tyr Asp Phe
 50 55 60
 Gly Asp Phe Ile Gln Thr Thr Pro Gly Ser Gly Val Pro Val Val Asp
 65 70 75 80
 Ser Tyr Tyr Ser Leu Leu Met Pro Ile Leu Thr
 85 90

<210> SEQ ID NO 443
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 443

Asp Leu Leu Lys Tyr Asp Phe Thr Glu Glu Arg Leu Lys Leu Phe Asp
 1 5 10 15
 Arg Tyr Phe Lys Tyr Trp Asp Gln Thr Tyr His Pro Asn Cys Val Asn
 20 25 30
 Cys Leu Asp Asp Arg Cys Ile Leu His Cys Ala Asn Phe Asn Val Leu
 35 40 45
 Phe Ser Thr Val Phe Pro Pro Thr Ser Phe Gly Pro Leu Val Arg Lys
 50 55 60
 Ile Phe Val Asp Gly Val Pro Phe Val Val Ser Thr Gly Tyr His Phe
 65 70 75 80

-continued

Arg Glu Leu Gly Val Val His Asn Gln Asp Val Asn Leu His Ser Ser
 85 90 95

Arg Leu Ser Phe Lys Glu Leu Leu Val Tyr Ala
 100 105

<210> SEQ ID NO 444
 <211> LENGTH: 341
 <212> TYPE: PRT
 <213> ORGANISM: Severe acute respiratory syndrome coronavirus 2

<400> SEQUENCE: 444

Ala Tyr Thr Lys Arg Asn Val Ile Pro Thr Ile Thr Gln Met Asn Leu
 1 5 10 15

Lys Tyr Ala Ile Ser Ala Lys Asn Arg Ala Arg Thr Val Ala Gly Val
 20 25 30

Ser Ile Cys Ser Thr Met Thr Asn Arg Gln Phe His Gln Lys Leu Leu
 35 40 45

Lys Ser Ile Ala Ala Thr Arg Gly Ala Thr Val Val Ile Gly Thr Ser
 50 55 60

Lys Phe Tyr Gly Gly Trp His Asn Met Leu Lys Thr Val Tyr Ser Asp
 65 70 75 80

Val Glu Asn Pro His Leu Met Gly Trp Asp Tyr Pro Lys Cys Asp Arg
 85 90 95

Ala Met Pro Asn Met Leu Arg Ile Met Ala Ser Leu Val Leu Ala Arg
 100 105 110

Lys His Thr Thr Cys Cys Ser Leu Ser His Arg Phe Tyr Arg Leu Ala
 115 120 125

Asn Glu Cys Ala Gln Val Leu Ser Glu Met Val Met Cys Gly Gly Ser
 130 135 140

Leu Tyr Val Lys Pro Gly Gly Thr Ser Ser Gly Asp Ala Thr Thr Ala
 145 150 155 160

Tyr Ala Asn Ser Val Phe Asn Ile Cys Gln Ala Val Thr Ala Asn Val
 165 170 175

Asn Ala Leu Leu Ser Thr Asp Gly Asn Lys Ile Ala Asp Lys Tyr Val
 180 185 190

Arg Asn Leu Gln His Arg Leu Tyr Glu Cys Leu Tyr Arg Asn Arg Asp
 195 200 205

Val Asp Thr Asp Phe Val Asn Glu Phe Tyr Ala Tyr Leu Arg Lys His
 210 215 220

Phe Ser Met Met Ile Leu Ser Asp Asp Ala Val Val Cys Phe Asn Ser
 225 230 235 240

Thr Tyr Ala Ser Gln Gly Leu Val Ala Ser Ile Lys Asn Phe Lys Ser
 245 250 255

Val Leu Tyr Tyr Gln Asn Asn Val Phe Met Ser Glu Ala Lys Cys Trp
 260 265 270

Thr Glu Thr Asp Leu Thr Lys Gly Pro His Glu Phe Cys Ser Gln His
 275 280 285

Thr Met Leu Val Lys Gln Gly Asp Asp Tyr Val Tyr Leu Pro Tyr Pro
 290 295 300

Asp Pro Ser Arg Ile Leu Gly Ala Gly Cys Phe Val Asp Asp Ile Val
 305 310 315 320

-continued

Lys Thr Asp Gly Thr Leu Met Ile Glu Arg Phe Val Ser Leu Ala Ile
 325 330 335

Asp Ala Tyr Pro Leu
 340

1. A composition comprising (a) a chimeric papillomavirus virus-like particle (VLP) comprising an L1 protein, having an amino acid insert inserted into the HI loop of the L1 protein and (b) a coronavirus antigen.

2. A composition according to claim 1 wherein said amino acid insert comprises a contiguous sequence of negatively charged amino acids and a terminal cysteine residue.

3. (canceled)

4. (canceled)

5. A composition according to claim 2, wherein said amino acid insert is selected from the inserts identified in Table 1.

6. A composition according to claim 1, wherein said amino acid insert is inserted into the HI loop of the L1 protein at a location between positions 344 and 357.

7. (canceled)

8. A composition according to claim 6, wherein said amino acid insert replaces the native amino acid sequences identified in Table 1.

9. A composition according to claim 1, wherein said coronavirus antigen is selected from the group consisting of an OC43 antigen, an HKU1 antigen, a 229E antigen, an NL63 antigen, a SARS-CoV-1 antigen, a MERS antigen, a SARS-CoV-2 antigen, and fusions thereof.

10. (canceled)

11. A composition according to claim 9, wherein said coronavirus antigen comprises a viral structural protein selected from the group consisting of the membrane protein (M), the nucleocapsid protein (N), and the S2 region of the spike(S) envelope protein from OC43, HKU1, 229E, NL63, SARS-CoV-1, MERS, and/or SARS-CoV-2 coronaviruses or a viral non-structural protein selected from the group consisting of nsp3, nsp4, nsp6, nsp7, and nsp12 proteins from OC43, HKU1, 229E, NL63, SARS-CoV-1, MERS, and/or SARS-CoV-2 coronaviruses.

12. (canceled)

13. (canceled)

14. (canceled)

15. A composition according to claim 9, wherein said coronavirus antigen is selected from one or more of the amino acid sequences in Tables 3-10.

16. (canceled)

17. (canceled)

18. (canceled)

19. (canceled)

20. (canceled)

21. (canceled)

22. (canceled)

23. A composition according to claim 9, wherein said coronavirus antigen comprises one or more viral proteins of

the OC43 coronavirus and/or HKU1 the coronavirus, wherein said one or more viral proteins comprise 70% or greater identity with an amino acid sequence of SARS-CoV-1 coronavirus, MERS coronavirus, and/or SARS-CoV-2 coronavirus.

24. (canceled)

25. A composition according to claim 9, wherein said OC43 and/or HKU1 coronavirus structural protein antigens is selected from one or more of the amino acid sequences in Tables 11-13 and/or said OC43 coronavirus nonstructural antigen is selected from one or more of the amino acid sequences in Tables 14-17.

26. (canceled)

27. (canceled)

28. (canceled)

29. A composition according to claim 9, wherein said coronavirus antigen comprises a first antigen selected from the structural proteins and peptides of OC43 and/or HKU1 coronavirus, a second antigen selected from the non-structural proteins and peptides of OC43, a third antigen selected from the structural proteins and peptides of SARS-CoV-2 coronavirus, and a fourth antigen selected from the non-structural proteins and peptides of SARS-CoV-2.

30. A composition according to claim 1, further comprising a TAG sequence linked at a first end to said amino acid insert and linked at a second end to said coronavirus antigen.

31. (canceled)

32. (canceled)

33. (canceled)

34. A composition according to claim 30, wherein said TAG sequence comprises an amino acid sequence identified in Table 2.

35. A composition according to claim 1, wherein said composition is effective to stimulate a cytotoxic T cell response in a mammal.

36. A method for stimulating a cytotoxic T cell response to a coronavirus in a mammal or for stimulating both therapeutic and protective immunity to a coronavirus in said mammal comprising administering to said mammal a composition according to any one of claims 1-14.

37. (canceled)

38. (canceled)

39. (canceled)

40. (canceled)

41. (canceled)

42. (canceled)

43. (canceled)

44. (canceled)

45. (canceled)

* * * * *